

Chemistry 2e

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CHAPTER 19

Transition Metals and Coordination Chemistry

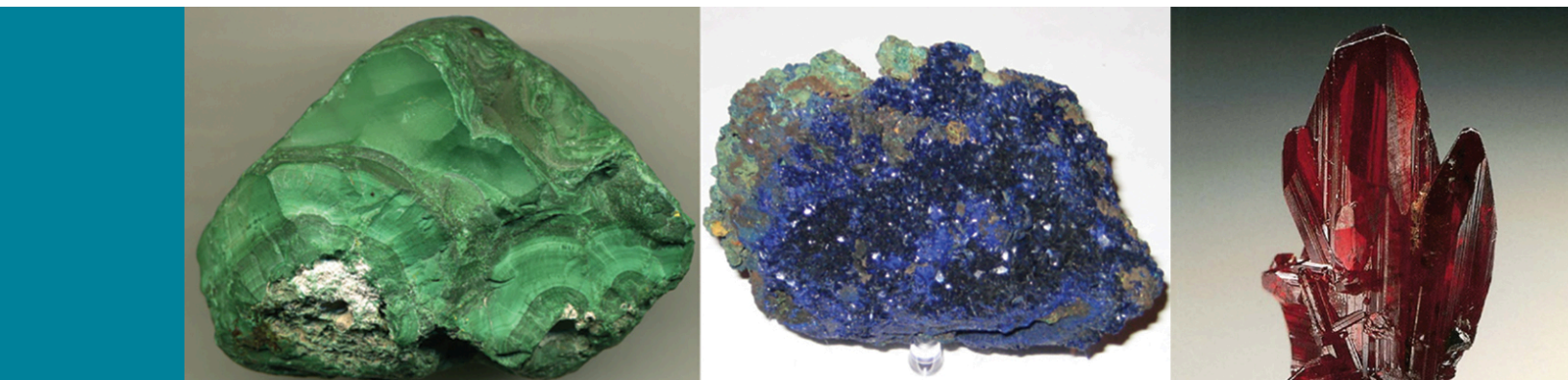


FIGURE 19.1 Transition metals often form vibrantly colored complexes. The minerals malachite (green), azurite (blue), and proustite (red) are some examples. (credit left: modification of work by James St. John; credit middle: modification of work by Stephanie Clifford; credit right: modification of work by Terry Wallace)

CHAPTER OUTLINE

19.1 Occurrence, Preparation, and Properties of Transition Metals and Their Compounds

19.2 Coordination Chemistry of Transition Metals

19.3 Spectroscopic and Magnetic Properties of Coordination Compounds

INTRODUCTION We have daily contact with many transition metals. Iron occurs everywhere—from the rings in your spiral notebook and the cutlery in your kitchen to automobiles, ships, buildings, and in the hemoglobin in your blood. Titanium is useful in the manufacture of lightweight, durable products such as bicycle frames, artificial hips, and jewelry. Chromium is useful as a protective plating on plumbing fixtures and automotive detailing.

In addition to being used in their pure elemental forms, many compounds containing transition metals have numerous other applications. Silver nitrate is used to create mirrors, zirconium silicate provides friction in automotive brakes, and many important cancer-fighting agents, like the drug cisplatin and related species, are platinum compounds.

The variety of properties exhibited by transition metals is due to their complex valence shells. Unlike most main group metals where one oxidation state is normally observed, the valence shell structure of transition metals means that they usually occur in several different stable oxidation states. In addition, electron transitions in these elements can correspond with absorption of photons in the visible electromagnetic spectrum, leading to colored compounds. Because of these behaviors, transition metals exhibit a rich and fascinating chemistry.

19.1 Occurrence, Preparation, and Properties of Transition Metals and Their Compounds

Transition metals are defined as those elements that have (or readily form) partially filled *d* orbitals. As shown in [Figure 19.2](#), the ***d*-block elements** in groups 3–11 are transition elements. The ***f*-block elements**, also called *inner transition metals* (the lanthanides and actinides), also meet this criterion because the *d* orbital is partially occupied before the *f* orbitals. The *d* orbitals fill with the copper family (group 11); for this reason, the next family (group 12) are technically not transition elements. However, the group 12 elements do display some of the same chemical properties and are commonly included in discussions of transition metals. Some chemists do treat the group 12 elements as transition metals.

Period

Group

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

1	1	1.008 hydrogen																	2	4.003 helium																
2	3	6.94 lithium	4	9.012 beryllium										5	10.81 boron	6	12.01 carbon	7	14.01 nitrogen	8	16.00 oxygen	9	19.00 fluorine	10	20.18 neon											
3	11	22.99 sodium	12	24.31 magnesium										13	26.98 aluminum	14	28.09 silicon	15	30.97 phosphorus	16	32.06 sulfur	17	35.45 chlorine	18	39.95 argon											
4	19	39.10 potassium	20	40.08 calcium	21	Sc 44.96 scandium	22	Ti 47.87 titanium	23	V 50.94 vanadium	24	Cr 52.00 chromium	25	Mn 54.94 manganese	26	Fe 55.85 iron	27	Co 58.93 cobalt	28	Ni 58.69 nickel	29	Cu 63.55 copper	30	Zn 65.38 zinc	31	Ga 69.72 gallium	32	Ge 72.63 germanium	33	As 74.92 arsenic	34	Se 78.97 selenium	35	Br 79.90 bromine	36	Kr 83.80 krypton
5	37	Rb 85.47 rubidium	38	Sr 87.62 strontium	39	Y 88.91 yttrium	40	Zr 91.22 zirconium	41	Nb 92.91 niobium	42	Mo 95.95 molybdenum	43	Tc [97] technetium	44	Ru 101.1 ruthenium	45	Rh 102.9 rhodium	46	Pd 106.4 palladium	47	Ag 107.9 silver	48	Cd 112.4 cadmium	49	In 114.8 indium	50	Sn 118.7 tin	51	Sb 121.8 antimony	52	Te 127.6 tellurium	53	I 126.9 iodine	54	Xe 131.3 xenon
6	55	Cs 132.9 cesium	56	Ba 137.3 barium	57-71	La-Lu *	72	Hf 178.5 hafnium	73	Ta 180.9 tantalum	74	W 183.8 tungsten	75	Re 186.2 rhenium	76	Os 190.2 osmium	77	Ir 192.2 iridium	78	Pt 195.1 platinum	79	Au 197.0 gold	80	Hg 200.6 mercury	81	Tl 204.4 thallium	82	Pb 207.2 lead	83	Bi 209.0 bismuth	84	Po [209] polonium	85	At [210] astatine	86	Rn [222] radon
7	87	Fr [223] francium	88	Ra [226] radium	89-103	Ac-Lr **	104	Rf [267] rutherfordium	105	Db [270] dubnium	106	Sg [271] seaborgium	107	Bh [277] bohrium	108	Hs [277] hassium	109	Mt [276] meitnerium	110	Ds [281] darmstadtium	111	Rg [282] roentgenium	112	Cn [285] copernicium	113	Nh [285] nihonium	114	Fl [289] flerovium	115	Mc [288] moscovium	116	Lv [293] livermorium	117	Ts [294] tennessine	118	Og [294] oganesson
					57	La 138.9 lanthanum	58	Ce 140.1 cerium	59	Pr 140.9 praseodymium	60	Nd 144.2 neodymium	61	Pm [145] promethium	62	Sm 150.4 samarium	63	Eu 152.0 europium	64	Gd 157.3 gadolinium	65	Tb 158.9 terbium	66	Dy 162.5 dysprosium	67	Ho 164.9 holmium	68	Er 167.3 erbium	69	Tm 168.9 thulium	70	Yb 173.1 ytterbium	71	Lu 175.0 lutetium		
					89	Ac [227] actinium	90	Th 232.0 thorium	91	Pa 231.0 protactinium	92	U 238.0 uranium	93	Np [237] neptunium	94	Pu [244] plutonium	95	Am [243] americium	96	Cm [247] curium	97	Bk [247] berkelium	98	Cf [251] californium	99	Es [252] einsteinium	100	Fm [257] fermium	101	Md [258] mendelevium	102	No [259] nobelium	103	Lr [262] lawrencium		

Atomic number → 1

H

1.008

hydrogen

Symbol

Atomic mass

Color Code

Metal

Metalloid

Nonmetal

Solid

Liquid

Gas

FIGURE 19.2 The transition metals are located in groups 3–11 of the periodic table. The inner transition metals are in the two rows below the body of the table.

The *d*-block elements are divided into the **first transition series** (the elements Sc through Cu), the **second transition series** (the elements Y through Ag), and the **third transition series** (the element La and the elements Hf through Au). Actinium, Ac, is the first member of the **fourth transition series**, which also includes Rf through Rg.

The *f*-block elements are the elements Ce through Lu, which constitute the **lanthanide series** (or **lanthanoid series**), and the elements Th through Lr, which constitute the **actinide series** (or **actinoid series**). Because lanthanum behaves very much like the lanthanide elements, it is considered a lanthanide element, even though its electron configuration makes it the first member of the third transition series. Similarly, the behavior of actinium means it is part of the actinide series, although its electron configuration makes it the first member of the fourth transition series.



EXAMPLE 19.1

Valence Electrons in Transition Metals

Review how to write electron configurations, covered in the chapter on electronic structure and periodic properties of elements. Recall that for the transition and inner transition metals, it is necessary to remove the *s* electrons before the *d* or *f* electrons. Then, for each ion, give the electron configuration:

- cerium(III)
- lead(II)
- Ti^{2+}

(d) Am^{3+}

(e) Pd^{2+}

For the examples that are transition metals, determine to which series they belong.

Solution

For ions, the s -valence electrons are lost prior to the d or f electrons.

(a) $\text{Ce}^{3+}[\text{Xe}]4f^1$; Ce^{3+} is an inner transition element in the lanthanide series.

(b) $\text{Pb}^{2+}[\text{Xe}]6s^25d^{10}4f^{14}$; the electrons are lost from the p orbital. This is a main group element.

(c) titanium(II) $[\text{Ar}]3d^2$; first transition series

(d) americium(III) $[\text{Rn}]5f^6$; actinide

(e) palladium(II) $[\text{Kr}]4d^8$; second transition series

Check Your Learning

Give an example of an ion from the first transition series with no d electrons.

Answer:

V^{5+} is one possibility. Other examples include Sc^{3+} , Ti^{4+} , Cr^{6+} , and Mn^{7+} .

CHEMISTRY IN EVERYDAY LIFE

Uses of Lanthanides in Devices

Lanthanides (elements 57–71) are fairly abundant in the earth's crust, despite their historic characterization as **rare earth elements**. Thulium, the rarest naturally occurring lanthanoid, is more common in the earth's crust than silver ($4.5 \times 10^{-5}\%$ versus $0.79 \times 10^{-5}\%$ by mass). There are 17 rare earth elements, consisting of the 15 lanthanoids plus scandium and yttrium. They are called rare because they were once difficult to extract economically, so it was rare to have a pure sample; due to similar chemical properties, it is difficult to separate any one lanthanide from the others. However, newer separation methods, such as ion exchange resins similar to those found in home water softeners, make the separation of these elements easier and more economical. Most ores that contain these elements have low concentrations of all the rare earth elements mixed together.

The commercial applications of lanthanides are growing rapidly. For example, europium is important in flat screen displays found in computer monitors, cell phones, and televisions. Neodymium is useful in laptop hard drives and in the processes that convert crude oil into gasoline (Figure 19.3). Holmium is found in dental and medical equipment. In addition, many alternative energy technologies rely heavily on lanthanoids. Neodymium and dysprosium are key components of hybrid vehicle engines and the magnets used in wind turbines.

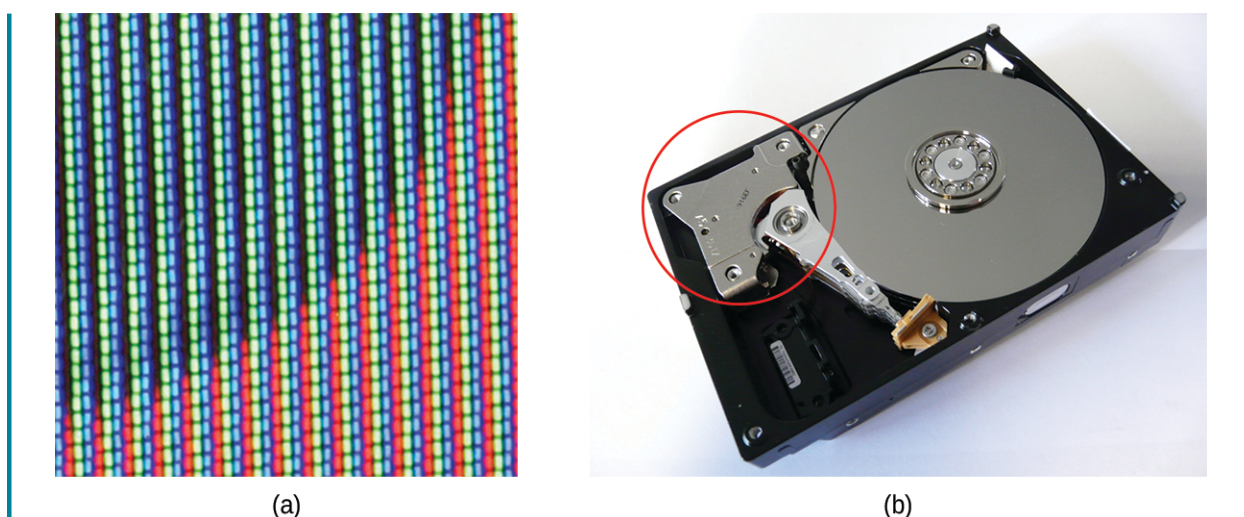


FIGURE 19.3 (a) Europium is used in display screens for televisions, computer monitors, and cell phones. (b) Neodymium magnets are commonly found in computer hard drives. (credit b: modification of work by “KUERT Datenrettung”/Flickr)

As the demand for lanthanide materials has increased faster than supply, prices have also increased. In 2008, dysprosium cost \$110/kg; by 2014, the price had increased to \$470/kg. Increasing the supply of lanthanoid elements is one of the most significant challenges facing the industries that rely on the optical and magnetic properties of these materials.

The transition elements have many properties in common with other metals. They are almost all hard, high-melting solids that conduct heat and electricity well. They readily form alloys and lose electrons to form stable cations. In addition, transition metals form a wide variety of stable **coordination compounds**, in which the central metal atom or ion acts as a Lewis acid and accepts one or more pairs of electrons. Many different molecules and ions can donate lone pairs to the metal center, serving as Lewis bases. In this chapter, we shall focus primarily on the chemical behavior of the elements of the first transition series.

Properties of the Transition Elements

Transition metals demonstrate a wide range of chemical behaviors. As can be seen from their reduction potentials (see [Appendix H](#)), some transition metals are strong reducing agents, whereas others have very low reactivity. For example, the lanthanides all form stable 3+ aqueous cations. The driving force for such oxidations is similar to that of alkaline earth metals such as Be or Mg, forming Be^{2+} and Mg^{2+} . On the other hand, materials like platinum and gold have much higher reduction potentials. Their ability to resist oxidation makes them useful materials for constructing circuits and jewelry.

Ions of the lighter *d*-block elements, such as Cr^{3+} , Fe^{3+} , and Co^{2+} , form colorful hydrated ions that are stable in water. However, ions in the period just below these (Mo^{3+} , Ru^{3+} , and Ir^{2+}) are unstable and react readily with oxygen from the air. The majority of simple, water-stable ions formed by the heavier *d*-block elements are oxoanions such as MoO_4^{2-} and ReO_4^- .

Ruthenium, osmium, rhodium, iridium, palladium, and platinum are the **platinum metals**. With difficulty, they form simple cations that are stable in water, and, unlike the earlier elements in the second and third transition series, they do not form stable oxoanions.

Both the *d*- and *f*-block elements react with nonmetals to form binary compounds; heating is often required. These elements react with halogens to form a variety of halides ranging in oxidation state from 1+ to 6+. On heating, oxygen reacts with all of the transition elements except palladium, platinum, silver, and gold. The oxides of these latter metals can be formed using other reactants, but they decompose upon heating. The *f*-block elements, the elements of group 3, and the elements of the first transition series except copper react with aqueous solutions of acids, forming hydrogen gas and solutions of the corresponding salts.

Transition metals can form compounds with a wide range of oxidation states. Some of the observed oxidation states

of the elements of the first transition series are shown in [Figure 19.4](#). As we move from left to right across the first transition series, we see that the number of common oxidation states increases at first to a maximum towards the middle of the table, then decreases. The values in the table are typical values; there are other known values, and it is possible to synthesize new additions. For example, in 2014, researchers were successful in synthesizing a new oxidation state of iridium (9+).

21	22	23	24	25	26	27	28	29	30
Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
								1+	
3+	3+	2+	2+	2+	2+	2+	2+	2+	2+
	4+	3+	3+	3+	3+	3+	3+	3+	
		4+	4+	4+					
		5+							
			6+	6+	6+				
				7+					

FIGURE 19.4 Transition metals of the first transition series can form compounds with varying oxidation states.

For the elements scandium through manganese (the first half of the first transition series), the highest oxidation state corresponds to the loss of all of the electrons in both the *s* and *d* orbitals of their valence shells. The titanium(IV) ion, for example, is formed when the titanium atom loses its two *3d* and two *4s* electrons. These highest oxidation states are the most stable forms of scandium, titanium, and vanadium. However, it is not possible to continue to remove all of the valence electrons from metals as we continue through the series. Iron is known to form oxidation states from 2+ to 6+, with iron(II) and iron(III) being the most common. Most of the elements of the first transition series form ions with a charge of 2+ or 3+ that are stable in water, although those of the early members of the series can be readily oxidized by air.

The elements of the second and third transition series generally are more stable in higher oxidation states than are the elements of the first series. In general, the atomic radius increases down a group, which leads to the ions of the second and third series being larger than are those in the first series. Removing electrons from orbitals that are located farther from the nucleus is easier than removing electrons close to the nucleus. For example, molybdenum and tungsten, members of group 6, are limited mostly to an oxidation state of 6+ in aqueous solution. Chromium, the lightest member of the group, forms stable Cr^{3+} ions in water and, in the absence of air, less stable Cr^{2+} ions. The sulfide with the highest oxidation state for chromium is Cr_2S_3 , which contains the Cr^{3+} ion. Molybdenum and tungsten form sulfides in which the metals exhibit oxidation states of 4+ and 6+.



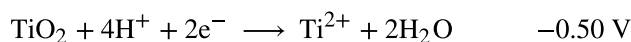
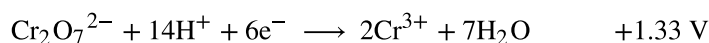
EXAMPLE 19.2

Activity of the Transition Metals

Which is the strongest oxidizing agent in acidic solution: dichromate ion, which contains chromium(VI), permanganate ion, which contains manganese(VII), or titanium dioxide, which contains titanium(IV)?

Solution

First, we need to look up the reduction half reactions (in [Appendix L](#)) for each oxide in the specified oxidation state:

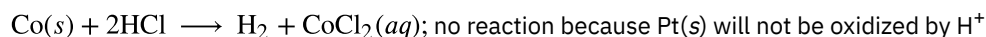


A larger reduction potential means that it is easier to reduce the reactant. Permanganate, with the largest reduction potential, is the strongest oxidizer under these conditions. Dichromate is next, followed by titanium dioxide as the weakest oxidizing agent (the hardest to reduce) of this set.

Check Your Learning

Predict what reaction (if any) will occur between HCl and Co(s), and between HBr and Pt(s). You will need to use the standard reduction potentials from [Appendix L](#).

Answer:



Preparation of the Transition Elements

Ancient civilizations knew about iron, copper, silver, and gold. The time periods in human history known as the Bronze Age and Iron Age mark the advancements in which societies learned to isolate certain metals and use them to make tools and goods. Naturally occurring ores of copper, silver, and gold can contain high concentrations of these metals in elemental form (Figure 19.5). Iron, on the other hand, occurs on earth almost exclusively in oxidized forms, such as rust (Fe_2O_3). The earliest known iron implements were made from iron meteorites. Surviving iron artifacts dating from approximately 4000 to 2500 BC are rare, but all known examples contain specific alloys of iron and nickel that occur only in extraterrestrial objects, not on earth. It took thousands of years of technological advances before civilizations developed iron **smelting**, the ability to extract a pure element from its naturally occurring ores and for iron tools to become common.

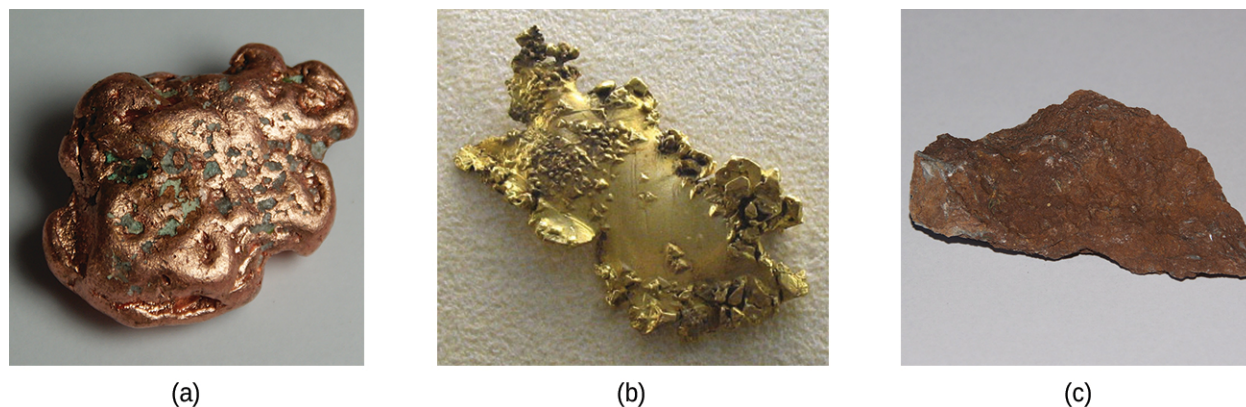


FIGURE 19.5 Transition metals occur in nature in various forms. Examples include (a) a nugget of copper, (b) a deposit of gold, and (c) an ore containing oxidized iron. (credit a: modification of work by <http://images-of-elements.com/copper-2.jpg>; credit c: modification of work by <http://images-of-elements.com/iron-ore.jpg>)

Generally, the transition elements are extracted from minerals found in a variety of ores. However, the ease of their recovery varies widely, depending on the concentration of the element in the ore, the identity of the other elements present, and the difficulty of reducing the element to the free metal.

In general, it is not difficult to reduce ions of the *d*-block elements to the free element. Carbon is a sufficiently strong reducing agent in most cases. However, like the ions of the more active main group metals, ions of the *f*-block elements must be isolated by electrolysis or by reduction with an active metal such as calcium.

We shall discuss the processes used for the isolation of iron, copper, and silver because these three processes illustrate the principal means of isolating most of the *d*-block metals. In general, each of these processes involves three principal steps: preliminary treatment, smelting, and refining.

1. Preliminary treatment. In general, there is an initial treatment of the ores to make them suitable for the extraction of the metals. This usually involves crushing or grinding the ore, concentrating the metal-bearing components, and sometimes treating these substances chemically to convert them into compounds that are easier to reduce to the metal.
2. Smelting. The next step is the extraction of the metal in the molten state, a process called smelting, which includes reduction of the metallic compound to the metal. Impurities may be removed by the addition of a compound that forms a slag—a substance with a low melting point that can be readily separated from the molten metal.
3. Refining. The final step in the recovery of a metal is refining the metal. Low boiling metals such as zinc and mercury can be refined by distillation. When fused on an inclined table, low melting metals like tin flow away from higher-melting impurities. Electrolysis is another common method for refining metals.

Isolation of Iron

The early application of iron to the manufacture of tools and weapons was possible because of the wide distribution of iron ores and the ease with which iron compounds in the ores could be reduced by carbon. For a long time, charcoal was the form of carbon used in the reduction process. The production and use of iron became much more widespread about 1620, when coke was introduced as the reducing agent. Coke is a form of carbon formed by

heating coal in the absence of air to remove impurities.

The first step in the metallurgy of iron is usually roasting the ore (heating the ore in air) to remove water, decomposing carbonates into oxides, and converting sulfides into oxides. The oxides are then reduced in a blast furnace that is 80–100 feet high and about 25 feet in diameter ([Figure 19.6](#)) in which the roasted ore, coke, and limestone (impure CaCO_3) are introduced continuously into the top. Molten iron and slag are withdrawn at the bottom. The entire stock in a furnace may weigh several hundred tons.

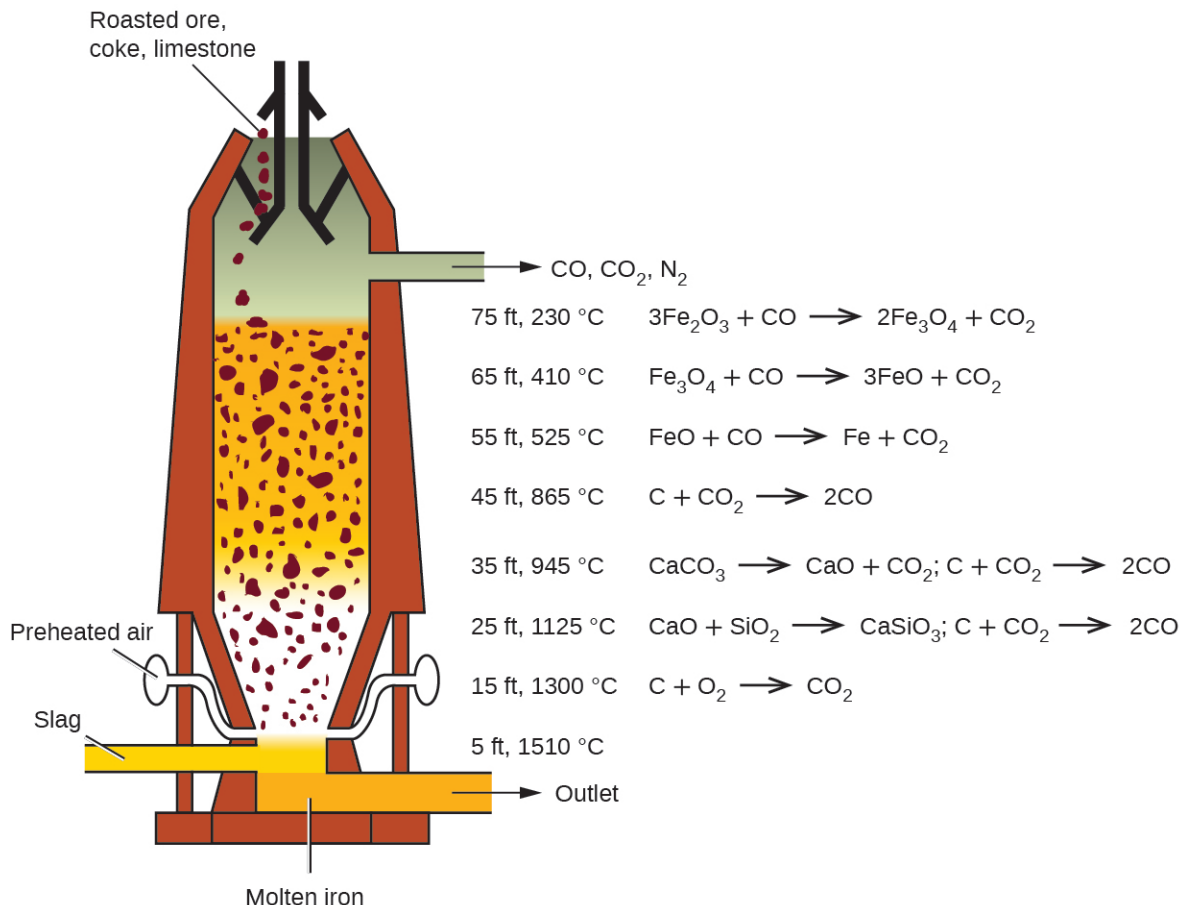
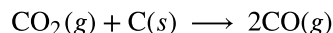


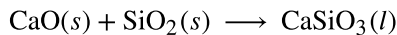
FIGURE 19.6 Within a blast furnace, different reactions occur in different temperature zones. Carbon monoxide is generated in the hotter bottom regions and rises upward to reduce the iron oxides to pure iron through a series of reactions that take place in the upper regions.

Near the bottom of a furnace are nozzles through which preheated air is blown into the furnace. As soon as the air enters, the coke in the region of the nozzles is oxidized to carbon dioxide with the liberation of a great deal of heat. The hot carbon dioxide passes upward through the overlying layer of white-hot coke, where it is reduced to carbon monoxide:



The carbon monoxide serves as the reducing agent in the upper regions of the furnace. The individual reactions are indicated in [Figure 19.6](#).

The iron oxides are reduced in the upper region of the furnace. In the middle region, limestone (calcium carbonate) decomposes, and the resulting calcium oxide combines with silica and silicates in the ore to form slag. The slag is mostly calcium silicate and contains most of the commercially unimportant components of the ore:



Just below the middle of the furnace, the temperature is high enough to melt both the iron and the slag. They collect in layers at the bottom of the furnace; the less dense slag floats on the iron and protects it from oxidation. Several times a day, the slag and molten iron are withdrawn from the furnace. The iron is transferred to casting machines or to a steelmaking plant ([Figure 19.7](#)).



FIGURE 19.7 Molten iron is shown being cast as steel. (credit: Clint Budd)

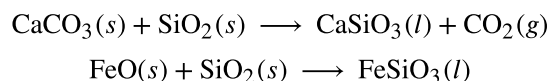
Much of the iron produced is refined and converted into steel. **Steel** is made from iron by removing impurities and adding substances such as manganese, chromium, nickel, tungsten, molybdenum, and vanadium to produce alloys with properties that make the material suitable for specific uses. Most steels also contain small but definite percentages of carbon (0.04%–2.5%). However, a large part of the carbon contained in iron must be removed in the manufacture of steel; otherwise, the excess carbon would make the iron brittle.

LINK TO LEARNING

You can watch an animation of [steelmaking \(http://openstax.org/l/16steelmaking\)](http://openstax.org/l/16steelmaking) that walks you through the process.

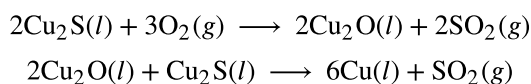
Isolation of Copper

The most important ores of copper contain copper sulfides (such as covellite, CuS), although copper oxides (such as tenorite, CuO) and copper hydroxycarbonates [such as malachite, $\text{Cu}_2(\text{OH})_2\text{CO}_3$] are sometimes found. In the production of copper metal, the concentrated sulfide ore is roasted to remove part of the sulfur as sulfur dioxide. The remaining mixture, which consists of Cu_2S , FeS , FeO , and SiO_2 , is mixed with limestone, which serves as a flux (a material that aids in the removal of impurities), and heated. Molten slag forms as the iron and silica are removed by Lewis acid-base reactions:



In these reactions, the silicon dioxide behaves as a Lewis acid, which accepts a pair of electrons from the Lewis base (the oxide ion).

Reduction of the Cu_2S that remains after smelting is accomplished by blowing air through the molten material. The air converts part of the Cu_2S into Cu_2O . As soon as copper(I) oxide is formed, it is reduced by the remaining copper(I) sulfide to metallic copper:



The copper obtained in this way is called blister copper because of its characteristic appearance, which is due to the air blisters it contains (Figure 19.8). This impure copper is cast into large plates, which are used as anodes in the

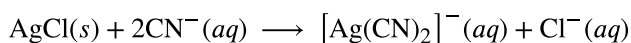
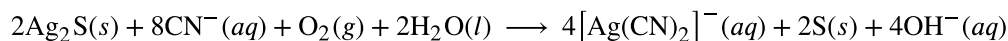
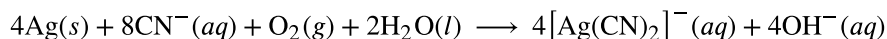
electrolytic refining of the metal (which is described in the chapter on electrochemistry).



FIGURE 19.8 Blister copper is obtained during the conversion of copper-containing ore into pure copper. (credit: “Tortie tude”/Wikimedia Commons)

Isolation of Silver

Silver sometimes occurs in large nuggets ([Figure 19.9](#)) but more frequently in veins and related deposits. At one time, panning was an effective method of isolating both silver and gold nuggets. Due to their low reactivity, these metals, and a few others, occur in deposits as nuggets. The discovery of platinum was due to Spanish explorers in Central America mistaking platinum nuggets for silver. When the metal is not in the form of nuggets, it often useful to employ a process called **hydrometallurgy** to separate silver from its ores. Hydrology involves the separation of a metal from a mixture by first converting it into soluble ions and then extracting and reducing them to precipitate the pure metal. In the presence of air, alkali metal cyanides readily form the soluble dicyanoargentate(I) ion, $[\text{Ag}(\text{CN})_2]^-$, from silver metal or silver-containing compounds such as Ag_2S and AgCl . Representative equations are:



(a)

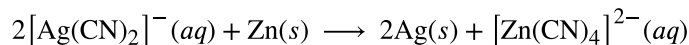


(b)

FIGURE 19.9 Naturally occurring free silver may be found as nuggets (a) or in veins (b). (credit a: modification of work by “Teravolt”/Wikimedia Commons; credit b: modification of work by James St. John)

The silver is precipitated from the cyanide solution by the addition of either zinc or iron(II) ions, which serves as the

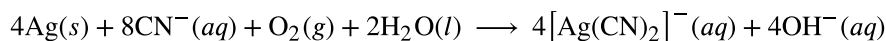
reducing agent:



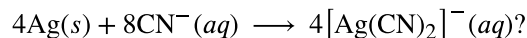
EXAMPLE 19.3

Refining Redox

One of the steps for refining silver involves converting silver into dicyanoargenate(I) ions:



Explain why oxygen must be present to carry out the reaction. Why does the reaction not occur as:



Solution

The charges, as well as the atoms, must balance in reactions. The silver atom is being oxidized from the 0 oxidation state to the 1+ state. Whenever something loses electrons, something must also gain electrons (be reduced) to balance the equation. Oxygen is a good oxidizing agent for these reactions because it can gain electrons to go from the 0 oxidation state to the 2− state.

Check Your Learning

During the refining of iron, carbon must be present in the blast furnace. Why is carbon necessary to convert iron oxide into iron?

Answer:

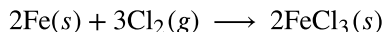
The carbon is converted into CO, which is the reducing agent that accepts electrons so that iron(III) can be reduced to iron(0).

Transition Metal Compounds

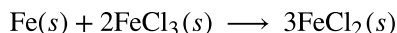
The bonding in the simple compounds of the transition elements ranges from ionic to covalent. In their lower oxidation states, the transition elements form ionic compounds; in their higher oxidation states, they form covalent compounds or polyatomic ions. The variation in oxidation states exhibited by the transition elements gives these compounds a metal-based, oxidation-reduction chemistry. The chemistry of several classes of compounds containing elements of the transition series follows.

Halides

Anhydrous halides of each of the transition elements can be prepared by the direct reaction of the metal with halogens. For example:



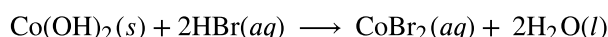
Heating a metal halide with additional metal can be used to form a halide of the metal with a lower oxidation state:



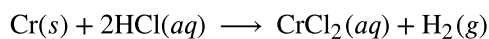
The stoichiometry of the metal halide that results from the reaction of the metal with a halogen is determined by the relative amounts of metal and halogen and by the strength of the halogen as an oxidizing agent. Generally, fluorine forms fluoride-containing metals in their highest oxidation states. The other halogens may not form analogous compounds.

In general, the preparation of stable water solutions of the halides of the metals of the first transition series is by the addition of a hydrohalic acid to carbonates, hydroxides, oxides, or other compounds that contain basic anions.

Sample reactions are:

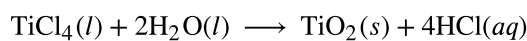
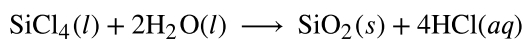


Most of the first transition series metals also dissolve in acids, forming a solution of the salt and hydrogen gas. For example:



The polarity of bonds with transition metals varies based not only upon the electronegativities of the atoms involved but also upon the oxidation state of the transition metal. Remember that bond polarity is a continuous spectrum with electrons being shared evenly (covalent bonds) at one extreme and electrons being transferred completely (ionic bonds) at the other. No bond is ever 100% ionic, and the degree to which the electrons are evenly distributed determines many properties of the compound. Transition metal halides with low oxidation numbers form more ionic bonds. For example, titanium(II) chloride and titanium(III) chloride (TiCl_2 and TiCl_3) have high melting points that are characteristic of ionic compounds, but titanium(IV) chloride (TiCl_4) is a volatile liquid, consistent with having covalent titanium-chlorine bonds. All halides of the heavier *d*-block elements have significant covalent characteristics.

The covalent behavior of the transition metals with higher oxidation states is exemplified by the reaction of the metal tetrahalides with water. Like covalent silicon tetrachloride, both the titanium and vanadium tetrahalides react with water to give solutions containing the corresponding hydrohalic acids and the metal oxides:

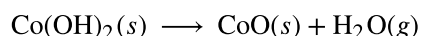
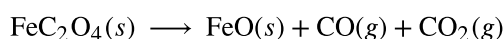


Oxides

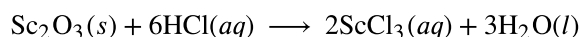
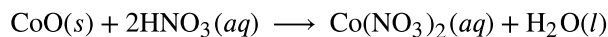
As with the halides, the nature of bonding in oxides of the transition elements is determined by the oxidation state of the metal. Oxides with low oxidation states tend to be more ionic, whereas those with higher oxidation states are more covalent. These variations in bonding are because the electronegativities of the elements are not fixed values. The electronegativity of an element increases with increasing oxidation state. Transition metals in low oxidation states have lower electronegativity values than oxygen; therefore, these metal oxides are ionic. Transition metals in very high oxidation states have electronegativity values close to that of oxygen, which leads to these oxides being covalent.

The oxides of the first transition series can be prepared by heating the metals in air. These oxides are Sc_2O_3 , TiO_2 , V_2O_5 , Cr_2O_3 , Mn_3O_4 , Fe_3O_4 , Co_3O_4 , NiO , and CuO .

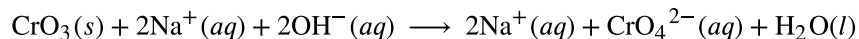
Alternatively, these oxides and other oxides (with the metals in different oxidation states) can be produced by heating the corresponding hydroxides, carbonates, or oxalates in an inert atmosphere. Iron(II) oxide can be prepared by heating iron(II) oxalate, and cobalt(II) oxide is produced by heating cobalt(II) hydroxide:



With the exception of CrO_3 and Mn_2O_7 , transition metal oxides are not soluble in water. They can react with acids and, in a few cases, with bases. Overall, oxides of transition metals with the lowest oxidation states are basic (and react with acids), the intermediate ones are amphoteric, and the highest oxidation states are primarily acidic. Basic metal oxides at a low oxidation state react with aqueous acids to form solutions of salts and water. Examples include the reaction of cobalt(II) oxide accepting protons from nitric acid, and scandium(III) oxide accepting protons from hydrochloric acid:



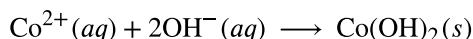
The oxides of metals with oxidation states of 4+ are amphoteric, and most are not soluble in either acids or bases. Vanadium(V) oxide, chromium(VI) oxide, and manganese(VII) oxide are acidic. They react with solutions of hydroxides to form salts of the oxyanions VO_4^{3-} , CrO_4^{2-} , and MnO_4^- . For example, the complete ionic equation for the reaction of chromium(VI) oxide with a strong base is given by:



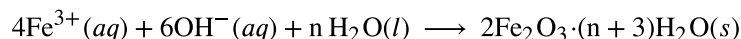
Chromium(VI) oxide and manganese(VII) oxide react with water to form the acids H_2CrO_4 and HMnO_4 , respectively.

Hydroxides

When a soluble hydroxide is added to an aqueous solution of a salt of a transition metal of the first transition series, a gelatinous precipitate forms. For example, adding a solution of sodium hydroxide to a solution of cobalt sulfate produces a gelatinous pink or blue precipitate of cobalt(II) hydroxide. The net ionic equation is:



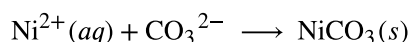
In this and many other cases, these precipitates are hydroxides containing the transition metal ion, hydroxide ions, and water coordinated to the transition metal. In other cases, the precipitates are hydrated oxides composed of the metal ion, oxide ions, and water of hydration:



These substances do not contain hydroxide ions. However, both the hydroxides and the hydrated oxides react with acids to form salts and water. When precipitating a metal from solution, it is necessary to avoid an excess of hydroxide ion, as this may lead to complex ion formation as discussed later in this chapter. The precipitated metal hydroxides can be separated for further processing or for waste disposal.

Carbonates

Many of the elements of the first transition series form insoluble carbonates. It is possible to prepare these carbonates by the addition of a soluble carbonate salt to a solution of a transition metal salt. For example, nickel carbonate can be prepared from solutions of nickel nitrate and sodium carbonate according to the following net ionic equation:

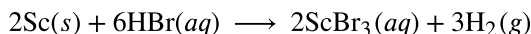


The reactions of the transition metal carbonates are similar to those of the active metal carbonates. They react with acids to form metal salts, carbon dioxide, and water. Upon heating, they decompose, forming the transition metal oxides.

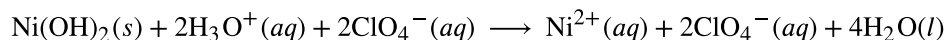
Other Salts

In many respects, the chemical behavior of the elements of the first transition series is very similar to that of the main group metals. In particular, the same types of reactions that are used to prepare salts of the main group metals can be used to prepare simple ionic salts of these elements.

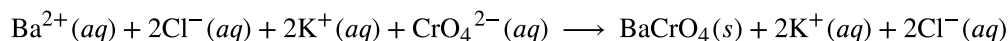
A variety of salts can be prepared from metals that are more active than hydrogen by reaction with the corresponding acids: Scandium metal reacts with hydrobromic acid to form a solution of scandium bromide:



The common compounds that we have just discussed can also be used to prepare salts. The reactions involved include the reactions of oxides, hydroxides, or carbonates with acids. For example:



Substitution reactions involving soluble salts may be used to prepare insoluble salts. For example:



In our discussion of oxides in this section, we have seen that reactions of the covalent oxides of the transition elements with hydroxides form salts that contain oxyanions of the transition elements.



HOW SCIENCES INTERCONNECT

High Temperature Superconductors

A **superconductor** is a substance that conducts electricity with no resistance. This lack of resistance means that there is no energy loss during the transmission of electricity. This would lead to a significant reduction in the cost of electricity.

Most currently used, commercial superconducting materials, such as NbTi and Nb₃Sn, do not become superconducting until they are cooled below 23 K (−250 °C). This requires the use of liquid helium, which has a boiling temperature of 4 K and is expensive and difficult to handle. The cost of liquid helium has deterred the widespread application of superconductors.

One of the most exciting scientific discoveries of the 1980s was the characterization of compounds that exhibit superconductivity at temperatures above 90 K. (Compared to liquid helium, 90 K is a high temperature.) Typical among the high-temperature superconducting materials are oxides containing yttrium (or one of several rare earth elements), barium, and copper in a 1:2:3 ratio. The formula of the ionic yttrium compound is YBa₂Cu₃O₇.

The new materials become superconducting at temperatures close to 90 K (Figure 19.10), temperatures that can be reached by cooling with liquid nitrogen (boiling temperature of 77 K). Not only are liquid nitrogen-cooled materials easier to handle, but the cooling costs are also about 1000 times lower than for liquid helium.

Further advances during the same period included materials that became superconducting at even higher temperatures and with a wider array of materials. The DuPont team led by Uma Chowdry and Arthur Sleight identified Bismuth-Strontium-Copper-Oxides that became superconducting at temperatures as high as 110 K and, importantly, did not contain rare earth elements. Advances continued through the subsequent decades until, in 2020, a team led by Ranga Dias at University of Rochester announced the development of a room-temperature superconductor, opening doors to widespread applications. More research and development is needed to realize the potential of these materials, but the possibilities are very promising.

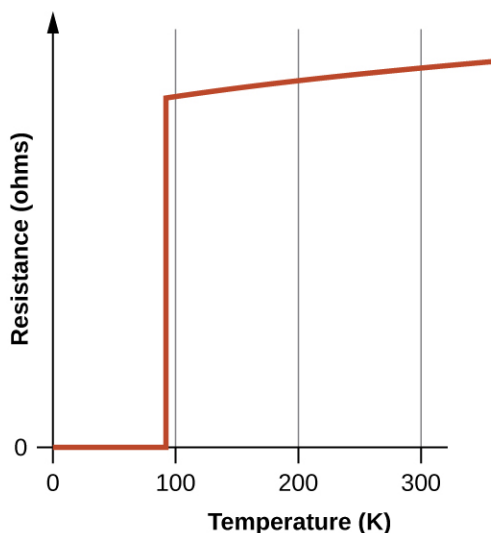


FIGURE 19.10 The resistance of the high-temperature superconductor YBa₂Cu₃O₇ varies with temperature. Note how the resistance falls to zero below 92 K, when the substance becomes superconducting.

Although the brittle, fragile nature of these materials presently hampers their commercial applications, they have tremendous potential that researchers are hard at work improving their processes to help realize. Superconducting transmission lines would carry current for hundreds of miles with no loss of power due to resistance in the wires. This could allow generating stations to be located in areas remote from population centers and near the natural resources necessary for power production. The first project demonstrating the viability of high-temperature superconductor power transmission was established in New York in 2008.

Researchers are also working on using this technology to develop other applications, such as smaller and more powerful microchips. In addition, high-temperature superconductors can be used to generate magnetic fields for applications such as medical devices, magnetic levitation trains, and containment fields for nuclear fusion reactors (Figure 19.11).

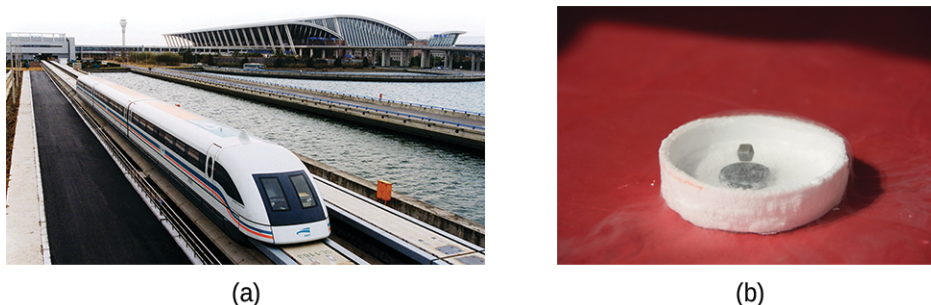


FIGURE 19.11 (a) This magnetic levitation train (or maglev) uses superconductor technology to move along its tracks. (b) A magnet can be levitated using a dish like this as a superconductor. (credit a: modification of work by Alex Needham; credit b: modification of work by Kevin Jarrett)

LINK TO LEARNING

Watch how a [high-temperature superconductor \(http://openstax.org/l/16supercond\)](http://openstax.org/l/16supercond) levitates around a magnetic racetrack in the video.

19.2 Coordination Chemistry of Transition Metals

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- List the defining traits of coordination compounds
- Describe the structures of complexes containing monodentate and polydentate ligands
- Use standard nomenclature rules to name coordination compounds
- Explain and provide examples of geometric and optical isomerism
- Identify several natural and technological occurrences of coordination compounds

The hemoglobin in your blood, the chlorophyll in green plants, vitamin B-12, and the catalyst used in the manufacture of polyethylene all contain coordination compounds. Ions of the metals, especially the transition metals, are likely to form complexes. Many of these compounds are highly colored ([Figure 19.12](#)). In the remainder of this chapter, we will consider the structure and bonding of these remarkable compounds.



FIGURE 19.12 Metal ions that contain partially filled d subshell usually form colored complex ions; ions with empty d subshell (d^0) or with filled d subshells (d^{10}) usually form colorless complexes. This figure shows, from left to right, solutions containing $[M(H_2O)_6]^{n+}$ ions with $M = Sc^{3+}(d^0)$, $Cr^{3+}(d^3)$, $Co^{2+}(d^7)$, $Ni^{2+}(d^8)$, $Cu^{2+}(d^9)$, and $Zn^{2+}(d^{10})$. (credit: Sahar Atwa)

Remember that in most main group element compounds, the valence electrons of the isolated atoms combine to form chemical bonds that satisfy the octet rule. For instance, the four valence electrons of carbon overlap with electrons from four hydrogen atoms to form CH_4 . The one valence electron leaves sodium and adds to the seven valence electrons of chlorine to form the ionic formula unit $NaCl$ ([Figure 19.13](#)). Transition metals do not normally bond in this fashion. They primarily form coordinate covalent bonds, a form of the Lewis acid-base interaction in which both of the electrons in the bond are contributed by a donor (Lewis base) to an electron acceptor (Lewis acid).

The Lewis acid in coordination complexes, often called a **central metal** ion (or atom), is often a transition metal or inner transition metal, although main group elements can also form **coordination compounds**. The Lewis base donors, called **ligands**, can be a wide variety of chemicals—atoms, molecules, or ions. The only requirement is that they have one or more electron pairs, which can be donated to the central metal. Most often, this involves a **donor atom** with a lone pair of electrons that can form a coordinate bond to the metal.

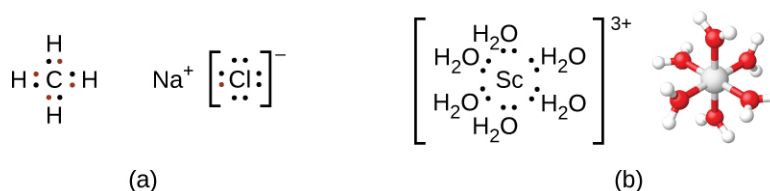


FIGURE 19.13 (a) Covalent bonds involve the sharing of electrons, and ionic bonds involve the transferring of electrons associated with each bonding atom, as indicated by the colored electrons. (b) However, coordinate covalent bonds involve electrons from a Lewis base being donated to a metal center. The lone pairs from six water molecules form bonds to the scandium ion to form an octahedral complex. (Only the donated pairs are shown.)

The **coordination sphere** consists of the central metal ion or atom plus its attached ligands. Brackets in a formula enclose the coordination sphere; species outside the brackets are not part of the coordination sphere. The **coordination number** of the central metal ion or atom is the number of donor atoms bonded to it. The coordination number for the silver ion in $[\text{Ag}(\text{NH}_3)_2]^+$ is two (Figure 19.14). For the copper(II) ion in $[\text{CuCl}_4]^{2-}$, the coordination number is four, whereas for the cobalt(II) ion in $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ the coordination number is six. Each of these ligands is **monodentate**, from the Greek for “one toothed,” meaning that they connect with the central metal through only one atom. In this case, the number of ligands and the coordination number are equal.

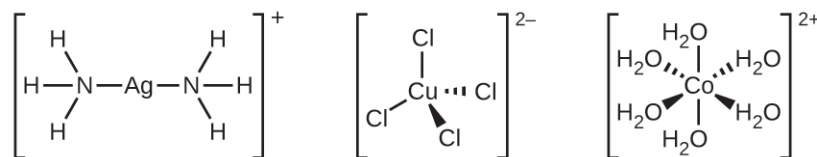


FIGURE 19.14 The complexes (a) $[\text{Ag}(\text{NH}_3)_2]^+$, (b) $[\text{CuCl}_4]^{2-}$, and (c) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ have coordination numbers of two, four, and six, respectively. The geometries of these complexes are the same as we have seen with VSEPR theory for main group elements: linear, tetrahedral, and octahedral.

Many other ligands coordinate to the metal in more complex fashions. **Bidentate ligands** are those in which two atoms coordinate to the metal center. For example, ethylenediamine (en, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$) contains two nitrogen atoms, each of which has a lone pair and can serve as a Lewis base (Figure 19.15). Both of the atoms can coordinate to a single metal center. In the complex $[\text{Co}(\text{en})_3]^{3+}$, there are three bidentate en ligands, and the coordination number of the cobalt(III) ion is six. The most common coordination numbers are two, four, and six, but examples of all coordination numbers from 1 to 15 are known.

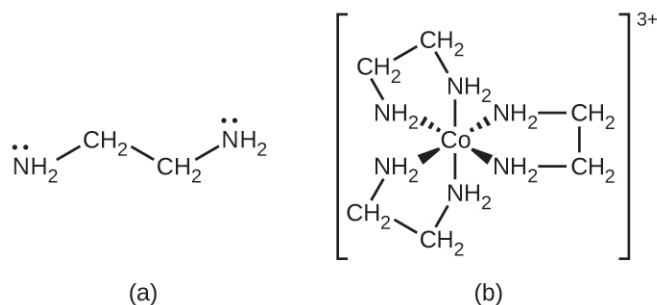


FIGURE 19.15 (a) The ethylenediamine (en) ligand contains two atoms with lone pairs that can coordinate to the metal center. (b) The cobalt(III) complex $[\text{Co}(\text{en})_3]^{3+}$ contains three of these ligands, each forming two bonds to the cobalt ion.

Any ligand that bonds to a central metal ion by more than one donor atom is a **polydentate ligand** (or “many teeth”) because it can bite into the metal center with more than one bond. The term **chelate** (pronounced “KEY-late”) from the Greek for “claw” is also used to describe this type of interaction. Many polydentate ligands are **chelating ligands**, and a complex consisting of one or more of these ligands and a central metal is a chelate. A chelating ligand is also known as a chelating agent. A chelating ligand holds the metal ion rather like a crab’s claw would hold a

marble. [Figure 19.15](#) showed one example of a chelate. The heme complex in hemoglobin is another important example ([Figure 19.16](#)). It contains a polydentate ligand with four donor atoms that coordinate to iron.

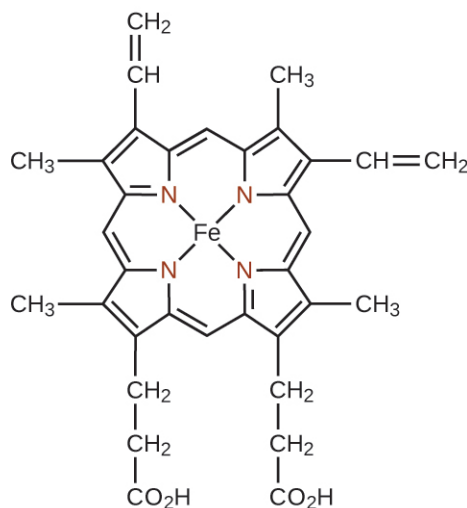


FIGURE 19.16 The single ligand heme contains four nitrogen atoms that coordinate to iron in hemoglobin to form a chelate.

Polydentate ligands are sometimes identified with prefixes that indicate the number of donor atoms in the ligand. As we have seen, ligands with one donor atom, such as NH_3 , Cl^- , and H_2O , are monodentate ligands. Ligands with two donor groups are bidentate ligands. Ethylenediamine, $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$, and the anion of the acid glycine, $\text{NH}_2\text{CH}_2\text{CO}_2^-$ ([Figure 19.17](#)) are examples of bidentate ligands. Tridentate ligands, tetradentate ligands, pentadentate ligands, and hexadentate ligands contain three, four, five, and six donor atoms, respectively. The ligand in heme ([Figure 19.16](#)) is a tetradentate ligand.

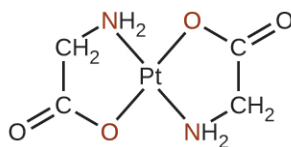


FIGURE 19.17 Each of the anionic ligands shown attaches in a bidentate fashion to platinum(II), with both a nitrogen and oxygen atom coordinating to the metal.

The Naming of Complexes

The nomenclature of the complexes is patterned after a system suggested by Alfred Werner, a Swiss chemist and Nobel laureate, whose outstanding work more than 100 years ago laid the foundation for a clearer understanding of these compounds. The following five rules are used for naming complexes:

1. If a coordination compound is ionic, name the cation first and the anion second, in accordance with the usual nomenclature.
2. Name the ligands first, followed by the central metal. Name the ligands alphabetically. Negative ligands (anions) have names formed by adding *-o* to the stem name of the group. For examples, see [Table 19.1](#). For most neutral ligands, the name of the molecule is used. The four common exceptions are *aqua* (H_2O), *ammine* (NH_3), *carbonyl* (CO), and *nitrosyl* (NO). For example, name $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$ as diamminetetrachloroplatinum(IV).

Examples of Anionic Ligands

Anionic Ligand	Name
F^-	fluoro
Cl^-	chloro

TABLE 19.1

Anionic Ligand	Name
Br^-	bromo
I^-	iodo
CN^-	cyano
NO_3^-	nitrate
OH^-	hydroxo
O^{2-}	oxo
$\text{C}_2\text{O}_4^{2-}$	oxalato
CO_3^{2-}	carbonato

TABLE 19.1

3. If more than one ligand of a given type is present, the number is indicated by the prefixes *di-* (for two), *tri-* (for three), *tetra-* (for four), *penta-* (for five), and *hexa-* (for six). Sometimes, the prefixes *bis-* (for two), *tris-* (for three), and *tetrakis-* (for four) are used when the name of the ligand already includes *di-*, *tri-*, or *tetra-*, or when the ligand name begins with a vowel. For example, the ion bis(bipyridyl)osmium(II) uses *bis-* to signify that there are two ligands attached to Os, and each bipyridyl ligand contains two pyridine groups ($\text{C}_5\text{H}_4\text{N}$).

When the complex is either a cation or a neutral molecule, the name of the central metal atom is spelled exactly like the name of the element and is followed by a Roman numeral in parentheses to indicate its oxidation state (Table 19.2 and Table 19.3). When the complex is an anion, the suffix *-ate* is added to the stem of the name of the metal, followed by the Roman numeral designation of its oxidation state (Table 19.4). Sometimes, the Latin name of the metal is used when the English name is clumsy. For example, *ferrate* is used instead of *ironate*, *plumbate* instead of *leadate*, and *stannate* instead of *tinate*. The oxidation state of the metal is determined based on the charges of each ligand and the overall charge of the coordination compound. For example, in $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Br}$, the coordination sphere (in brackets) has a charge of 1+ to balance the bromide ion. The water ligands are neutral, and the chloride ligands are anionic with a charge of 1– each. To determine the oxidation state of the metal, we set the overall charge equal to the sum of the ligands and the metal: $+1 = -2 + x$, so the oxidation state (x) is equal to 3+.

Examples in Which the Complex Is a Cation

$[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$	hexaamminecobalt(III) chloride
$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+}$	tetraamminedichloroplatinum(IV) ion
$[\text{Ag}(\text{NH}_3)_2]^+$	diamminesilver(I) ion
$[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}$	tetraaquadichlorochromium(III) chloride
$[\text{Co}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_3]_2(\text{SO}_4)_3$	tris(ethylenediamine)cobalt(III) sulfate

TABLE 19.2

Examples in Which the Complex Is Neutral

$[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$	diamminetetrachloroplatinum(IV)
$[\text{Ni}(\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2)_2\text{Cl}_2]$	dichlorobis(ethylenediamine)nickel(II)

TABLE 19.3

Examples in Which the Complex Is an Anion

$[\text{PtCl}_6]^{2-}$	hexachloroplatinate(IV) ion
$\text{Na}_2[\text{SnCl}_6]$	sodium hexachlorostannate(IV)

TABLE 19.4



LINK TO LEARNING

Do you think you understand naming coordination complexes? You can look over more examples and test yourself with [online quizzes \(http://openstax.org/l/16namingcomps\)](http://openstax.org/l/16namingcomps) at the University of Sydney's site.



EXAMPLE 19.4

Coordination Numbers and Oxidation States

Determine the name of the following complexes and give the coordination number of the central metal atom.

- (a) $\text{Na}_2[\text{PtCl}_6]$
 (b) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
 (c) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$

Solution

(a) There are two Na^+ ions, so the coordination sphere has a negative two charge: $[\text{PtCl}_6]^{2-}$. There are six anionic chloride ligands, so $-2 = -6 + x$, and the oxidation state of the platinum is 4+. The name of the complex is sodium hexachloroplatinate(IV), and the coordination number is six. (b) The coordination sphere has a charge of 3– (based on the potassium) and the oxalate ligands each have a charge of 2–, so the metal oxidation state is given by $-3 = -6 + x$, and this is an iron(III) complex. The name is potassium trisoxalatoferrate(III) (note that tris is used instead of tri because the ligand name starts with a vowel). Because oxalate is a bidentate ligand, this complex has a coordination number of six. (c) In this example, the coordination sphere has a cationic charge of 2+. The NH_3 ligand is neutral, but the chloro ligand has a charge of 1–. The oxidation state is found by $+2 = -1 + x$ and is 3+, so the complex is pentaamminechlorocobalt(III) chloride and the coordination number is six.

Check Your Learning

The complex potassium dicyanoargentate(I) is used to make antiseptic compounds. Give the formula and coordination number.

Answer:

$\text{K}[\text{Ag}(\text{CN})_2]$; coordination number two

The Structures of Complexes

For transition metal complexes, the coordination number determines the geometry around the central metal ion. The most common structures of the complexes in coordination compounds are square planar, tetrahedral, and octahedral, corresponding to coordination numbers of four, four, and six, respectively. Coordination numbers greater

than six are less common and yield a variety of structures (see [Figure 19.18](#) and [Table 19.5](#)):

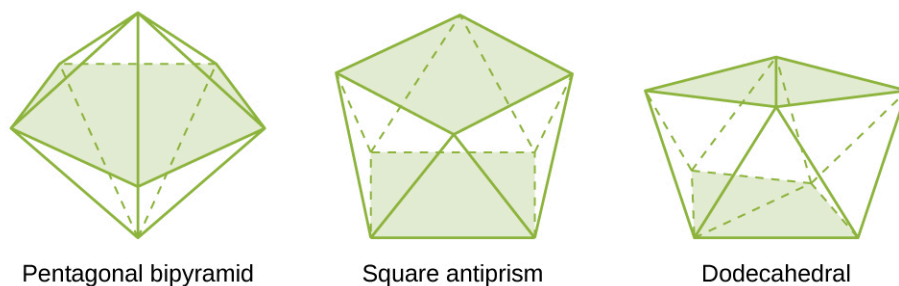


FIGURE 19.18 These are geometries of some complexes with coordination numbers of seven and eight.

Coordination Numbers and Molecular Geometry

Coordination Number	Molecular Geometry	Example
2	linear	$[\text{Ag}(\text{NH}_3)_2]^+$
3	trigonal planar	$[\text{Cu}(\text{CN})_3]^{2-}$
4	tetrahedral(d^0 or d^{10}), low oxidation states for M	$[\text{Ni}(\text{CO})_4]$
4	square planar (d^8)	$[\text{Ni}(\text{CN})_4]^{2-}$
5	trigonal bipyramidal	$[\text{CoCl}_5]^{2-}$
5	square pyramidal	$[\text{VO}(\text{CN})_4]^{2-}$
6	octahedral	$[\text{CoCl}_6]^{3-}$
7	pentagonal bipyramid	$[\text{ZrF}_7]^{3-}$
8	square antiprism	$[\text{ReF}_8]^{2-}$
8	dodecahedron	$[\text{Mo}(\text{CN})_8]^{4-}$
9 and above	more complicated structures	$[\text{ReH}_9]^{2-}$

TABLE 19.5

Unlike main group atoms in which both the bonding and nonbonding electrons determine the molecular shape, the nonbonding d -electrons do not change the arrangement of the ligands. Octahedral complexes have a coordination number of six, and the six donor atoms are arranged at the corners of an octahedron around the central metal ion. Examples are shown in [Figure 19.19](#). The chloride and nitrate anions in $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$ and $[\text{Cr}(\text{en})_3](\text{NO}_3)_3$, and the potassium cations in $\text{K}_2[\text{PtCl}_6]$, are outside the brackets and are not bonded to the metal ion.

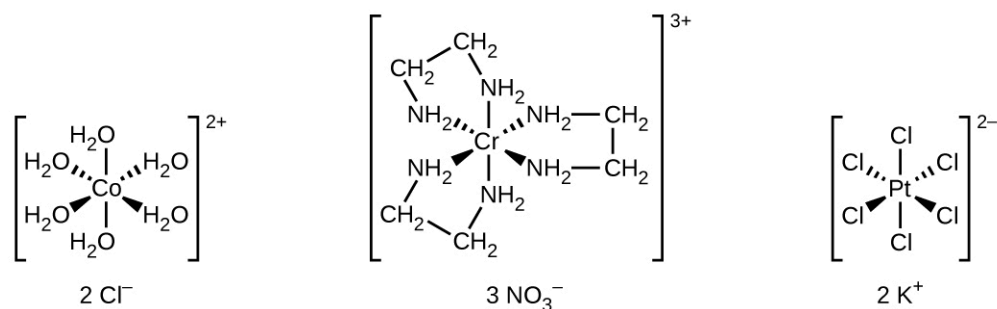


FIGURE 19.19 Many transition metal complexes adopt octahedral geometries, with six donor atoms forming bond angles of 90° about the central atom with adjacent ligands. Note that only ligands within the coordination sphere affect the geometry around the metal center.

For transition metals with a coordination number of four, two different geometries are possible: tetrahedral or square planar. Unlike main group elements, where these geometries can be predicted from VSEPR theory, a more detailed discussion of transition metal orbitals (discussed in the section on Crystal Field Theory) is required to predict which complexes will be tetrahedral and which will be square planar. In tetrahedral complexes such as $[\text{Zn}(\text{CN})_4]^{2-}$ (Figure 19.20), each of the ligand pairs forms an angle of 109.5° . In square planar complexes, such as $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, each ligand has two other ligands at 90° angles (called the *cis* positions) and one additional ligand at an 180° angle, in the *trans* position.

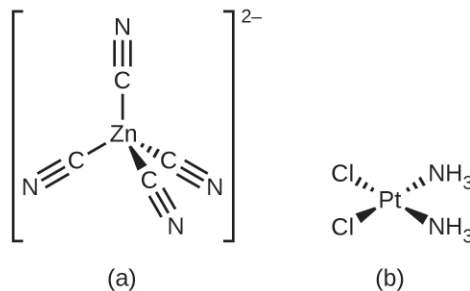


FIGURE 19.20 Transition metals with a coordination number of four can adopt a tetrahedral geometry (a) as in $\text{K}_2[\text{Zn}(\text{CN})_4]$ or a square planar geometry (b) as shown in $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$.

Isomerism in Complexes

Isomers are different chemical species that have the same chemical formula. Transition metal complexes often exist as **geometric isomers**, in which the same atoms are connected through the same types of bonds but with differences in their orientation in space. Coordination complexes with two different ligands in the *cis* and *trans* positions from a ligand of interest form isomers. For example, the octahedral $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ ion has two isomers. In the ***cis* configuration**, the two chloride ligands are adjacent to each other (Figure 19.21). The other isomer, the ***trans* configuration**, has the two chloride ligands directly across from one another.

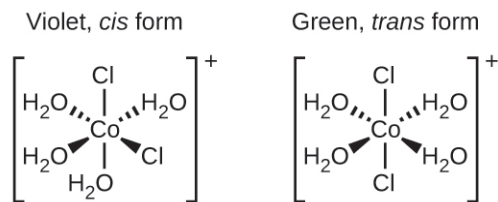


FIGURE 19.21 The *cis* and *trans* isomers of $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]^+$ contain the same ligands attached to the same metal ion, but the spatial arrangement causes these two compounds to have very different properties.

Different geometric isomers of a substance are different chemical compounds. They exhibit different properties, even though they have the same formula. For example, the two isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_3$ differ in color; the *cis* form is violet, and the *trans* form is green. Furthermore, these isomers have different dipole moments, solubilities, and reactivities. As an example of how the arrangement in space can influence the molecular properties, consider the polarity of the two $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_3$ isomers. Remember that the polarity of a molecule or ion is determined by the bond dipoles (which are due to the difference in electronegativity of the bonding atoms) and their arrangement in space. In one isomer, *cis* chloride ligands cause more electron density on one side of the molecule than on the

other, making it polar. For the *trans* isomer, each ligand is directly across from an identical ligand, so the bond dipoles cancel out, and the molecule is nonpolar.

EXAMPLE 19.5

Geometric Isomers

Identify which geometric isomer of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is shown in Figure 19.20. Draw the other geometric isomer and give its full name.

Solution

In the Figure 19.20, the two chlorine ligands occupy *cis* positions. The other form is shown in Figure 19.22. When naming specific isomers, the descriptor is listed in front of the name. Therefore, this complex is *trans*-diamminedichloroplatinum(II).

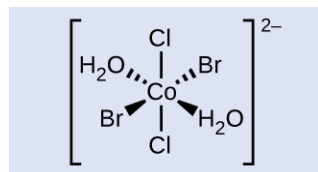


FIGURE 19.22 The *trans* isomer of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ has each ligand directly across from an adjacent ligand.

Check Your Learning

Draw the ion *trans*-diaqua-*trans*-dibromo-*trans*-dichlorocobalt(II).

Answer:



Another important type of isomers are **optical isomers**, or **enantiomers**, in which two objects are exact mirror images of each other but cannot be lined up so that all parts match. This means that optical isomers are nonsuperimposable mirror images. A classic example of this is a pair of hands, in which the right and left hand are mirror images of one another but cannot be superimposed. Optical isomers are very important in organic and biochemistry because living systems often incorporate one specific optical isomer and not the other. Unlike geometric isomers, pairs of optical isomers have identical properties (boiling point, polarity, solubility, etc.). Optical isomers differ only in the way they affect polarized light and how they react with other optical isomers. For coordination complexes, many coordination compounds such as $[\text{M}(\text{en})_3]^{n+}$ [in which M^{n+} is a central metal ion such as iron(III) or cobalt(II)] form enantiomers, as shown in Figure 19.23. These two isomers will react differently with other optical isomers. For example, DNA helices are optical isomers, and the form that occurs in nature (right-handed DNA) will bind to only one isomer of $[\text{M}(\text{en})_3]^{n+}$ and not the other.

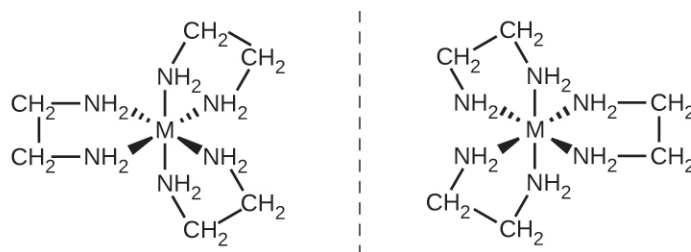


FIGURE 19.23 The complex $[\text{M}(\text{en})_3]^{n+}$ (M^{n+} = a metal ion, en = ethylenediamine) has a nonsuperimposable mirror image.

The $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ ion exhibits geometric isomerism (*cis/trans*), and its *cis* isomer exists as a pair of optical isomers (Figure 19.24).

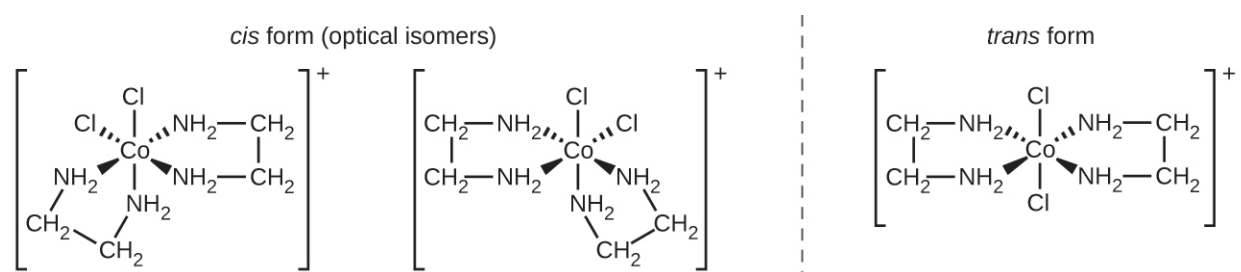


FIGURE 19.24 Three isomeric forms of $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ exist. The *trans* isomer, formed when the chlorines are positioned at a 180° angle, has very different properties from the *cis* isomers. The mirror images of the *cis* isomer form a pair of optical isomers, which have identical behavior except when reacting with other enantiomers.

Linkage isomers occur when the coordination compound contains a ligand that can bind to the transition metal center through two different atoms. For example, the CN ligand can bind through the carbon atom (cyano) or through the nitrogen atom (isocyano). Similarly, SCN^- can be bound through the sulfur or nitrogen atom, affording two distinct compounds ($[\text{Co}(\text{NH}_3)_5\text{SCN}]^{2+}$ or $[\text{Co}(\text{NH}_3)_5\text{NCS}]^{2+}$).

Ionization isomers (or **coordination isomers**) occur when one anionic ligand in the inner coordination sphere is replaced with the counter ion from the outer coordination sphere. A simple example of two ionization isomers are $[\text{CoCl}_6][\text{Br}]$ and $[\text{CoCl}_5\text{Br}][\text{Cl}]$.

Coordination Complexes in Nature and Technology

Chlorophyll, the green pigment in plants, is a complex that contains magnesium (Figure 19.25). This is an example of a main group element in a coordination complex. Plants appear green because chlorophyll absorbs red and purple light; the reflected light consequently appears green. The energy resulting from the absorption of light is used in photosynthesis.

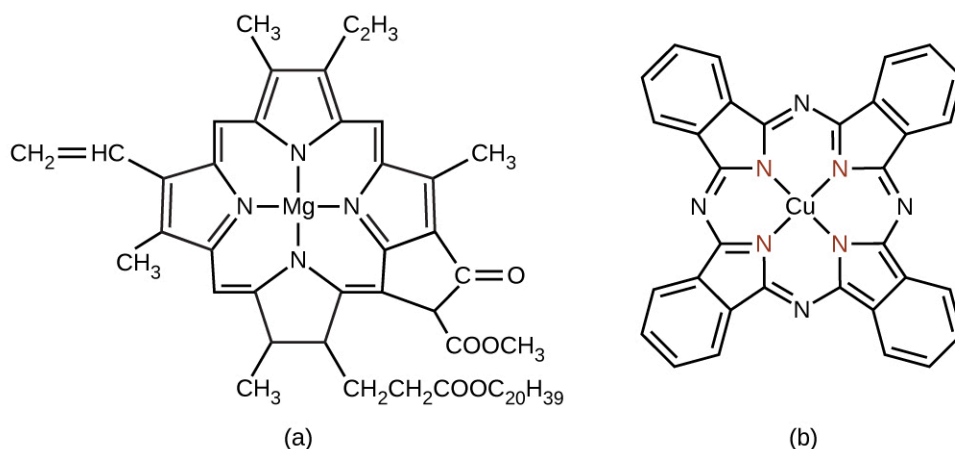


FIGURE 19.25 (a) Chlorophyll comes in several different forms, which all have the same basic structure around the magnesium center. (b) Copper phthalocyanine blue, a square planar copper complex, is present in some blue dyes.

CHEMISTRY IN EVERYDAY LIFE

Transition Metal Catalysts

One of the most important applications of transition metals is as industrial catalysts. As you recall from the chapter on kinetics, a catalyst increases the rate of reaction by lowering the activation energy and is regenerated in the catalytic cycle. Over 90% of all manufactured products are made with the aid of one or more catalysts. The ability to bind ligands and change oxidation states makes transition metal catalysts well suited for catalytic applications. Vanadium oxide is used to produce 230,000,000 tons of sulfuric acid worldwide each year, which in turn is used to make everything from fertilizers to cans for food. Plastics are made with the aid of transition metal catalysts, along with detergents, fertilizers, paints, and more (see Figure 19.26). Very complicated pharmaceuticals are manufactured with catalysts that are selective, reacting with one specific bond out of a

large number of possibilities. Catalysts allow processes to be more economical and more environmentally friendly. Developing new catalysts and better understanding of existing systems are important areas of current research.

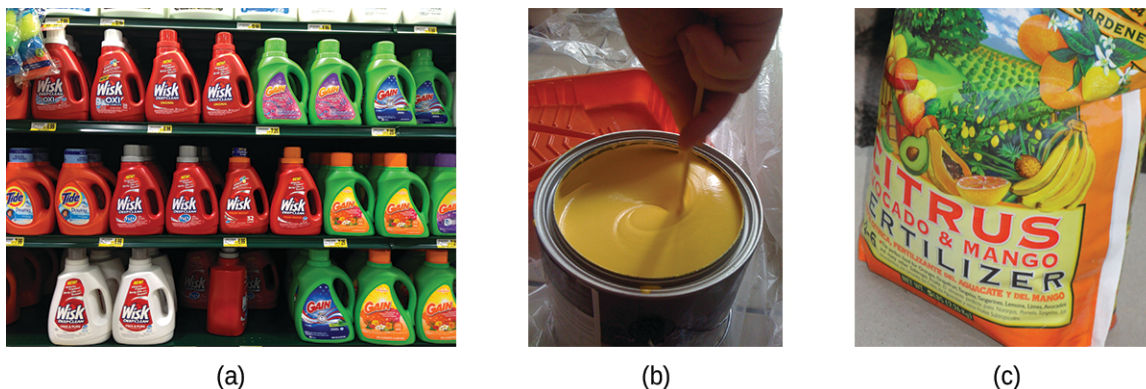


FIGURE 19.26 (a) Detergents, (b) paints, and (c) fertilizers are all made using transition metal catalysts. (credit a: modification of work by “Mr. Brian”/Flickr; credit b: modification of work by Ewen Roberts; credit c: modification of work by “osseous”/Flickr)

PORTRAIT OF A CHEMIST

Deanna D'Alessandro

Dr. Deanna D'Alessandro develops new metal-containing materials that demonstrate unique electronic, optical, and magnetic properties. Her research combines the fields of fundamental inorganic and physical chemistry with materials engineering. She is working on many different projects that rely on transition metals. For example, one type of compound she is developing captures carbon dioxide waste from power plants and catalytically converts it into useful products (see [Figure 19.27](#)).

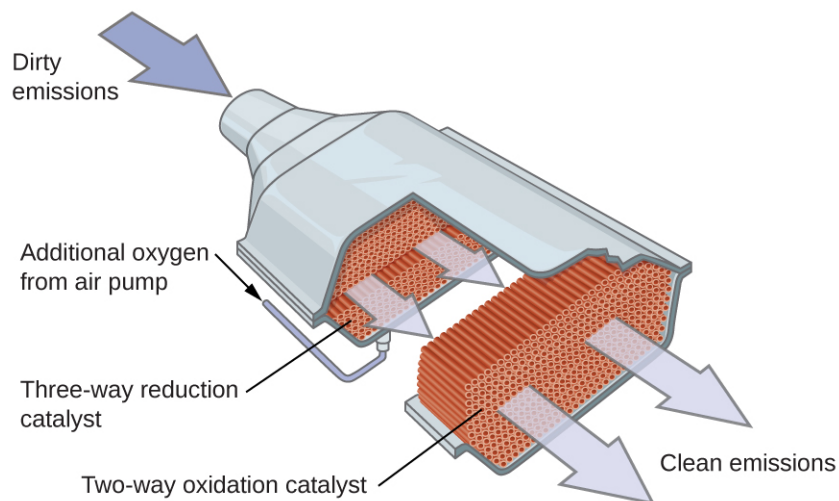


FIGURE 19.27 Catalytic converters change carbon dioxide emissions from power plants into useful products, and, like the one shown here, are also found in cars.

Another project involves the development of porous, sponge-like materials that are “photoactive.” The absorption of light causes the pores of the sponge to change size, allowing gas diffusion to be controlled. This has many potential useful applications, from powering cars with hydrogen fuel cells to making better electronics components. Although not a complex, self-darkening sunglasses are an example of a photoactive substance.

Watch this [video \(http://openstax.org/l/16DeannaD\)](http://openstax.org/l/16DeannaD) to learn more about this research and listen to Dr. D'Alessandro (shown in [Figure 19.28](#)) describe what it is like being a research chemist.



FIGURE 19.28 Dr. Deanna D'Alessandro is a functional materials researcher. Her work combines the inorganic and physical chemistry fields with engineering, working with transition metals to create new systems to power cars and convert energy (credit: image courtesy of Deanna D'Alessandro).

Many other coordination complexes are also brightly colored. The square planar copper(II) complex phthalocyanine blue (from [Figure 19.25](#)) is one of many complexes used as pigments or dyes. This complex is used in blue ink, blue jeans, and certain blue paints.

The structure of heme ([Figure 19.29](#)), the iron-containing complex in hemoglobin, is very similar to that in chlorophyll. In hemoglobin, the red heme complex is bonded to a large protein molecule (globin) by the attachment of the protein to the heme ligand. Oxygen molecules are transported by hemoglobin in the blood by being bound to the iron center. When the hemoglobin loses its oxygen, the color changes to a bluish red. Hemoglobin will only transport oxygen if the iron is Fe^{2+} ; oxidation of the iron to Fe^{3+} prevents oxygen transport.

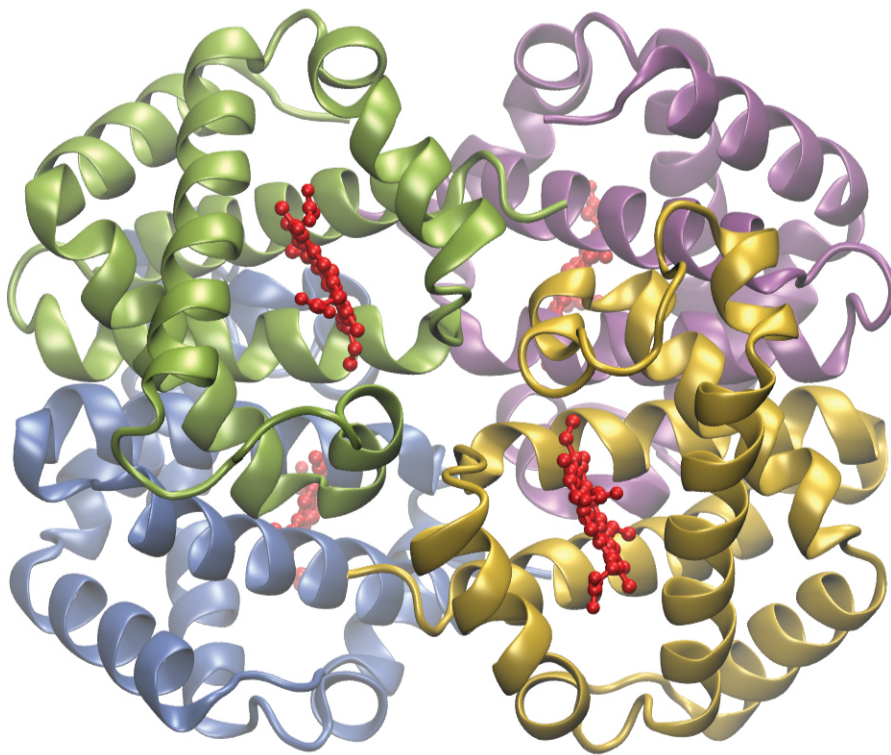


FIGURE 19.29 Hemoglobin contains four protein subunits, each of which has an iron center attached to a heme ligand (shown in red), which is coordinated to a globin protein. Each subunit is shown in a different color.

Complexing agents often are used for water softening because they tie up such ions as Ca^{2+} , Mg^{2+} , and Fe^{2+} , which make water hard. Many metal ions are also undesirable in food products because these ions can catalyze reactions that change the color of food. Coordination complexes are useful as preservatives. For example, the ligand EDTA, $(\text{HO}_2\text{CCH}_2)_2\text{NCH}_2\text{CH}_2\text{N}(\text{CH}_2\text{CO}_2\text{H})_2$, coordinates to metal ions through six donor atoms and prevents the metals from reacting (Figure 19.30). This ligand also is used to sequester metal ions in paper production, textiles, and detergents, and has pharmaceutical uses.

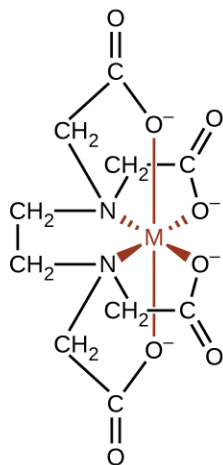


FIGURE 19.30 The ligand EDTA binds tightly to a variety of metal ions by forming hexadentate complexes.

Complexing agents that tie up metal ions are also used as drugs. British Anti-Lewisite (BAL), $\text{HSCH}_2\text{CH}(\text{SH})\text{CH}_2\text{OH}$, is a drug developed during World War I as an antidote for the arsenic-based war gas Lewisite. BAL is now used to treat poisoning by heavy metals, such as arsenic, mercury, thallium, and chromium. The drug is a ligand and functions by making a water-soluble chelate of the metal; the kidneys eliminate this metal chelate (Figure 19.31). Another polydentate ligand, enterobactin, which is isolated from certain bacteria, is used to form complexes of iron and thereby to control the severe iron buildup found in patients suffering from blood diseases such as Cooley's anemia, who require frequent transfusions. As the transfused blood breaks down, the usual metabolic processes that remove iron are overloaded, and excess iron can build up to fatal levels. Enterobactin forms a water-soluble complex with excess iron, and the body can safely eliminate this complex.

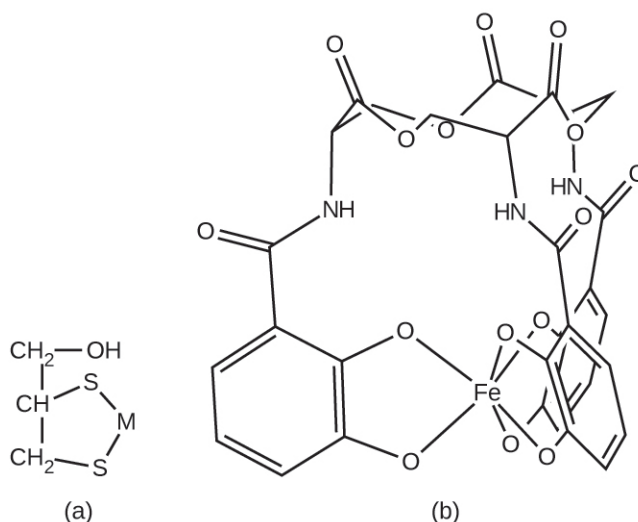


FIGURE 19.31 Coordination complexes are used as drugs. (a) British Anti-Lewisite is used to treat heavy metal poisoning by coordinating metals (M), and enterobactin (b) allows excess iron in the blood to be removed.

**EXAMPLE 19.6****Chelation Therapy**

Ligands like BAL and enterobactin are important in medical treatments for heavy metal poisoning. However, chelation therapies can disrupt the normal concentration of ions in the body, leading to serious side effects, so researchers are searching for new chelation drugs. One drug that has been developed is dimercaptosuccinic acid (DMSA), shown in [Figure 19.32](#). Identify which atoms in this molecule could act as donor atoms.

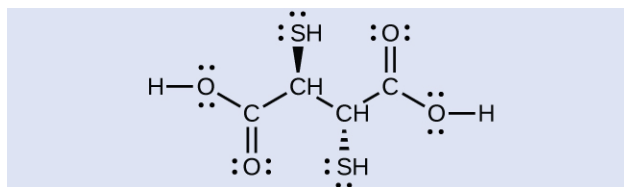


FIGURE 19.32 Dimercaptosuccinic acid is used to treat heavy metal poisoning.

Solution

All of the oxygen and sulfur atoms have lone pairs of electrons that can be used to coordinate to a metal center, so there are six possible donor atoms. Geometrically, only two of these atoms can be coordinated to a metal at once. The most common binding mode involves the coordination of one sulfur atom and one oxygen atom, forming a five-member ring with the metal.

Check Your Learning

Some alternative medicine practitioners recommend chelation treatments for ailments that are not clearly related to heavy metals, such as cancer and autism, although the practice is discouraged by many scientific organizations.¹ Identify at least two biologically important metals that could be disrupted by chelation therapy.

Answer:

Ca, Fe, Zn, and Cu

Ligands are also used in the electroplating industry. When metal ions are reduced to produce thin metal coatings, metals can clump together to form clusters and nanoparticles. When metal coordination complexes are used, the ligands keep the metal atoms isolated from each other. It has been found that many metals plate out as a smoother, more uniform, better-looking, and more adherent surface when plated from a bath containing the metal as a complex ion. Thus, complexes such as $[\text{Ag}(\text{CN})_2]^-$ and $[\text{Au}(\text{CN})_2]^-$ are used extensively in the electroplating industry.

In 1965, scientists at Michigan State University discovered that there was a platinum complex that inhibited cell division in certain microorganisms. Later work showed that the complex was *cis*-diamminedichloroplatinum(II), $[\text{Pt}(\text{NH}_3)_2(\text{Cl})_2]$, and that the *trans* isomer was not effective. The inhibition of cell division indicated that this square planar compound could be an anticancer agent. In 1978, the US Food and Drug Administration approved this compound, known as cisplatin, for use in the treatment of certain forms of cancer. Since that time, many similar platinum compounds have been developed for the treatment of cancer. In all cases, these are the *cis* isomers and never the *trans* isomers. The diammine $(\text{NH}_3)_2$ portion is retained with other groups, replacing the dichloro $[(\text{Cl})_2]$ portion. The newer drugs include carboplatin, oxaliplatin, and satraplatin.

19.3 Spectroscopic and Magnetic Properties of Coordination Compounds

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Outline the basic premise of crystal field theory (CFT)
- Identify molecular geometries associated with various d-orbital splitting patterns
- Predict electron configurations of split d orbitals for selected transition metal atoms or ions
- Explain spectral and magnetic properties in terms of CFT concepts

1 National Council against Health Fraud, *NCAHF Policy Statement on Chelation Therapy*, (Peabody, MA, 2002).

The behavior of coordination compounds cannot be adequately explained by the same theories used for main group element chemistry. The observed geometries of coordination complexes are not consistent with hybridized orbitals on the central metal overlapping with ligand orbitals, as would be predicted by valence bond theory. The observed colors indicate that the d orbitals often occur at different energy levels rather than all being degenerate, that is, of equal energy, as are the three p orbitals. To explain the stabilities, structures, colors, and magnetic properties of transition metal complexes, a different bonding model has been developed. Just as valence bond theory explains many aspects of bonding in main group chemistry, crystal field theory is useful in understanding and predicting the behavior of transition metal complexes.

Crystal Field Theory

To explain the observed behavior of transition metal complexes (such as how colors arise), a model involving electrostatic interactions between the electrons from the ligands and the electrons in the unhybridized d orbitals of the central metal atom has been developed. This electrostatic model is **crystal field theory** (CFT). It allows us to understand, interpret, and predict the colors, magnetic behavior, and some structures of coordination compounds of transition metals.

CFT focuses on the nonbonding electrons on the central metal ion in coordination complexes not on the metal-ligand bonds. Like valence bond theory, CFT tells only part of the story of the behavior of complexes. However, it tells the part that valence bond theory does not. In its pure form, CFT ignores any covalent bonding between ligands and metal ions. Both the ligand and the metal are treated as infinitesimally small point charges.

All electrons are negative, so the electrons donated from the ligands will repel the electrons of the central metal. Let us consider the behavior of the electrons in the unhybridized d orbitals in an octahedral complex. The five d orbitals consist of lobe-shaped regions and are arranged in space, as shown in [Figure 19.33](#). In an octahedral complex, the six ligands coordinate along the axes.

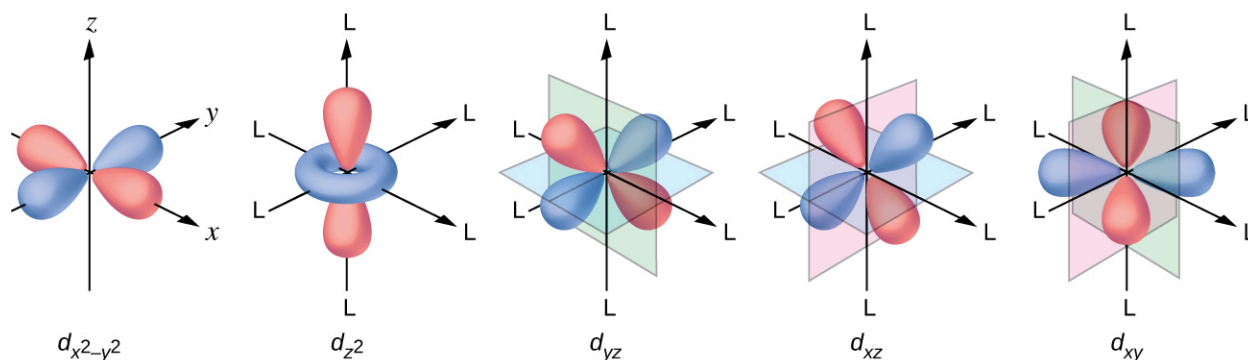


FIGURE 19.33 The directional characteristics of the five d orbitals are shown here. The shaded portions indicate the phase of the orbitals. The ligands (L) coordinate along the axes. For clarity, the ligands have been omitted from the $d_{x^2-y^2}$ orbital so that the axis labels could be shown.

In an uncomplexed metal ion in the gas phase, the electrons are distributed among the five d orbitals in accord with Hund's rule because the orbitals all have the same energy. However, when ligands coordinate to a metal ion, the energies of the d orbitals are no longer the same.

In octahedral complexes, the lobes in two of the five d orbitals, the d_{z^2} and $d_{x^2-y^2}$ orbitals, point toward the ligands ([Figure 19.33](#)). These two orbitals are called the **e_g orbitals** (the symbol actually refers to the symmetry of the orbitals, but we will use it as a convenient name for these two orbitals in an octahedral complex). The other three orbitals, the d_{xy} , d_{xz} , and d_{yz} orbitals, have lobes that point between the ligands and are called the **t_{2g} orbitals** (again, the symbol really refers to the symmetry of the orbitals). As six ligands approach the metal ion along the axes of the octahedron, their point charges repel the electrons in the d orbitals of the metal ion. However, the repulsions between the electrons in the e_g orbitals (the d_{z^2} and $d_{x^2-y^2}$ orbitals) and the ligands are greater than the repulsions between the electrons in the t_{2g} orbitals (the d_{xy} , d_{xz} , and d_{yz} orbitals) and the ligands. This is because the lobes of the e_g orbitals point directly at the ligands, whereas the lobes of the t_{2g} orbitals point between them. Thus, electrons in the e_g orbitals of the metal ion in an octahedral complex have higher potential energies

than those of electrons in the t_{2g} orbitals. The difference in energy may be represented as shown in [Figure 19.34](#).

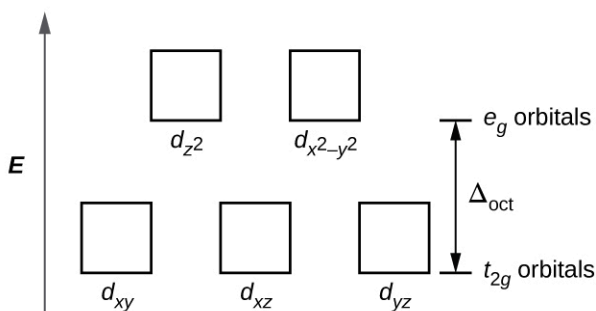
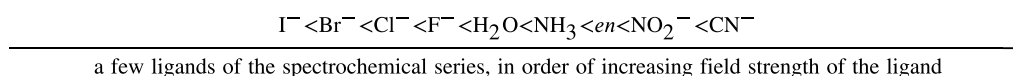


FIGURE 19.34 In octahedral complexes, the e_g orbitals are destabilized (higher in energy) compared to the t_{2g} orbitals because the ligands interact more strongly with the d orbitals at which they are pointed directly.

The difference in energy between the e_g and the t_{2g} orbitals is called the **crystal field splitting** and is symbolized by Δ_{oct} , where oct stands for octahedral.

The magnitude of Δ_{oct} depends on many factors, including the nature of the six ligands located around the central metal ion, the charge on the metal, and whether the metal is using $3d$, $4d$, or $5d$ orbitals. Different ligands produce different crystal field splittings. The increasing crystal field splitting produced by ligands is expressed in the **spectrochemical series**, a short version of which is given here:



In this series, ligands on the left cause small crystal field splittings and are **weak-field ligands**, whereas those on the right cause larger splittings and are **strong-field ligands**. Thus, the Δ_{oct} value for an octahedral complex with iodide ligands (I^-) is much smaller than the Δ_{oct} value for the same metal with cyanide ligands (CN^-).

Electrons in the d orbitals follow the aufbau (“filling up”) principle, which says that the orbitals will be filled to give the lowest total energy, just as in main group chemistry. When two electrons occupy the same orbital, the like charges repel each other. The energy needed to pair up two electrons in a single orbital is called the **pairing energy (P)**. Electrons will always singly occupy each orbital in a degenerate set before pairing. P is similar in magnitude to Δ_{oct} . When electrons fill the d orbitals, the relative magnitudes of Δ_{oct} and P determine which orbitals will be occupied.

In $[\text{Fe}(\text{CN})_6]^{4-}$, the strong field of six cyanide ligands produces a large Δ_{oct} . Under these conditions, the electrons require less energy to pair than they require to be excited to the e_g orbitals ($\Delta_{\text{oct}} > P$). The six $3d$ electrons of the Fe^{2+} ion pair in the three t_{2g} orbitals ([Figure 19.35](#)). Complexes in which the electrons are paired because of the large crystal field splitting are called **low-spin complexes** because the number of unpaired electrons (spins) is minimized.

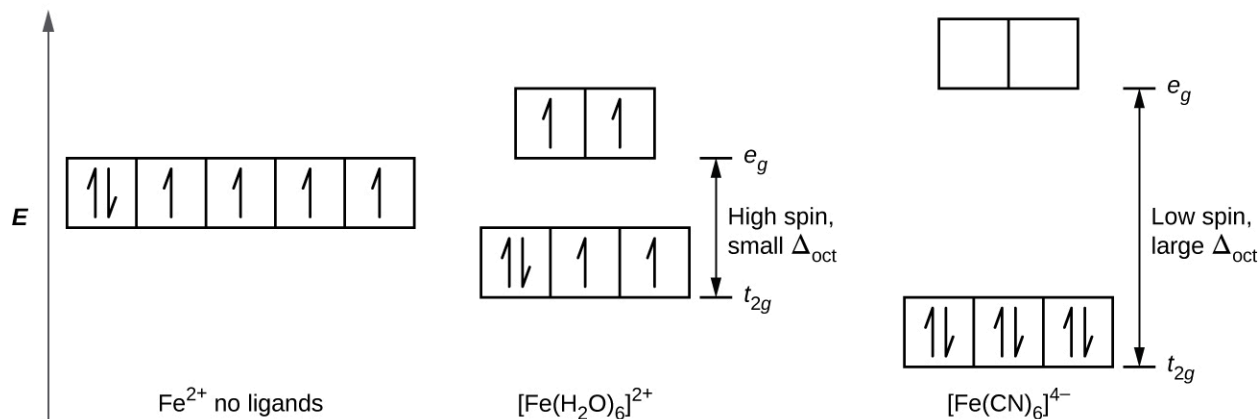


FIGURE 19.35 Iron(II) complexes have six electrons in the $5d$ orbitals. In the absence of a crystal field, the orbitals are degenerate. For coordination complexes with strong-field ligands such as $[\text{Fe}(\text{CN})_6]^{4-}$, Δ_{oct} is greater than P, and the electrons pair in the lower energy t_{2g}

orbitals before occupying the e_g orbitals. With weak-field ligands such as H_2O , the ligand field splitting is less than the pairing energy, Δ_{oct} less than P , so the electrons occupy all d orbitals singly before any pairing occurs.

In $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, on the other hand, the weak field of the water molecules produces only a small crystal field splitting ($\Delta_{\text{oct}} < P$). Because it requires less energy for the electrons to occupy the e_g orbitals than to pair together, there will be an electron in each of the five $3d$ orbitals before pairing occurs. For the six d electrons on the iron(II) center in $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$, there will be one pair of electrons and four unpaired electrons (Figure 19.35). Complexes such as the $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ ion, in which the electrons are unpaired because the crystal field splitting is not large enough to cause them to pair, are called **high-spin complexes** because the number of unpaired electrons (spins) is maximized.

A similar line of reasoning shows why the $[\text{Fe}(\text{CN})_6]^{3-}$ ion is a low-spin complex with only one unpaired electron, whereas both the $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{FeF}_6]^{3-}$ ions are high-spin complexes with five unpaired electrons.



EXAMPLE 19.7

High- and Low-Spin Complexes

Predict the number of unpaired electrons.

- (a) $\text{K}_3[\text{CrI}_6]$
- (b) $[\text{Cu}(\text{en})_2(\text{H}_2\text{O})_2]\text{Cl}_2$
- (c) $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$

Solution

The complexes are octahedral.

- (a) Cr^{3+} has a d^3 configuration. These electrons will all be unpaired.
- (b) Cu^{2+} is d^9 , so there will be one unpaired electron.
- (c) Co^{3+} has d^6 valence electrons, so the crystal field splitting will determine how many are paired. Nitrite is a strong-field ligand, so the complex will be low spin. Six electrons will go in the t_{2g} orbitals, leaving 0 unpaired.

Check Your Learning

The size of the crystal field splitting only influences the arrangement of electrons when there is a choice between pairing electrons and filling the higher-energy orbitals. For which d -electron configurations will there be a difference between high- and low-spin configurations in octahedral complexes?

Answer:

d^4 , d^5 , d^6 , and d^7



EXAMPLE 19.8

CFT for Other Geometries

CFT is applicable to molecules in geometries other than octahedral. In octahedral complexes, remember that the lobes of the e_g set point directly at the ligands. For tetrahedral complexes, the d orbitals remain in place, but now we have only four ligands located between the axes (Figure 19.36). None of the orbitals points directly at the tetrahedral ligands. However, the e_g set (along the Cartesian axes) overlaps with the ligands less than does the t_{2g} set. By analogy with the octahedral case, predict the energy diagram for the d orbitals in a tetrahedral crystal field. To avoid confusion, the octahedral e_g set becomes a tetrahedral e set, and the octahedral t_{2g} set becomes a t_2 set.

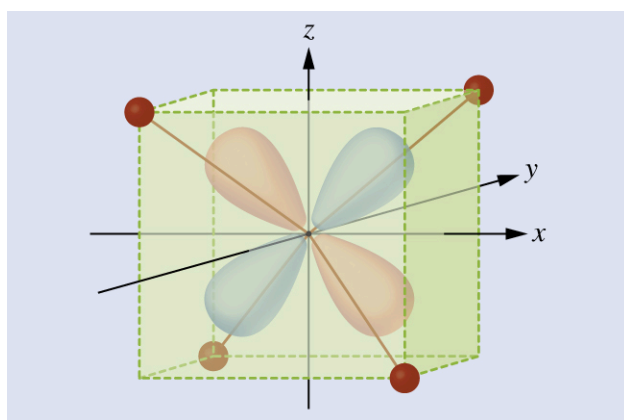
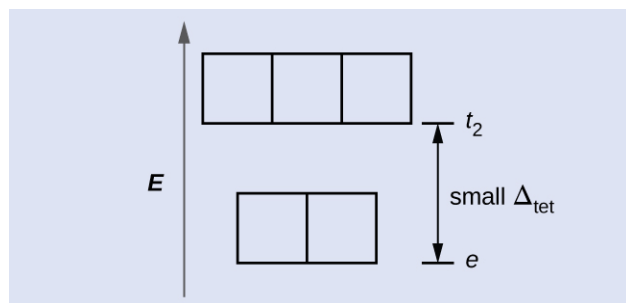


FIGURE 19.36 This diagram shows the orientation of the tetrahedral ligands with respect to the axis system for the orbitals.

Solution

Since CFT is based on electrostatic repulsion, the orbitals closer to the ligands will be destabilized and raised in energy relative to the other set of orbitals. The splitting is less than for octahedral complexes because the overlap is less, so Δ_{tet} is usually small ($\Delta_{\text{tet}} = \frac{4}{9} \Delta_{\text{oct}}$):



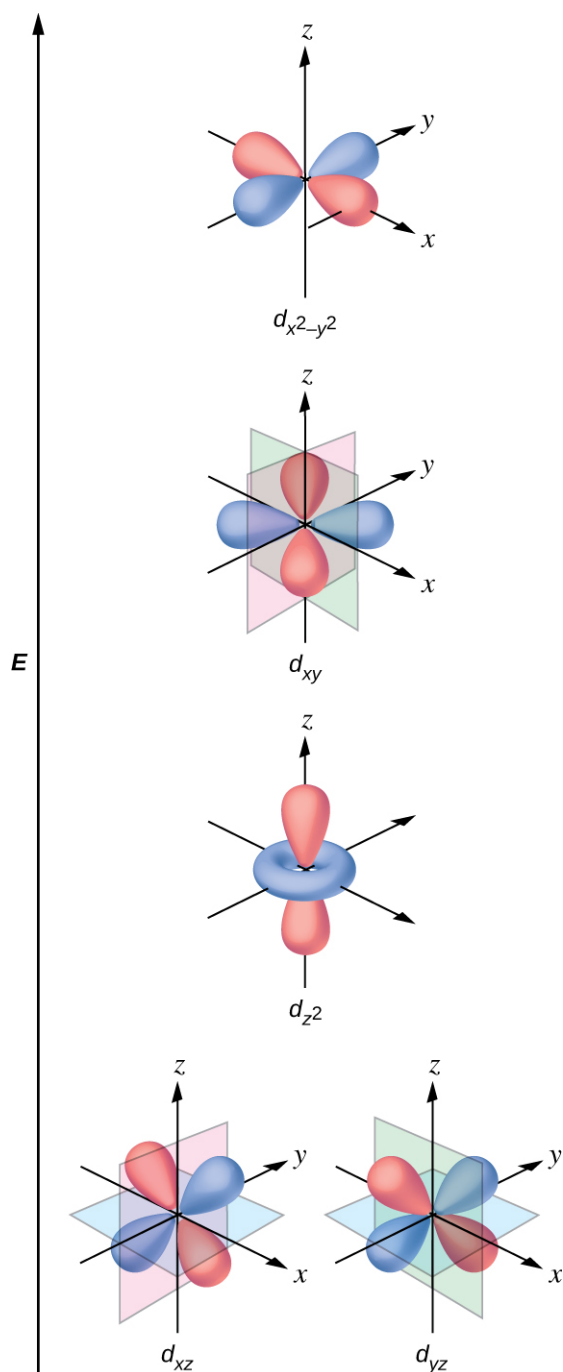
Check Your Learning

Explain how many unpaired electrons a tetrahedral d^4 ion will have.

Answer:

4; because Δ_{tet} is small, all tetrahedral complexes are high spin and the electrons go into the t_2 orbitals before pairing

The other common geometry is square planar. It is possible to consider a square planar geometry as an octahedral structure with a pair of *trans* ligands removed. The removed ligands are assumed to be on the z-axis. This changes the distribution of the d orbitals, as orbitals on or near the z-axis become more stable, and those on or near the x- or y-axes become less stable. This results in the octahedral t_{2g} and the e_g sets splitting and gives a more complicated pattern, as depicted below:



Magnetic Moments of Molecules and Ions

Experimental evidence of magnetic measurements supports the theory of high- and low-spin complexes. Remember that molecules such as O_2 that contain unpaired electrons are paramagnetic. Paramagnetic substances are attracted to magnetic fields. Many transition metal complexes have unpaired electrons and hence are paramagnetic. Molecules such as N_2 and ions such as Na^+ and $[Fe(CN)_6]^{4-}$ that contain no unpaired electrons are diamagnetic. Diamagnetic substances have a slight tendency to be repelled by magnetic fields.

When an electron in an atom or ion is unpaired, the magnetic moment due to its spin makes the entire atom or ion paramagnetic. The size of the magnetic moment of a system containing unpaired electrons is related directly to the number of such electrons: the greater the number of unpaired electrons, the larger the magnetic moment. Therefore, the observed magnetic moment is used to determine the number of unpaired electrons present. The measured magnetic moment of low-spin d^6 $[Fe(CN)_6]^{4-}$ confirms that iron is diamagnetic, whereas high-spin d^6

$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ has four unpaired electrons with a magnetic moment that confirms this arrangement.

Colors of Transition Metal Complexes

When atoms or molecules absorb light at the proper frequency, their electrons are excited to higher-energy orbitals. For many main group atoms and molecules, the absorbed photons are in the ultraviolet range of the electromagnetic spectrum, which cannot be detected by the human eye. For coordination compounds, the energy difference between the d orbitals often allows photons in the visible range to be absorbed.

The human eye perceives a mixture of all the colors, in the proportions present in sunlight, as white light. Complementary colors, those located across from each other on a color wheel, are also used in color vision. The eye perceives a mixture of two complementary colors, in the proper proportions, as white light. Likewise, when a color is missing from white light, the eye sees its complement. For example, when red photons are absorbed from white light, the eyes see the color green. When violet photons are removed from white light, the eyes see lemon yellow. The blue color of the $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion results because this ion absorbs orange and red light, leaving the complementary colors of blue and green (Figure 19.37).

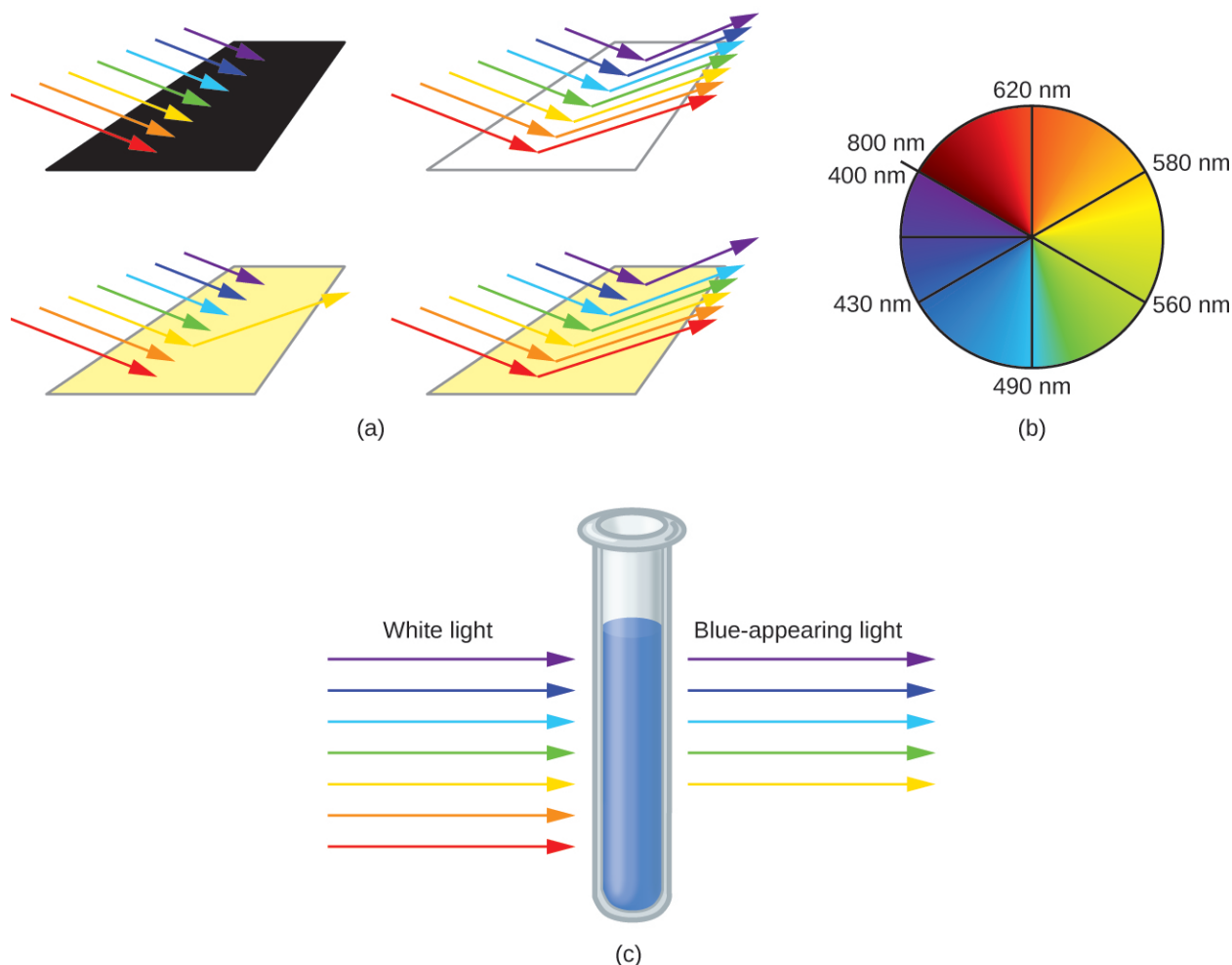


FIGURE 19.37 (a) An object is black if it absorbs all colors of light. If it reflects all colors of light, it is white. An object has a color if it absorbs all colors except one, such as this yellow strip. The strip also appears yellow if it absorbs the complementary color from white light (in this case, indigo). (b) Complementary colors are located directly across from one another on the color wheel. (c) A solution of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ions absorbs red and orange light, so the transmitted light appears as the complementary color, blue.



EXAMPLE 19.9

Colors of Complexes

The octahedral complex $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ has a single d electron. To excite this electron from the ground state t_{2g} orbital to the e_g orbital, this complex absorbs light from 450 to 600 nm. The maximum absorbance corresponds to Δ_{oct} and occurs at 499 nm. Calculate the value of Δ_{oct} in Joules and predict what color the solution will appear.

Solution

Using Planck's equation (refer to the section on electromagnetic energy), we calculate:

$$\nu = \frac{c}{\lambda} \text{ so } \frac{3.00 \times 10^8 \text{ m/s}}{\frac{499 \text{ nm} \times 1 \text{ m}}{10^9 \text{ nm}}} = 6.01 \times 10^{14} \text{ Hz}$$

$$E = h\nu \text{ so } 6.63 \times 10^{-34} \text{ J}\cdot\text{s} \times 6.01 \times 10^{14} \text{ Hz} = 3.99 \times 10^{-19} \text{ Joules/ion}$$

Because the complex absorbs 600 nm (orange) through 450 (blue), the indigo, violet, and red wavelengths will be transmitted, and the complex will appear purple.

Check Your Learning

A complex that appears green, absorbs photons of what wavelengths?

Answer:

red, 620–800 nm

Small changes in the relative energies of the orbitals that electrons are transitioning between can lead to drastic shifts in the color of light absorbed. Therefore, the colors of coordination compounds depend on many factors. As shown in [Figure 19.38](#), different aqueous metal ions can have different colors. In addition, different oxidation states of one metal can produce different colors, as shown for the vanadium complexes in the link below.



FIGURE 19.38 The partially filled d orbitals of the stable ions $\text{Cr}^{3+}(\text{aq})$, $\text{Fe}^{3+}(\text{aq})$, and $\text{Co}^{2+}(\text{aq})$ (left, center and right, respectively) give rise to various colors. (credit: Sahar Atwa)

The specific ligands coordinated to the metal center also influence the color of coordination complexes. For example, the iron(II) complex $[\text{Fe}(\text{H}_2\text{O})_6]\text{SO}_4$ appears blue-green because the high-spin complex absorbs photons in the red wavelengths ([Figure 19.39](#)). In contrast, the low-spin iron(II) complex $\text{K}_4[\text{Fe}(\text{CN})_6]$ appears pale yellow because it absorbs higher-energy violet photons.

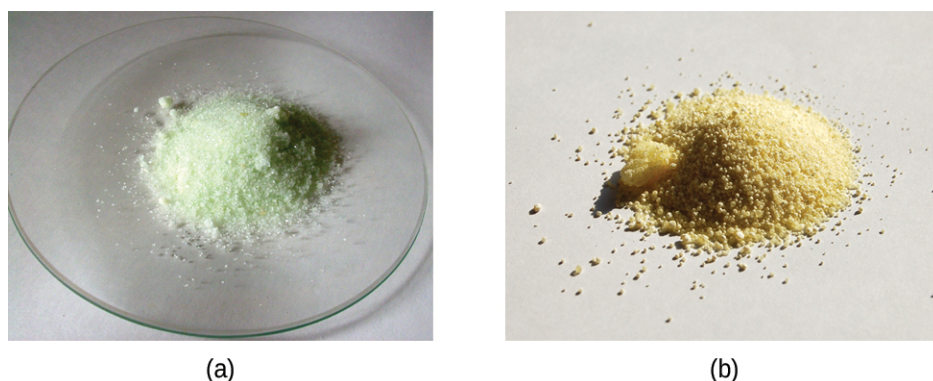


FIGURE 19.39 Both (a) hexaaquairon(II) sulfate and (b) potassium hexacyanoferrate(II) contain d^6 iron(II) octahedral metal centers, but they absorb photons in different ranges of the visible spectrum.

LINK TO LEARNING

Watch this [video \(http://openstax.org/l/16vanadium\)](http://openstax.org/l/16vanadium) of the reduction of vanadium complexes to observe the colorful effect of changing oxidation states.

In general, strong-field ligands cause a large split in the energies of d orbitals of the central metal atom (large Δ_{oct}). Transition metal coordination compounds with these ligands are yellow, orange, or red because they absorb higher-energy violet or blue light. On the other hand, coordination compounds of transition metals with weak-field ligands are often blue-green, blue, or indigo because they absorb lower-energy yellow, orange, or red light.

A coordination compound of the Cu^+ ion has a d^{10} configuration, and all the e_g orbitals are filled. To excite an electron to a higher level, such as the $4p$ orbital, photons of very high energy are necessary. This energy corresponds to very short wavelengths in the ultraviolet region of the spectrum. No visible light is absorbed, so the eye sees no change, and the compound appears white or colorless. A solution containing $[\text{Cu}(\text{CN})_2]^-$, for example, is colorless. On the other hand, octahedral Cu^{2+} complexes have a vacancy in the e_g orbitals, and electrons can be excited to this level. The wavelength (energy) of the light absorbed corresponds to the visible part of the spectrum, and Cu^{2+} complexes are almost always colored—blue, blue-green violet, or yellow (Figure 19.40). Although CFT successfully describes many properties of coordination complexes, molecular orbital explanations (beyond the introductory scope provided here) are required to understand fully the behavior of coordination complexes.

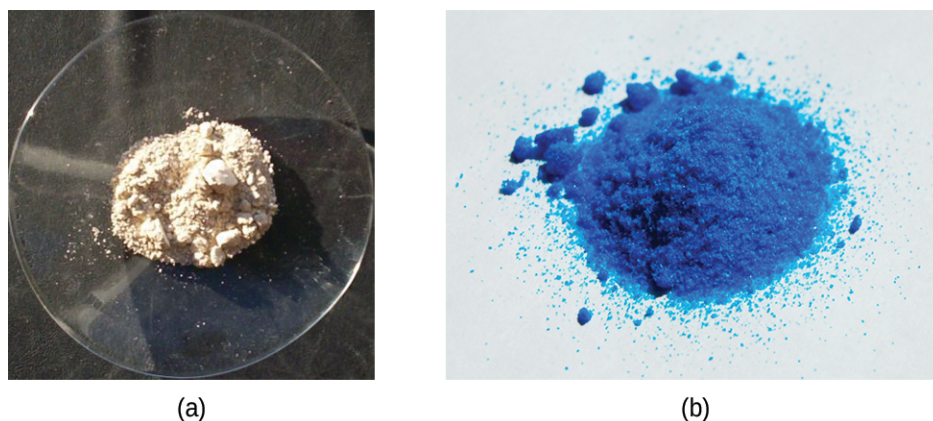


FIGURE 19.40 (a) Copper(I) complexes with d^{10} configurations such as CuI tend to be colorless, whereas (b) d^9 copper(II) complexes such as $\text{Cu}(\text{NO}_3)_2 \cdot 5\text{H}_2\text{O}$ are brightly colored.

Key Terms

actinide series (also, actinoid series) actinium and the elements in the second row or the *f*-block, atomic numbers 89–103

bidentate ligand ligand that coordinates to one central metal through coordinate bonds from two different atoms

central metal ion or atom to which one or more ligands is attached through coordinate covalent bonds

chelate complex formed from a polydentate ligand attached to a central metal

chelating ligand ligand that attaches to a central metal ion by bonds from two or more donor atoms

cis configuration configuration of a geometrical isomer in which two similar groups are on the same side of an imaginary reference line on the molecule

coordination compound stable compound in which the central metal atom or ion acts as a Lewis acid and accepts one or more pairs of electrons

coordination compound substance consisting of atoms, molecules, or ions attached to a central atom through Lewis acid-base interactions

coordination number number of coordinate covalent bonds to the central metal atom in a complex or the number of closest contacts to an atom in a crystalline form

coordination sphere central metal atom or ion plus the attached ligands of a complex

crystal field splitting (Δ_{oct}) difference in energy between the t_{2g} and e_g sets or t and e sets of orbitals

crystal field theory model that explains the energies of the orbitals in transition metals in terms of electrostatic interactions with the ligands but does not include metal ligand bonding

***d*-block element** one of the elements in groups 3–11 with valence electrons in *d* orbitals

donor atom atom in a ligand with a lone pair of electrons that forms a coordinate covalent bond to a central metal

e_g orbitals set of two *d* orbitals that are oriented on the Cartesian axes for coordination complexes; in octahedral complexes, they are higher in energy than the t_{2g} orbitals

***f*-block element** (also, inner transition element) one of the elements with atomic numbers 58–71 or 90–103 that have valence electrons in *f* orbitals; they are frequently shown offset below the periodic table

first transition series transition elements in the fourth period of the periodic table (first row of the *d*-block), atomic numbers 21–29

fourth transition series transition elements in the seventh period of the periodic table (fourth row of the *d*-block), atomic numbers 89 and 104–111

geometric isomers isomers that differ in the way in which atoms are oriented in space relative to each other, leading to different physical and chemical properties

high-spin complex complex in which the electrons maximize the total electron spin by singly populating all of the orbitals before pairing two electrons into the lower-energy orbitals

hydrometallurgy process in which a metal is separated from a mixture by first converting it into soluble ions, extracting the ions, and then reducing the ions to precipitate the pure metal

ionization isomer (or coordination isomer) isomer in which an anionic ligand is replaced by the counter ion in the inner coordination sphere

lanthanide series (also, lanthanoid series) lanthanum and the elements in the first row or the *f*-block, atomic numbers 57–71

ligand ion or neutral molecule attached to the central metal ion in a coordination compound

linkage isomer coordination compound that possesses a ligand that can bind to the transition metal in two different ways (CN^- vs. NC^-)

low-spin complex complex in which the electrons minimize the total electron spin by pairing in the lower-energy orbitals before populating the higher-energy orbitals

monodentate ligand that attaches to a central metal through just one coordinate covalent bond

optical isomer (also, enantiomer) molecule that is a nonsuperimposable mirror image with identical chemical and physical properties, except when it reacts with other optical isomers

pairing energy (*P*) energy required to place two electrons with opposite spins into a single orbital

platinum metals group of six transition metals consisting of ruthenium, osmium, rhodium, iridium, palladium, and platinum that tend to occur in the same minerals and demonstrate similar chemical properties

polydentate ligand ligand that is attached to a central metal ion by bonds from two or more donor atoms, named with prefixes specifying how many donors are present (e.g., hexadentate = six coordinate bonds formed)

rare earth element collection of 17 elements including the lanthanides, scandium, and yttrium that often occur together and have similar chemical properties, making separation difficult

second transition series transition elements in the fifth period of the periodic table (second row of the *d*-block), atomic numbers 39–47

smelting process of extracting a pure metal from a molten ore

spectrochemical series ranking of ligands according to the magnitude of the crystal field splitting they induce

steel material made from iron by removing impurities in the iron and adding substances that produce alloys with properties suitable for specific uses

strong-field ligand ligand that causes larger crystal field splittings

superconductor material that conducts electricity

with no resistance

t_{2g} orbitals set of three *d* orbitals aligned between the Cartesian axes for coordination complexes; in octahedral complexes, they are lowered in energy compared to the e_g orbitals according to CFT

third transition series transition elements in the sixth period of the periodic table (third row of the *d*-block), atomic numbers 57 and 72–79

trans configuration configuration of a geometrical isomer in which two similar groups are on opposite sides of an imaginary reference line on the molecule

weak-field ligand ligand that causes small crystal field splittings

Summary

19.1 Occurrence, Preparation, and Properties of Transition Metals and Their Compounds

The transition metals are elements with partially filled *d* orbitals, located in the *d*-block of the periodic table. The reactivity of the transition elements varies widely from very active metals such as scandium and iron to almost inert elements, such as the platinum metals. The type of chemistry used in the isolation of the elements from their ores depends upon the concentration of the element in its ore and the difficulty of reducing ions of the elements to the metals. Metals that are more active are more difficult to reduce.

Transition metals exhibit chemical behavior typical of metals. For example, they oxidize in air upon heating and react with elemental halogens to form halides. Those elements that lie above hydrogen in the activity series react with acids, producing salts and hydrogen gas. Oxides, hydroxides, and carbonates of transition metal compounds in low oxidation states are basic. Halides and other salts are generally stable in water, although oxygen must be excluded in some cases. Most transition metals form a variety of stable oxidation states, allowing them to demonstrate a wide range of chemical reactivity.

19.2 Coordination Chemistry of Transition Metals

The transition elements and main group elements can form coordination compounds, or complexes, in which a central metal atom or ion is bonded to one or more

ligands by coordinate covalent bonds. Ligands with more than one donor atom are called polydentate ligands and form chelates. The common geometries found in complexes are tetrahedral and square planar (both with a coordination number of four) and octahedral (with a coordination number of six). *Cis* and *trans* configurations are possible in some octahedral and square planar complexes. In addition to these geometrical isomers, optical isomers (molecules or ions that are mirror images but not superimposable) are possible in certain octahedral complexes. Coordination complexes have a wide variety of uses including oxygen transport in blood, water purification, and pharmaceutical use.

19.3 Spectroscopic and Magnetic Properties of Coordination Compounds

Crystal field theory treats interactions between the electrons on the metal and the ligands as a simple electrostatic effect. The presence of the ligands near the metal ion changes the energies of the metal *d* orbitals relative to their energies in the free ion. Both the color and the magnetic properties of a complex can be attributed to this crystal field splitting. The magnitude of the splitting (Δ_{oct}) depends on the nature of the ligands bonded to the metal. Strong-field ligands produce large splitting and favor low-spin complexes, in which the t_{2g} orbitals are completely filled before any electrons occupy the e_g orbitals. Weak-field ligands favor formation of high-spin complexes. The t_{2g} and the e_g orbitals are singly occupied before any are doubly occupied.

Exercises

19.1 Occurrence, Preparation, and Properties of Transition Metals and Their Compounds

- Write the electron configurations for each of the following elements:
 - Sc
 - Ti
 - Cr
 - Fe
 - Ru
- Write the electron configurations for each of the following elements and its ions:
 - Ti
 - Ti²⁺
 - Ti³⁺
 - Ti⁴⁺
- Write the electron configurations for each of the following elements and its 3+ ions:
 - La
 - Sm
 - Lu
- Why are the lanthanoid elements not found in nature in their elemental forms?
- Which of the following elements is most likely to be used to prepare La by the reduction of La₂O₃: Al, C, or Fe? Why?
- Which of the following is the strongest oxidizing agent: VO₄³⁻, CrO₄²⁻, or MnO₄⁻?
- Which of the following elements is most likely to form an oxide with the formula MO₃: Zr, Nb, or Mo?
- The following reactions all occur in a blast furnace. Which of these are redox reactions?
 - $3\text{Fe}_2\text{O}_3(s) + \text{CO}(g) \longrightarrow 2\text{Fe}_3\text{O}_4(s) + \text{CO}_2(g)$
 - $\text{Fe}_3\text{O}_4(s) + \text{CO}(g) \longrightarrow 3\text{FeO}(s) + \text{CO}_2(g)$
 - $\text{FeO}(s) + \text{CO}(g) \longrightarrow \text{Fe}(l) + \text{CO}_2(g)$
 - $\text{C}(s) + \text{O}_2(g) \longrightarrow \text{CO}_2(g)$
 - $\text{C}(s) + \text{CO}_2(g) \longrightarrow 2\text{CO}(g)$
 - $\text{CaCO}_3(s) \longrightarrow \text{CaO}(s) + \text{CO}_2(g)$
 - $\text{CaO}(s) + \text{SiO}_2(s) \longrightarrow \text{CaSiO}_3(l)$
- Why is the formation of slag useful during the smelting of iron?
- Would you expect an aqueous manganese(VII) oxide solution to have a pH greater or less than 7.0? Justify your answer.
- Iron(II) can be oxidized to iron(III) by dichromate ion, which is reduced to chromium(III) in acid solution. A 2.5000-g sample of iron ore is dissolved and the iron converted into iron(II). Exactly 19.17 mL of 0.0100 M Na₂Cr₂O₇ is required in the titration. What percentage of the ore sample was iron?
- How many cubic feet of air at a pressure of 760 torr and 0 °C is required per ton of Fe₂O₃ to convert that Fe₂O₃ into iron in a blast furnace? For this exercise, assume air is 19% oxygen by volume.
- Find the potentials of the following electrochemical cell:
Cd | Cd²⁺, M = 0.10 || Ni²⁺, M = 0.50 | Ni
- A 2.5624-g sample of a pure solid alkali metal chloride is dissolved in water and treated with excess silver nitrate. The resulting precipitate, filtered and dried, weighs 3.03707 g. What was the percent by mass of chloride ion in the original compound? What is the identity of the salt?
- The standard reduction potential for the reaction $[\text{Co}(\text{H}_2\text{O})_6]^{3+}(aq) + e^- \longrightarrow [\text{Co}(\text{H}_2\text{O})_6]^{2+}(aq)$ is about 1.8 V. The reduction potential for the reaction $[\text{Co}(\text{NH}_3)_6]^{3+}(aq) + e^- \longrightarrow [\text{Co}(\text{NH}_3)_6]^{2+}(aq)$ is +0.1 V.

Calculate the cell potentials to show whether the complex ions, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and/or $[\text{Co}(\text{NH}_3)_6]^{2+}$, can be oxidized to the corresponding cobalt(III) complex by oxygen.

16. Predict the products of each of the following reactions. (Note: In addition to using the information in this chapter, also use the knowledge you have accumulated at this stage of your study, including information on the prediction of reaction products.)
 - (a) $\text{MnCO}_3(s) + \text{HI}(aq) \longrightarrow$
 - (b) $\text{CoO}(s) + \text{O}_2(g) \longrightarrow$
 - (c) $\text{La}(s) + \text{O}_2(g) \longrightarrow$
 - (d) $\text{V}(s) + \text{VCl}_4(s) \longrightarrow$
 - (e) $\text{Co}(s) + x\text{F}_2(g) \longrightarrow$
 - (f) $\text{CrO}_3(s) + \text{CsOH}(aq) \longrightarrow$
17. Predict the products of each of the following reactions. (Note: In addition to using the information in this chapter, also use the knowledge you have accumulated at this stage of your study, including information on the prediction of reaction products.)
 - (a) $\text{Fe}(s) + \text{H}_2\text{SO}_4(aq) \longrightarrow$
 - (b) $\text{FeCl}_3(aq) + \text{NaOH}(aq) \longrightarrow$
 - (c) $\text{Mn}(\text{OH})_2(s) + \text{HBr}(aq) \longrightarrow$
 - (d) $\text{Cr}(s) + \text{O}_2(g) \longrightarrow$
 - (e) $\text{Mn}_2\text{O}_3(s) + \text{HCl}(aq) \longrightarrow$
 - (f) $\text{Ti}(s) + x\text{F}_2(g) \longrightarrow$
18. Describe the electrolytic process for refining copper.
19. Predict the products of the following reactions and balance the equations.
 - (a) Zn is added to a solution of $\text{Cr}_2(\text{SO}_4)_3$ in acid.
 - (b) FeCl_2 is added to a solution containing an excess of $\text{Cr}_2\text{O}_7^{2-}$ in hydrochloric acid.
 - (c) Cr^{2+} is added to $\text{Cr}_2\text{O}_7^{2-}$ in acid solution.
 - (d) Mn is heated with CrO_3 .
 - (e) CrO is added to 2HNO_3 in water.
 - (f) FeCl_3 is added to an aqueous solution of NaOH.
20. What is the gas produced when iron(II) sulfide is treated with a nonoxidizing acid?
21. Predict the products of each of the following reactions and then balance the chemical equations.
 - (a) Fe is heated in an atmosphere of steam.
 - (b) NaOH is added to a solution of $\text{Fe}(\text{NO}_3)_3$.
 - (c) FeSO_4 is added to an acidic solution of KMnO_4 .
 - (d) Fe is added to a dilute solution of H_2SO_4 .
 - (e) A solution of $\text{Fe}(\text{NO}_3)_2$ and HNO_3 is allowed to stand in air.
 - (f) FeCO_3 is added to a solution of HClO_4 .
 - (g) Fe is heated in air.
22. Balance the following equations by oxidation-reduction methods; note that *three* elements change oxidation state.

$$\text{Co}(\text{NO}_3)_2(s) \longrightarrow \text{Co}_2\text{O}_3(s) + \text{NO}_2(g) + \text{O}_2(g)$$
23. Dilute sodium cyanide solution is slowly dripped into a slowly stirred silver nitrate solution. A white precipitate forms temporarily but dissolves as the addition of sodium cyanide continues. Use chemical equations to explain this observation. Silver cyanide is similar to silver chloride in its solubility.
24. Predict which will be more stable, $[\text{CrO}_4]^{2-}$ or $[\text{WO}_4]^{2-}$, and explain.
25. Give the oxidation state of the metal for each of the following oxides of the first transition series. (Hint: Oxides of formula M_3O_4 are examples of *mixed valence compounds* in which the metal ion is present in more than one oxidation state. It is possible to write these compound formulas in the equivalent format $\text{MO} \cdot \text{M}_2\text{O}_3$, to permit estimation of the metal's two oxidation states.)

- (a) Sc_2O_3
- (b) TiO_2
- (c) V_2O_5
- (d) CrO_3
- (e) MnO_2
- (f) Fe_3O_4
- (g) Co_3O_4
- (h) NiO
- (i) Cu_2O

19.2 Coordination Chemistry of Transition Metals

- 26.** Indicate the coordination number for the central metal atom in each of the following coordination compounds:
- (a) $[\text{Pt}(\text{H}_2\text{O})_2\text{Br}_2]$
 - (b) $[\text{Pt}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$ (py = pyridine, $\text{C}_5\text{H}_5\text{N}$)
 - (c) $[\text{Zn}(\text{NH}_3)_2\text{Cl}_2]$
 - (d) $[\text{Zn}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$
 - (e) $[\text{Ni}(\text{H}_2\text{O})_4\text{Cl}_2]$
 - (f) $[\text{Fe}(\text{en})_2(\text{CN})_2]^+$ (en = ethylenediamine, $\text{C}_2\text{H}_8\text{N}_2$)
- 27.** Give the coordination numbers and write the formulas for each of the following, including all isomers where appropriate:
- (a) tetrahydroxozincate(II) ion (tetrahedral)
 - (b) hexacyanopalladate(IV) ion
 - (c) dichloroaurate(I) ion (note that *aurum* is Latin for "gold")
 - (d) diamminedichloroplatinum(II)
 - (e) potassium diamminetetrahydrochlorochromate(III)
 - (f) hexaamminecobalt(III) hexacyanochromate(III)
 - (g) dibromobis(ethylenediamine) cobalt(III) nitrate
- 28.** Give the coordination number for each metal ion in the following compounds:
- (a) $[\text{Co}(\text{CO}_3)_3]^{3-}$ (note that CO_3^{2-} is bidentate in this complex)
 - (b) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 - (c) $[\text{Co}(\text{NH}_3)_4\text{Br}_2](\text{SO}_4)$
 - (d) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$
 - (e) $[\text{Cr}(\text{en})_3](\text{NO}_3)_3$
 - (f) $[\text{Pd}(\text{NH}_3)_2\text{Br}_2]$ (square planar)
 - (g) $\text{K}_3[\text{Cu}(\text{Cl})_5]$
 - (h) $[\text{Zn}(\text{NH}_3)_2\text{Cl}_2]$
- 29.** Sketch the structures of the following complexes. Indicate any *cis*, *trans*, and optical isomers.
- (a) $[\text{Pt}(\text{H}_2\text{O})_2\text{Br}_2]$ (square planar)
 - (b) $[\text{Pt}(\text{NH}_3)(\text{py})(\text{Cl})(\text{Br})]$ (square planar, py = pyridine, $\text{C}_5\text{H}_5\text{N}$)
 - (c) $[\text{Zn}(\text{NH}_3)_3\text{Cl}]^+$ (tetrahedral)
 - (d) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^+$ (square planar)
 - (e) $[\text{Ni}(\text{H}_2\text{O})_4\text{Cl}_2]$
 - (f) $[\text{Co}(\text{C}_2\text{O}_4)_2\text{Cl}_2]^{3-}$ (note that $\text{C}_2\text{O}_4^{2-}$ is the bidentate oxalate ion, $-\text{O}_2\text{CCO}_2^-$)
- 30.** Draw diagrams for any *cis*, *trans*, and optical isomers that could exist for the following (en is ethylenediamine):
- (a) $[\text{Co}(\text{en})_2(\text{NO}_2)\text{Cl}]^+$
 - (b) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 - (c) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
 - (d) $[\text{Cr}(\text{en})_3]^{3+}$
 - (e) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

31. Name each of the compounds or ions given in [Exercise 19.28](#), including the oxidation state of the metal.
32. Name each of the compounds or ions given in [Exercise 19.30](#).
33. Specify whether the following complexes have isomers.
 - (a) tetrahedral $[\text{Ni}(\text{CO})_2(\text{Cl})_2]$
 - (b) trigonal bipyramidal $[\text{Mn}(\text{CO})_4\text{NO}]$
 - (c) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$
34. Predict whether the carbonate ligand CO_3^{2-} will coordinate to a metal center as a monodentate, bidentate, or tridentate ligand.
35. Draw the geometric, linkage, and ionization isomers for $[\text{CoCl}_5\text{CN}][\text{CN}]$.

19.3 Spectroscopic and Magnetic Properties of Coordination Compounds

36. Determine the number of unpaired electrons expected for $[\text{Fe}(\text{NO}_2)_6]^{3-}$ and for $[\text{FeF}_6]^{3-}$ in terms of crystal field theory.
37. Draw the crystal field diagrams for $[\text{Fe}(\text{NO}_2)_6]^{4-}$ and $[\text{FeF}_6]^{3-}$. State whether each complex is high spin or low spin, paramagnetic or diamagnetic, and compare Δ_{oct} to P for each complex.
38. Give the oxidation state of the metal, number of *d* electrons, and the number of unpaired electrons predicted for $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$.
39. The anhydrous solid CoCl_2 is blue in color. Because it readily absorbs water from the air, it is used as a humidity indicator to monitor if equipment (such as a cell phone) has been exposed to excessive levels of moisture. Predict what product is formed by this reaction, and how many unpaired electrons this complex will have.
40. Is it possible for a complex of a metal in the transition series to have six unpaired electrons? Explain.
41. How many unpaired electrons are present in each of the following?
 - (a) $[\text{CoF}_6]^{3-}$ (high spin)
 - (b) $[\text{Mn}(\text{CN})_6]^{3-}$ (low spin)
 - (c) $[\text{Mn}(\text{CN})_6]^{4-}$ (low spin)
 - (d) $[\text{MnCl}_6]^{4-}$ (high spin)
 - (e) $[\text{RhCl}_6]^{3-}$ (low spin)
42. Explain how the diphosphate ion, $[\text{O}_3\text{P}-\text{O}-\text{PO}_3]^{4-}$, can function as a water softener that prevents the precipitation of Fe^{2+} as an insoluble iron salt.
43. For complexes of the same metal ion with no change in oxidation number, the stability increases as the number of electrons in the t_{2g} orbitals increases. Which complex in each of the following pairs of complexes is more stable?
 - (a) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ or $[\text{Fe}(\text{CN})_6]^{4-}$
 - (b) $[\text{Co}(\text{NH}_3)_6]^{3+}$ or $[\text{CoF}_6]^{3-}$
 - (c) $[\text{Mn}(\text{CN})_6]^{4-}$ or $[\text{MnCl}_6]^{4-}$
44. Trimethylphosphine, $\text{P}(\text{CH}_3)_3$, can act as a ligand by donating the lone pair of electrons on the phosphorus atom. If trimethylphosphine is added to a solution of nickel(II) chloride in acetone, a blue compound that has a molecular mass of approximately 270 g and contains 21.5% Ni, 26.0% Cl, and 52.5% $\text{P}(\text{CH}_3)_3$ can be isolated. This blue compound does not have any isomeric forms. What are the geometry and molecular formula of the blue compound?
45. Would you expect the complex $[\text{Co}(\text{en})_3]\text{Cl}_3$ to have any unpaired electrons? Any isomers?
46. Would you expect the $\text{Mg}_3[\text{Cr}(\text{CN})_6]_2$ to be diamagnetic or paramagnetic? Explain your reasoning.
47. Would you expect salts of the gold(I) ion, Au^+ , to be colored? Explain.
48. $[\text{CuCl}_4]^{2-}$ is green. $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ is blue. Which absorbs higher-energy photons? Which is predicted to have a

larger crystal field splitting?

CHAPTER 20

Organic Chemistry



FIGURE 20.1 All organic compounds contain carbon and most are formed by living things, although they are also formed by geological and artificial processes. (credit left: modification of work by Jon Sullivan; credit left middle: modification of work by Deb Tremper; credit right middle: modification of work by “annszyp”/Wikimedia Commons; credit right: modification of work by George Shuklin)

CHAPTER OUTLINE

20.1 Hydrocarbons

20.2 Alcohols and Ethers

20.3 Aldehydes, Ketones, Carboxylic Acids, and Esters

20.4 Amines and Amides

INTRODUCTION All living things on earth are formed mostly of carbon compounds. The prevalence of carbon compounds in living things has led to the epithet “carbon-based” life. The truth is we know of no other kind of life. Early chemists regarded substances isolated from *organisms* (plants and animals) as a different type of matter that could not be synthesized artificially, and these substances were thus known as *organic compounds*. The widespread belief called vitalism held that organic compounds were formed by a vital force present only in living organisms. The German chemist Friedrich Wohler was one of the early chemists to refute this aspect of vitalism, when, in 1828, he reported the synthesis of urea, a component of many body fluids, from nonliving materials. Since then, it has been recognized that organic molecules obey the same natural laws as inorganic substances, and the category of organic compounds has evolved to include both natural and synthetic compounds that contain carbon. Some carbon-containing compounds are *not* classified as organic, for example, carbonates and cyanides, and simple oxides, such as CO and CO₂. Although a single, precise definition has yet to be identified by the chemistry community, most agree that a defining trait of organic molecules is the presence of carbon as the principal element, bonded to hydrogen and other carbon atoms.

Today, organic compounds are key components of plastics, soaps, perfumes, sweeteners, fabrics, pharmaceuticals, and many other substances that we use every day. The value to us of organic compounds ensures that organic chemistry is an important discipline within the general field of chemistry. In this chapter, we discuss why the element carbon gives rise to a vast number and variety of compounds, how those compounds are classified, and the role of organic compounds in representative biological and industrial settings.

20.1 Hydrocarbons

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Explain the importance of hydrocarbons and the reason for their diversity
- Name saturated and unsaturated hydrocarbons, and molecules derived from them
- Describe the reactions characteristic of saturated and unsaturated hydrocarbons
- Identify structural and geometric isomers of hydrocarbons

The largest database¹ of organic compounds lists about 10 million substances, which include compounds originating from living organisms and those synthesized by chemists. The number of potential organic compounds has been estimated² at 10^{60} —an astronomically high number. The existence of so many organic molecules is a consequence of the ability of carbon atoms to form up to four strong bonds to other carbon atoms, resulting in chains and rings of many different sizes, shapes, and complexities.

The simplest **organic compounds** contain only the elements carbon and hydrogen, and are called hydrocarbons. Even though they are composed of only two types of atoms, there is a wide variety of hydrocarbons because they may consist of varying lengths of chains, branched chains, and rings of carbon atoms, or combinations of these structures. In addition, hydrocarbons may differ in the types of carbon-carbon bonds present in their molecules. Many hydrocarbons are found in plants, animals, and their fossils; other hydrocarbons have been prepared in the laboratory. We use hydrocarbons every day, mainly as fuels, such as natural gas, acetylene, propane, butane, and the principal components of gasoline, diesel fuel, and heating oil. The familiar plastics polyethylene, polypropylene, and polystyrene are also hydrocarbons. We can distinguish several types of hydrocarbons by differences in the bonding between carbon atoms. This leads to differences in geometries and in the hybridization of the carbon orbitals.

Alkanes

Alkanes, or **saturated hydrocarbons**, contain only single covalent bonds between carbon atoms. Each of the carbon atoms in an alkane has sp^3 hybrid orbitals and is bonded to four other atoms, each of which is either carbon or hydrogen. The Lewis structures and models of methane, ethane, and pentane are illustrated in Figure 20.2. Carbon chains are usually drawn as straight lines in Lewis structures, but one has to remember that Lewis structures are not intended to indicate the geometry of molecules. Notice that the carbon atoms in the structural models (the ball-and-stick and space-filling models) of the pentane molecule do not lie in a straight line. Because of the sp^3 hybridization, the bond angles in carbon chains are close to 109.5° , giving such chains in an alkane a zigzag shape.

The structures of alkanes and other organic molecules may also be represented in a less detailed manner by condensed structural formulas (or simply, *condensed formulas*). Instead of the usual format for chemical formulas in which each element symbol appears just once, a condensed formula is written to suggest the bonding in the molecule. These formulas have the appearance of a Lewis structure from which most or all of the bond symbols have been removed. Condensed structural formulas for ethane and pentane are shown at the bottom of Figure 20.2, and several additional examples are provided in the exercises at the end of this chapter.

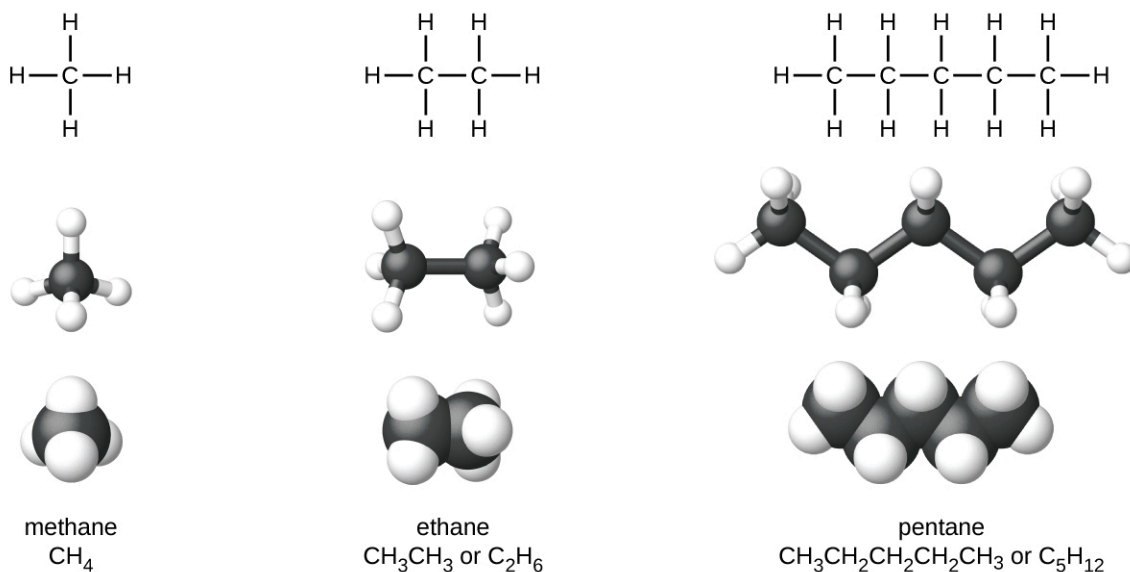


FIGURE 20.2 Pictured are the Lewis structures, ball-and-stick models, and space-filling models for molecules of methane, ethane, and pentane.

1 This is the Beilstein database, now available through the Reaxys site (www.elsevier.com/online-tools/reaxys).

2 Peplow, Mark. "Organic Synthesis: The Robo-Chemist," *Nature* 512 (2014): 20–2.

A common method used by organic chemists to simplify the drawings of larger molecules is to use a **skeletal structure** (also called a line-angle structure). In this type of structure, carbon atoms are not symbolized with a C, but represented by each end of a line or bend in a line. Hydrogen atoms are not drawn if they are attached to a carbon. Other atoms besides carbon and hydrogen are represented by their elemental symbols. [Figure 20.3](#) shows three different ways to draw the same structure.

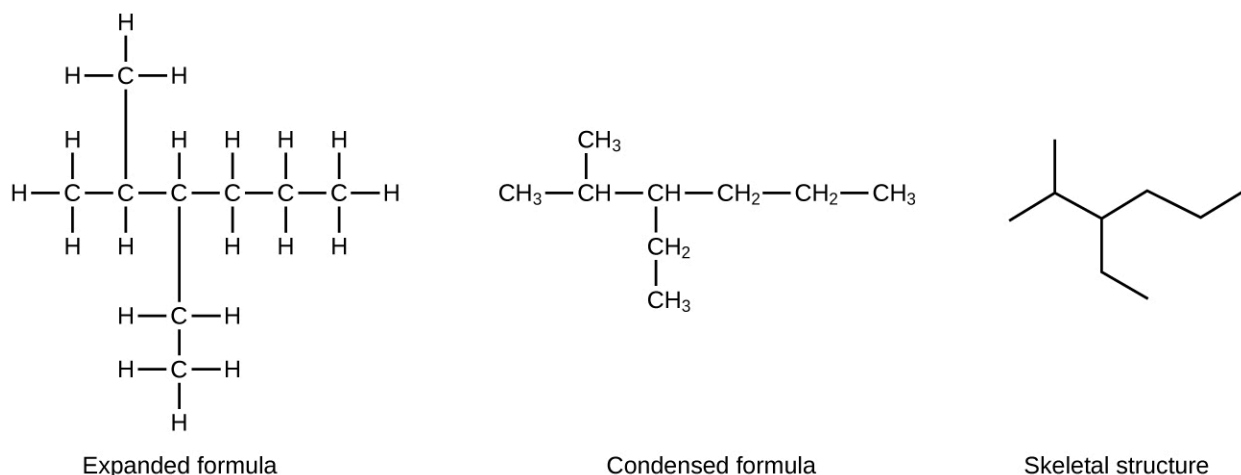


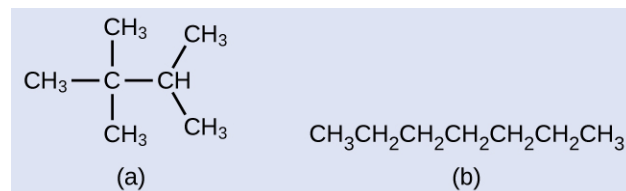
FIGURE 20.3 The same structure can be represented three different ways: an expanded formula, a condensed formula, and a skeletal structure.



EXAMPLE 20.1

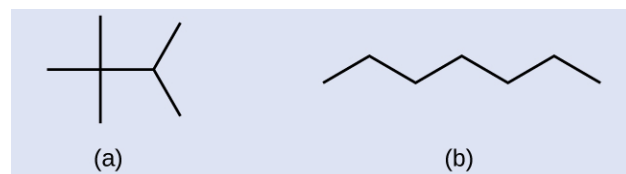
Drawing Skeletal Structures

Draw the skeletal structures for these two molecules:



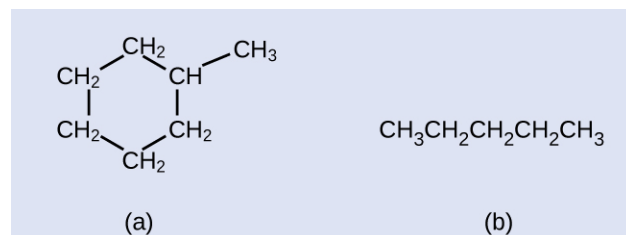
Solution

Each carbon atom is converted into the end of a line or the place where lines intersect. All hydrogen atoms attached to the carbon atoms are left out of the structure (although we still need to recognize they are there):

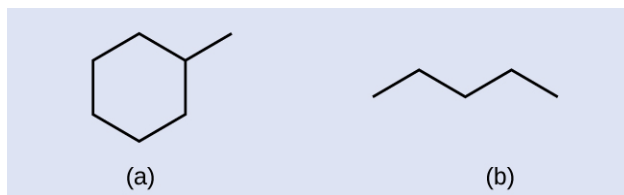


Check Your Learning

Draw the skeletal structures for these two molecules:



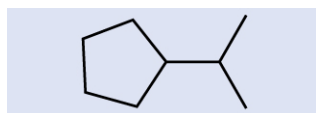
Answer:



EXAMPLE 20.2

Interpreting Skeletal Structures

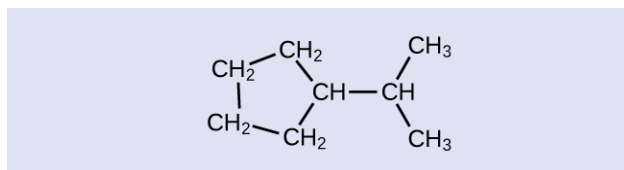
Identify the chemical formula of the molecule represented here:



Solution

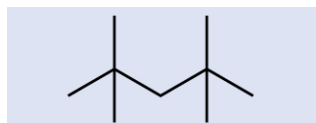
There are eight places where lines intersect or end, meaning that there are eight carbon atoms in the molecule. Since we know that carbon atoms tend to make four bonds, each carbon atom will have the number of hydrogen atoms that are required for four bonds. This compound contains 16 hydrogen atoms for a molecular formula of C_8H_{16} .

Location of the hydrogen atoms:



Check Your Learning

Identify the chemical formula of the molecule represented here:



Answer:

C_9H_{20}

All alkanes are composed of carbon and hydrogen atoms, and have similar bonds, structures, and formulas; noncyclic alkanes all have a formula of C_nH_{2n+2} . The number of carbon atoms present in an alkane has no limit. Greater numbers of atoms in the molecules will lead to stronger intermolecular attractions (dispersion forces) and correspondingly different physical properties of the molecules. Properties such as melting point and boiling point ([Table 20.1](#)) usually change smoothly and predictably as the number of carbon and hydrogen atoms in the molecules change.

Properties of Some Alkanes³

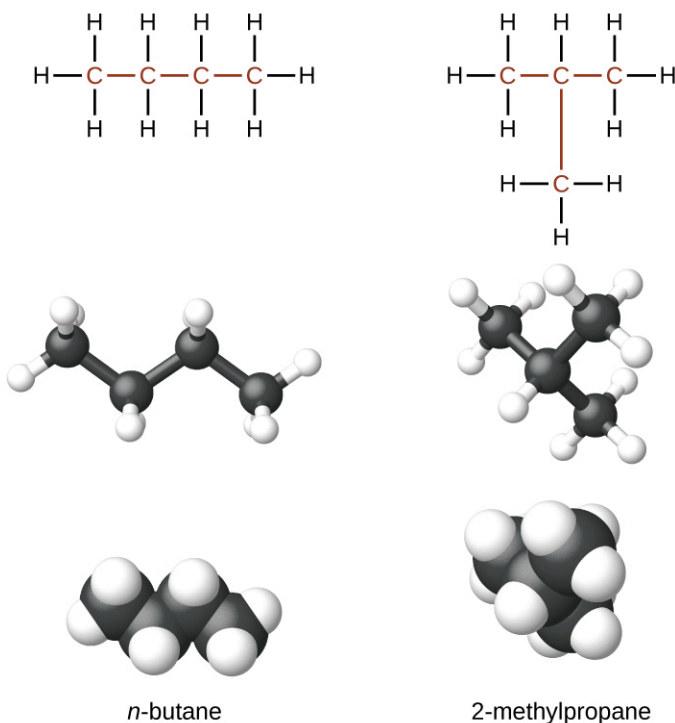
Alkane	Molecular Formula	Melting Point (°C)	Boiling Point (°C)	Phase at STP ⁴	Number of Structural Isomers
methane	CH ₄	−182.5	−161.5	gas	1
ethane	C ₂ H ₆	−183.3	−88.6	gas	1
propane	C ₃ H ₈	−187.7	−42.1	gas	1
butane	C ₄ H ₁₀	−138.3	−0.5	gas	2
pentane	C ₅ H ₁₂	−129.7	36.1	liquid	3
hexane	C ₆ H ₁₄	−95.3	68.7	liquid	5
heptane	C ₇ H ₁₆	−90.6	98.4	liquid	9
octane	C ₈ H ₁₈	−56.8	125.7	liquid	18
nonane	C ₉ H ₂₀	−53.6	150.8	liquid	35
decane	C ₁₀ H ₂₂	−29.7	174.0	liquid	75
tetradecane	C ₁₄ H ₃₀	5.9	253.5	solid	1858
octadecane	C ₁₈ H ₃₈	28.2	316.1	solid	60,523

TABLE 20.1

Hydrocarbons with the same formula, including alkanes, can have different structures. For example, two alkanes have the formula C₄H₁₀: They are called *n*-butane and 2-methylpropane (or isobutane), and have the following Lewis structures:

³ Physical properties for C₄H₁₀ and heavier molecules are those of the *normal isomer*, *n*-butane, *n*-pentane, etc.

⁴ STP indicates a temperature of 0 °C and a pressure of 1 atm.



The compounds *n*-butane and 2-methylpropane are structural isomers (the term constitutional isomers is also commonly used). Constitutional isomers have the same molecular formula but different spatial arrangements of the atoms in their molecules. The *n*-butane molecule contains an *unbranched chain*, meaning that no carbon atom is bonded to more than two other carbon atoms. We use the term *normal*, or the prefix *n*, to refer to a chain of carbon atoms without branching. The compound 2-methylpropane has a branched chain (the carbon atom in the center of the Lewis structure is bonded to three other carbon atoms)

Identifying isomers from Lewis structures is not as easy as it looks. Lewis structures that look different may actually represent the same isomers. For example, the three structures in [Figure 20.4](#) all represent the same molecule, *n*-butane, and hence are not different isomers. They are identical because each contains an unbranched chain of four carbon atoms.

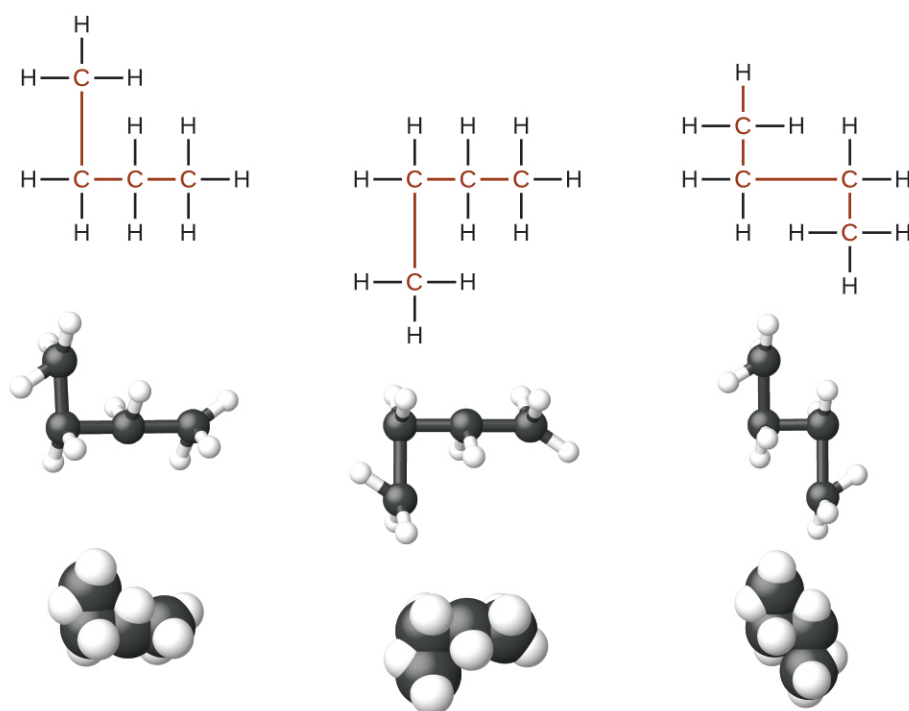
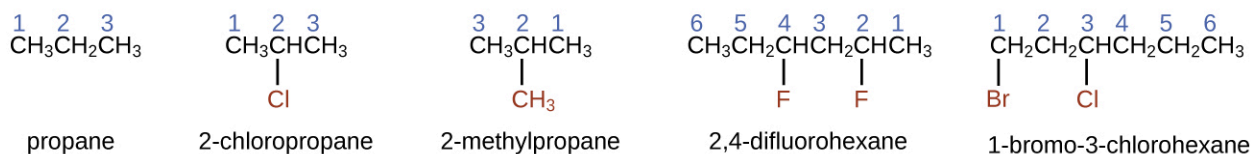


FIGURE 20.4 These three representations of the structure of n-butane are not isomers because they all contain the same arrangement of atoms and bonds.

The Basics of Organic Nomenclature: Naming Alkanes

The International Union of Pure and Applied Chemistry (IUPAC) has devised a system of nomenclature that begins with the names of the alkanes and can be adjusted from there to account for more complicated structures. The nomenclature for alkanes is based on two rules:

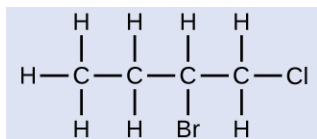
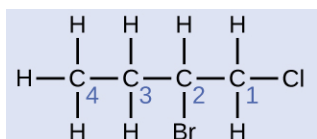
1. To name an alkane, first identify the longest chain of carbon atoms in its structure. A two-carbon chain is called ethane; a three-carbon chain, propane; and a four-carbon chain, butane. Longer chains are named as follows: pentane (five-carbon chain), hexane (6), heptane (7), octane (8), nonane (9), and decane (10). These prefixes can be seen in the names of the alkanes described in [Table 20.1](#).
2. Add prefixes to the name of the longest chain to indicate the positions and names of **substituents**. Substituents are branches or functional groups that replace hydrogen atoms on a chain. The position of a substituent or branch is identified by the number of the carbon atom it is bonded to in the chain. We number the carbon atoms in the chain by counting from the end of the chain nearest the substituents. Multiple substituents are named individually and placed in alphabetical order at the front of the name.



When more than one substituent is present, either on the same carbon atom or on different carbon atoms, the substituents are listed alphabetically. Because the carbon atom numbering begins at the end closest to a substituent, the longest chain of carbon atoms is numbered in such a way as to produce the lowest number for the substituents. The ending *-o* replaces *-ide* at the end of the name of an electronegative substituent (in ionic compounds, the negatively charged ion ends with *-ide* like chloride; in organic compounds, such atoms are treated as substituents and the *-o* ending is used). The number of substituents of the same type is indicated by the prefixes *di-* (two), *tri-* (three), *tetra-* (four), and so on (for example, *difluoro-* indicates two fluoride substituents).

**EXAMPLE 20.3****Naming Halogen-substituted Alkanes**

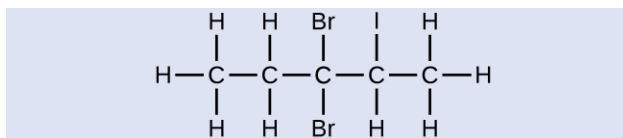
Name the molecule whose structure is shown here:

**Solution**

The four-carbon chain is numbered from the end with the chlorine atom. This puts the substituents on positions 1 and 2 (numbering from the other end would put the substituents on positions 3 and 4). Four carbon atoms means that the base name of this compound will be butane. The bromine at position 2 will be described by adding 2-bromo-; this will come at the beginning of the name, since bromo- comes before chloro- alphabetically. The chlorine at position 1 will be described by adding 1-chloro-, resulting in the name of the molecule being 2-bromo-1-chlorobutane.

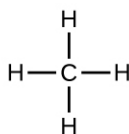
Check Your Learning

Name the following molecule:

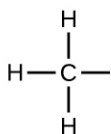
**Answer:**

3,3-dibromo-2-iodopentane

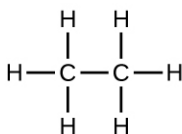
We call a substituent that contains one less hydrogen than the corresponding alkane an alkyl group. The name of an **alkyl group** is obtained by dropping the suffix *-ane* of the alkane name and adding *-yl*:



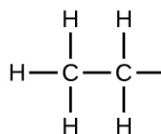
methane



methyl group



ethane

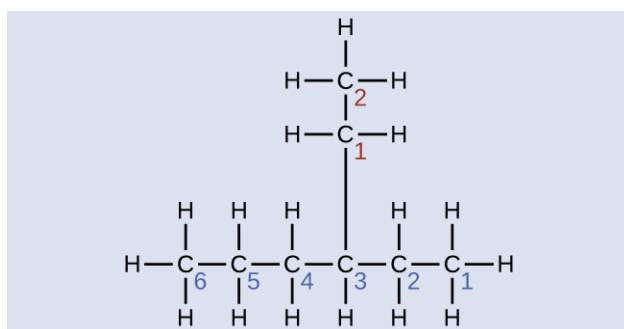


ethyl group

The open bonds in the methyl and ethyl groups indicate that these alkyl groups are bonded to another atom.

**EXAMPLE 20.4****Naming Substituted Alkanes**

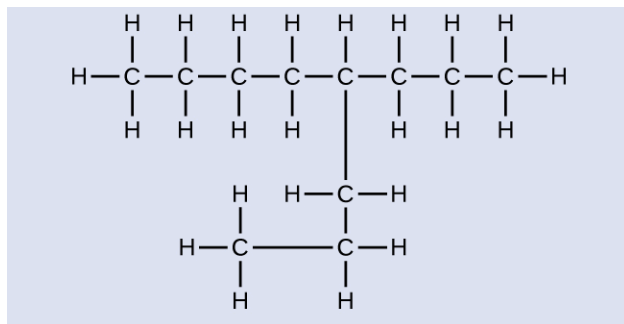
Name the molecule whose structure is shown here:

**Solution**

The longest carbon chain runs horizontally across the page and contains six carbon atoms (this makes the base of the name hexane, but we will also need to incorporate the name of the branch). In this case, we want to number from right to left (as shown by the blue numbers) so the branch is connected to carbon 3 (imagine the numbers from left to right—this would put the branch on carbon 4, violating our rules). The branch attached to position 3 of our chain contains two carbon atoms (numbered in red)—so we take our name for two carbons *eth*- and attach *-yl* at the end to signify we are describing a branch. Putting all the pieces together, this molecule is 3-ethylhexane.

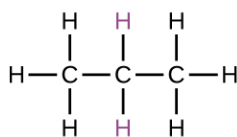
Check Your Learning

Name the following molecule:

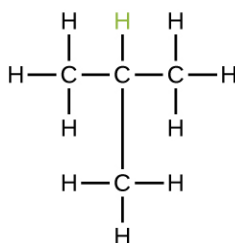
**Answer:**

4-propyloctane

Some hydrocarbons can form more than one type of alkyl group when the hydrogen atoms that would be removed have different “environments” in the molecule. This diversity of possible alkyl groups can be identified in the following way: The four hydrogen atoms in a methane molecule are equivalent; they all have the same environment. They are equivalent because each is bonded to a carbon atom (the same carbon atom) that is bonded to three hydrogen atoms. (It may be easier to see the equivalency in the ball and stick models in [Figure 20.2](#). Removal of any one of the four hydrogen atoms from methane forms a methyl group. Likewise, the six hydrogen atoms in ethane are equivalent ([Figure 20.2](#)) and removing any one of these hydrogen atoms produces an ethyl group. Each of the six hydrogen atoms is bonded to a carbon atom that is bonded to two other hydrogen atoms and a carbon atom. However, in both propane and 2-methylpropane, there are hydrogen atoms in two different environments, distinguished by the adjacent atoms or groups of atoms:



propane



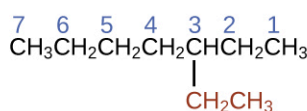
2-methylpropane

Each of the six equivalent hydrogen atoms of the first type in propane and each of the nine equivalent hydrogen atoms of that type in 2-methylpropane (all shown in black) are bonded to a carbon atom that is bonded to only one other carbon atom. The two purple hydrogen atoms in propane are of a second type. They differ from the six hydrogen atoms of the first type in that they are bonded to a carbon atom bonded to two other carbon atoms. The green hydrogen atom in 2-methylpropane differs from the other nine hydrogen atoms in that molecule and from the purple hydrogen atoms in propane. The green hydrogen atom in 2-methylpropane is bonded to a carbon atom bonded to three other carbon atoms. Two different alkyl groups can be formed from each of these molecules, depending on which hydrogen atom is removed. The names and structures of these and several other alkyl groups are listed in [Figure 20.5](#).

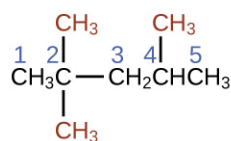
Alkyl Group	Structure
methyl	$\text{CH}_3\text{—}$
ethyl	$\text{CH}_3\text{CH}_2\text{—}$
<i>n</i> -propyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{—}$
isopropyl	$\begin{array}{c} \\ \text{CH}_3\text{CHCH}_3 \end{array}$
<i>n</i> -butyl	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{—}$
<i>sec</i> -butyl	$\begin{array}{c} \\ \text{CH}_3\text{CH}_2\text{CHCH}_3 \end{array}$
isobutyl	$\begin{array}{c} \text{CH}_3\text{CHCH}_2\text{—} \\ \\ \text{CH}_3 \end{array}$
<i>tert</i> -butyl	$\begin{array}{c} \\ \text{CH}_3\text{CCH}_3 \\ \\ \text{CH}_3 \end{array}$

FIGURE 20.5 This listing gives the names and formulas for various alkyl groups formed by the removal of hydrogen atoms from different locations.

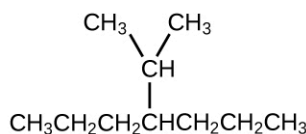
Note that alkyl groups do not exist as stable independent entities. They are always a part of some larger molecule. The location of an alkyl group on a hydrocarbon chain is indicated in the same way as any other substituent:



3-ethylheptane

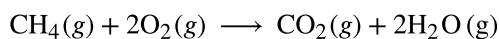


2,2,4-trimethylpentane



4-isopropylheptane

Alkanes are relatively stable molecules, but heat or light will activate reactions that involve the breaking of C–H or C–C single bonds. Combustion is one such reaction:



Alkanes burn in the presence of oxygen, a highly exothermic oxidation-reduction reaction that produces carbon dioxide and water. As a consequence, alkanes are excellent fuels. For example, methane, CH_4 , is the principal component of natural gas. Butane, C_4H_{10} , used in camping stoves and lighters is an alkane. Gasoline is a liquid mixture of continuous- and branched-chain alkanes, each containing from five to nine carbon atoms, plus various additives to improve its performance as a fuel. Kerosene, diesel oil, and fuel oil are primarily mixtures of alkanes with higher molecular masses. The main source of these liquid alkane fuels is crude oil, a complex mixture that is separated by fractional distillation. Fractional distillation takes advantage of differences in the boiling points of the components of the mixture (see [Figure 20.6](#)). You may recall that boiling point is a function of intermolecular interactions, which was discussed in the chapter on solutions and colloids.

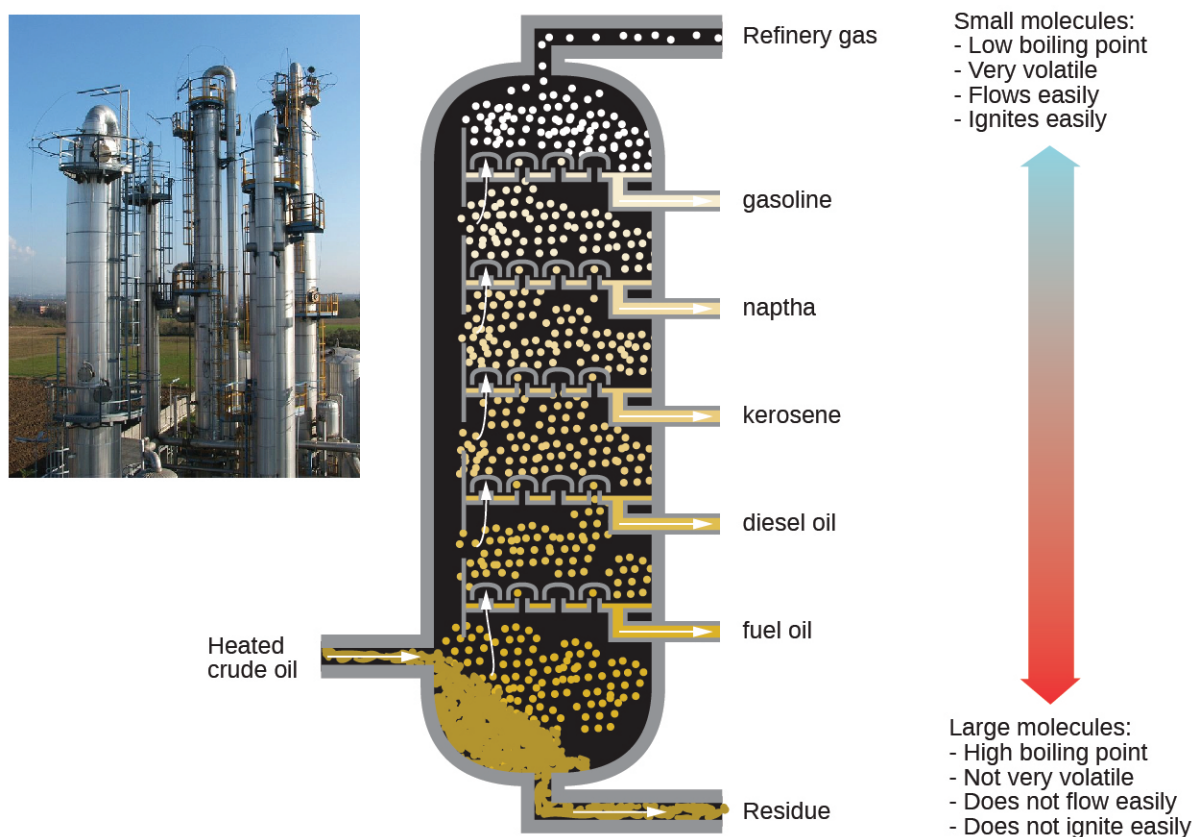
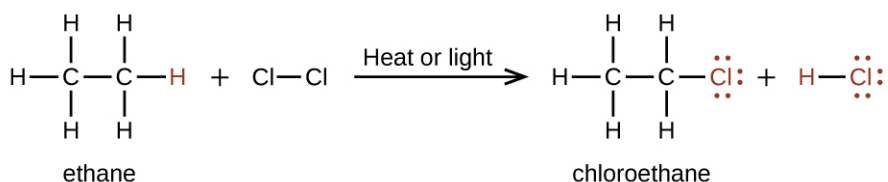


FIGURE 20.6 In a column for the fractional distillation of crude oil, oil heated to about 425 °C in the furnace vaporizes when it enters the base of the tower. The vapors rise through bubble caps in a series of trays in the tower. As the vapors gradually cool, fractions of higher, then of lower, boiling points condense to liquids and are drawn off. (credit left: modification of work by Luigi Chiesa)

In a **substitution reaction**, another typical reaction of alkanes, one or more of the alkane's hydrogen atoms is replaced with a different atom or group of atoms. No carbon-carbon bonds are broken in these reactions, and the hybridization of the carbon atoms does not change. For example, the reaction between ethane and molecular chlorine depicted here is a substitution reaction:



The C-Cl portion of the chloroethane molecule is an example of a **functional group**, the part or moiety of a molecule

that imparts a specific chemical reactivity. The types of functional groups present in an organic molecule are major determinants of its chemical properties and are used as a means of classifying organic compounds as detailed in the remaining sections of this chapter.

LINK TO LEARNING

Want more practice naming alkanes? Watch this brief [video tutorial \(http://openstax.org/l/16alkanes\)](http://openstax.org/l/16alkanes) to review the nomenclature process.

Alkenes

Organic compounds that contain one or more double or triple bonds between carbon atoms are described as unsaturated. You have likely heard of unsaturated fats. These are complex organic molecules with long chains of carbon atoms, which contain at least one double bond between carbon atoms. Unsaturated hydrocarbon molecules that contain one or more double bonds are called **alkenes**. Carbon atoms linked by a double bond are bound together by two bonds, one σ bond and one π bond. Double and triple bonds give rise to a different geometry around the carbon atom that participates in them, leading to important differences in molecular shape and properties. The differing geometries are responsible for the different properties of unsaturated versus saturated fats.

Ethene, C_2H_4 , is the simplest alkene. Each carbon atom in ethene, commonly called ethylene, has a trigonal planar structure. The second member of the series is propene (propylene) (Figure 20.7); the butene isomers follow in the series. Four carbon atoms in the chain of butene allows for the formation of isomers based on the position of the double bond, as well as a new form of isomerism.

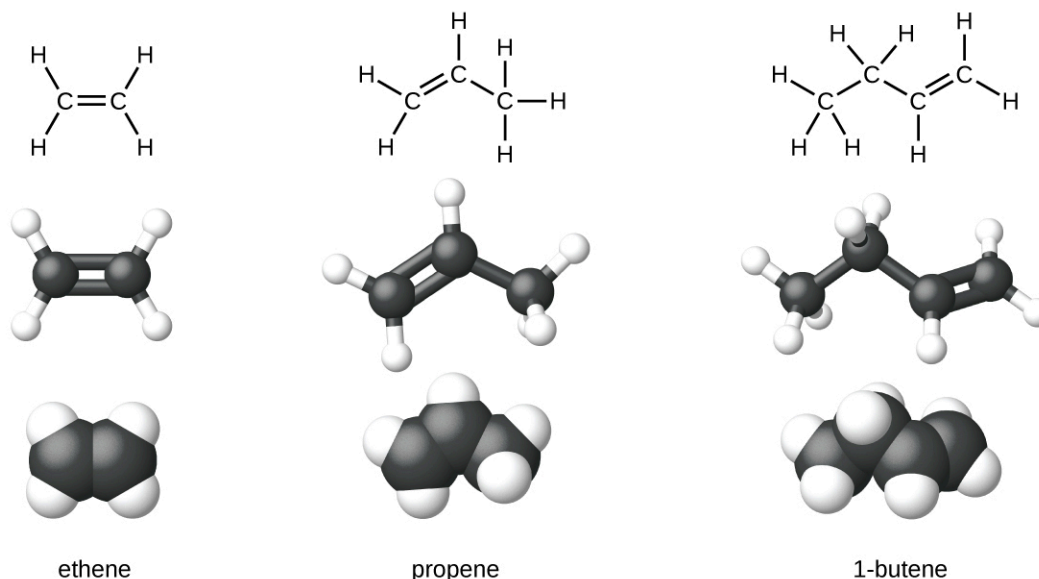


FIGURE 20.7 Expanded structures, ball-and-stick structures, and space-filling models for the alkenes ethene, propene, and 1-butene are shown.

Ethylene (the common industrial name for ethene) is a basic raw material in the production of polyethylene and other important compounds. Over 135 million tons of ethylene were produced worldwide in 2010 for use in the polymer, petrochemical, and plastic industries. Ethylene is produced industrially in a process called cracking, in which the long hydrocarbon chains in a petroleum mixture are broken into smaller molecules.

CHEMISTRY IN EVERYDAY LIFE

Recycling Plastics

Polymers (from Greek words *poly* meaning “many” and *mer* meaning “parts”) are large molecules made up of repeating units, referred to as monomers. Polymers can be natural (starch is a polymer of sugar residues and

proteins are polymers of amino acids) or synthetic [like polyethylene, polyvinyl chloride (PVC), and polystyrene]. The variety of structures of polymers translates into a broad range of properties and uses that make them integral parts of our everyday lives. Adding functional groups to the structure of a polymer can result in significantly different properties (see the discussion about Kevlar later in this chapter).

An example of a polymerization reaction is shown in [Figure 20.8](#). The monomer ethylene (C_2H_4) is a gas at room temperature, but when polymerized, using a transition metal catalyst, it is transformed into a solid material made up of long chains of $-\text{CH}_2-$ units called polyethylene. Polyethylene is a commodity plastic used primarily for packaging (bags and films).

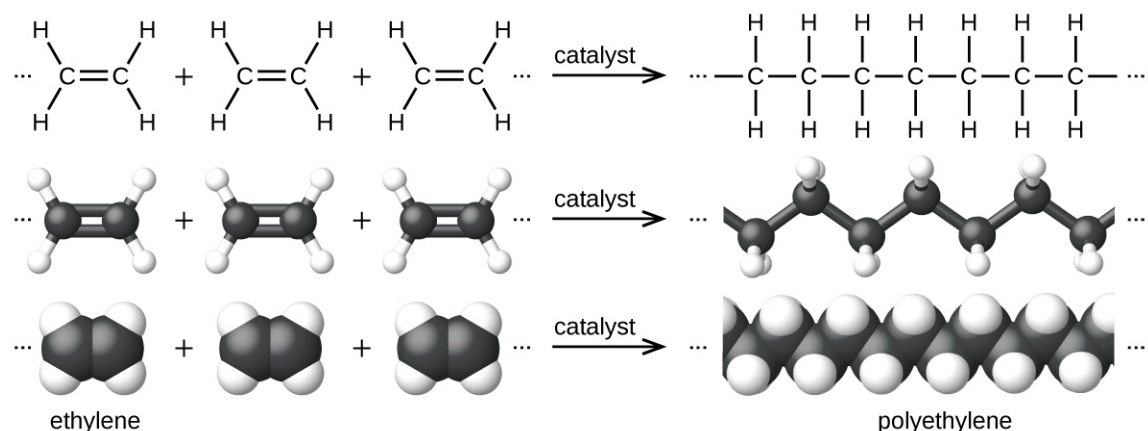


FIGURE 20.8 The reaction for the polymerization of ethylene to polyethylene is shown.

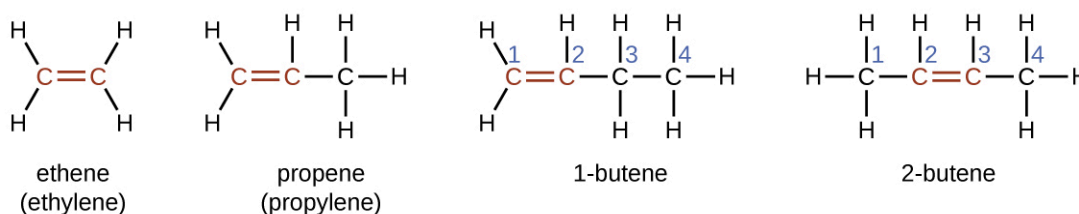
Polyethylene is a member of one subset of synthetic polymers classified as plastics. Plastics are synthetic organic solids that can be molded; they are typically organic polymers with high molecular masses. Most of the monomers that go into common plastics (ethylene, propylene, vinyl chloride, styrene, and ethylene terephthalate) are derived from petrochemicals and are not very biodegradable, making them candidate materials for recycling. Recycling plastics helps minimize the need for using more of the petrochemical supplies and also minimizes the environmental damage caused by throwing away these nonbiodegradable materials.

Plastic recycling is the process of recovering waste, scrap, or used plastics, and reprocessing the material into useful products. For example, polyethylene terephthalate (soft drink bottles) can be melted down and used for plastic furniture, in carpets, or for other applications. Other plastics, like polyethylene (bags) and polypropylene (cups, plastic food containers), can be recycled or reprocessed to be used again. Many areas of the country have recycling programs that focus on one or more of the commodity plastics that have been assigned a recycling code (see [Figure 20.9](#)). These operations have been in effect since the 1970s and have made the production of some plastics among the most efficient industrial operations today.

 polyethylene terephthalate (PETE)	Soda bottles and oven-ready food trays
 high-density polyethylene (HDPE)	Bottles for milk and dishwashing liquids
 polyvinyl chloride (PVC)	Food trays, plastic wrap, bottles for mineral water and shampoo
 low density polyethylene (LDPE)	Shopping bags and garbage bags
 polypropylene (PP)	Margarine tubs, microwaveable food trays
 polystyrene (PS)	Yogurt tubs, foam meat trays, egg cartons, vending cups, plastic cutlery, packaging for electronics and toys
 any other plastics (OTHER)	Plastics that do not fall into any of the above categories One example is melamine resin (plastic plates, plastic cups)

FIGURE 20.9 Each type of recyclable plastic is imprinted with a code for easy identification.

The name of an alkene is derived from the name of the alkane with the same number of carbon atoms. The presence of the double bond is signified by replacing the suffix *-ane* with the suffix *-ene*. The location of the double bond is identified by naming the smaller of the numbers of the carbon atoms participating in the double bond:



Isomers of Alkenes

Molecules of 1-butene and 2-butene are structural isomers; the arrangement of the atoms in these two molecules differs. As an example of arrangement differences, the first carbon atom in 1-butene is bonded to two hydrogen atoms; the first carbon atom in 2-butene is bonded to three hydrogen atoms.

The compound 2-butene and some other alkenes also form a second type of isomer called a geometric isomer. In a set of geometric isomers, the same types of atoms are attached to each other in the same order, but the geometries of the two molecules differ. Geometric isomers of alkenes differ in the orientation of the groups on either side of a $C = C$ bond.

Carbon atoms are free to rotate around a single bond but not around a double bond; a double bond is rigid. This makes it possible to have two isomers of 2-butene, one with both methyl groups on the same side of the double bond and one with the methyl groups on opposite sides. When structures of butene are drawn with 120° bond

angles around the sp^2 -hybridized carbon atoms participating in the double bond, the isomers are apparent. The 2-butene isomer in which the two methyl groups are on the same side is called a *cis*-isomer; the one in which the two methyl groups are on opposite sides is called a *trans*-isomer (Figure 20.10). The different geometries produce different physical properties, such as boiling point, that may make separation of the isomers possible:

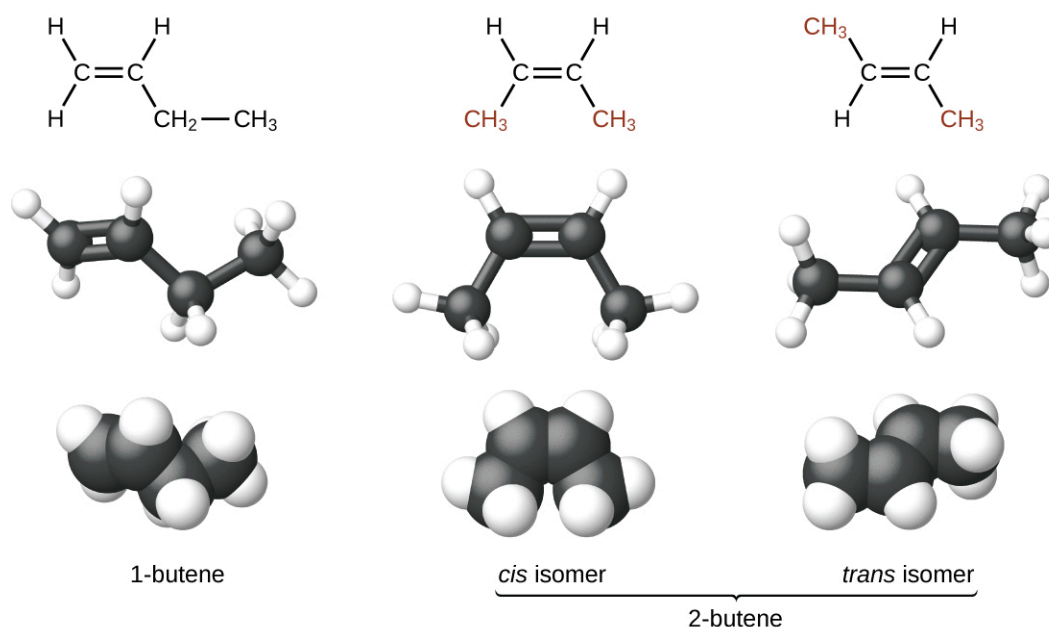
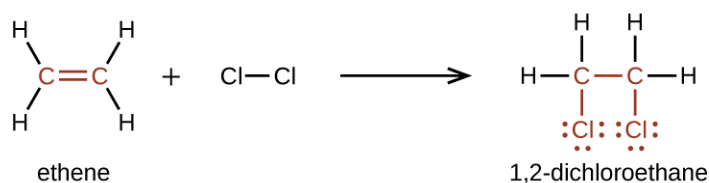


FIGURE 20.10 These molecular models show the structural and geometric isomers of butene.

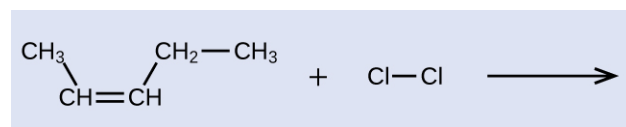
Alkenes are much more reactive than alkanes because the $C = C$ moiety is a reactive functional group. A π bond, being a weaker bond, is disrupted much more easily than a σ bond. Thus, alkenes undergo a characteristic reaction in which the π bond is broken and replaced by two σ bonds. This reaction is called an **addition reaction**. The hybridization of the carbon atoms in the double bond in an alkene changes from sp^2 to sp^3 during an addition reaction. For example, halogens add to the double bond in an alkene instead of replacing hydrogen, as occurs in an alkane:



EXAMPLE 20.5

Alkene Reactivity and Naming

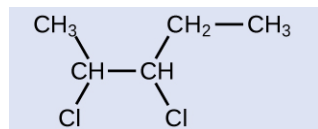
Provide the IUPAC names for the reactant and product of the halogenation reaction shown here:



Solution

The reactant is a five-carbon chain that contains a carbon-carbon double bond, so the base name will be pentene. We begin counting at the end of the chain closest to the double bond—in this case, from the left—the double bond spans carbons 2 and 3, so the name becomes 2-pentene. Since there are two carbon-containing groups attached to the two carbon atoms in the double bond—and they are on the same side of the double bond—this molecule is the *cis*-isomer, making the name of the starting alkene *cis*-2-pentene. The product of the halogenation reaction will

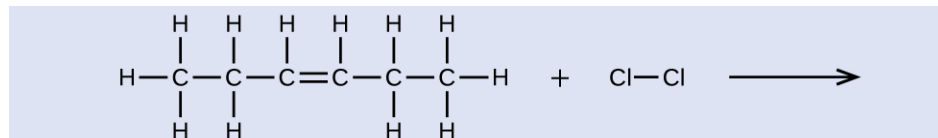
have two chlorine atoms attached to the carbon atoms that were a part of the carbon-carbon double bond:



This molecule is now a substituted alkane and will be named as such. The base of the name will be pentane. We will count from the end that numbers the carbon atoms where the chlorine atoms are attached as 2 and 3, making the name of the product 2,3-dichloropentane.

Check Your Learning

Provide names for the reactant and product of the reaction shown:



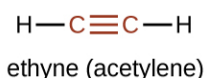
Answer:

reactant: cis-3-hexene product: 3,4-dichlorohexane

Alkynes

Hydrocarbon molecules with one or more triple bonds are called **alkynes**; they make up another series of unsaturated hydrocarbons. Two carbon atoms joined by a triple bond are bound together by one σ bond and two π bonds. The sp -hybridized carbons involved in the triple bond have bond angles of 180° , giving these types of bonds a linear, rod-like shape.

The simplest member of the alkyne series is ethyne, C_2H_2 , commonly called acetylene. The Lewis structure for ethyne, a linear molecule, is:



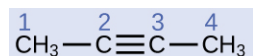
The IUPAC nomenclature for alkynes is similar to that for alkenes except that the suffix *-yne* is used to indicate a triple bond in the chain. For example, $CH_3CH_2C \equiv CH$ is called 1-butyne.



EXAMPLE 20.6

Structure of Alkynes

Describe the geometry and hybridization of the carbon atoms in the following molecule:

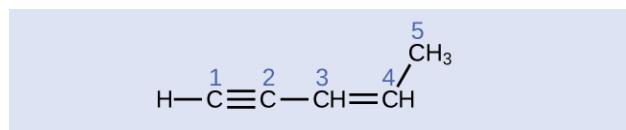


Solution

Carbon atoms 1 and 4 have four single bonds and are thus tetrahedral with sp^3 hybridization. Carbon atoms 2 and 3 are involved in the triple bond, so they have linear geometries and would be classified as sp hybrids.

Check Your Learning

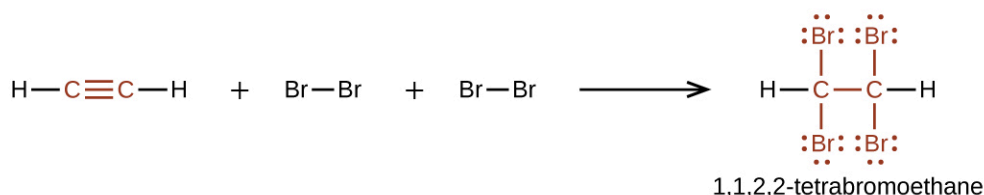
Identify the hybridization and bond angles at the carbon atoms in the molecule shown:



Answer:

carbon 1: sp , 180° ; carbon 2: sp , 180° ; carbon 3: sp^2 , 120° ; carbon 4: sp^2 , 120° ; carbon 5: sp^3 , 109.5°

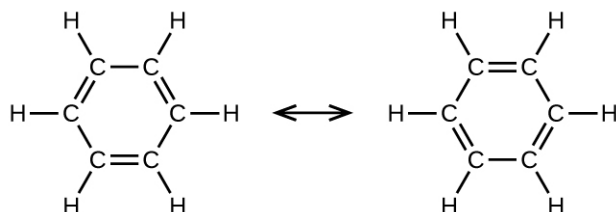
Chemically, the alkynes are similar to the alkenes. Since the $C \equiv C$ functional group has two π bonds, alkynes typically react even more readily, and react with twice as much reagent in addition reactions. The reaction of acetylene with bromine is a typical example:



Acetylene and the other alkynes also burn readily. An acetylene torch takes advantage of the high heat of combustion for acetylene.

Aromatic Hydrocarbons

Benzene, C_6H_6 , is the simplest member of a large family of hydrocarbons, called **aromatic hydrocarbons**. These compounds contain ring structures and exhibit bonding that must be described using the resonance hybrid concept of valence bond theory or the delocalization concept of molecular orbital theory. (To review these concepts, refer to the earlier chapters on chemical bonding). The resonance structures for benzene, C_6H_6 , are:



Valence bond theory describes the benzene molecule and other planar aromatic hydrocarbon molecules as hexagonal rings of sp^2 -hybridized carbon atoms with the unhybridized p orbital of each carbon atom perpendicular to the plane of the ring. Three valence electrons in the sp^2 hybrid orbitals of each carbon atom and the valence electron of each hydrogen atom form the framework of σ bonds in the benzene molecule. The fourth valence electron of each carbon atom is shared with an adjacent carbon atom in their unhybridized p orbitals to yield the π bonds. Benzene does not, however, exhibit the characteristics typical of an alkene. Each of the six bonds between its carbon atoms is equivalent and exhibits properties that are intermediate between those of a C–C single bond and a C = C double bond. To represent this unique bonding, structural formulas for benzene and its derivatives are typically drawn with single bonds between the carbon atoms and a circle within the ring as shown in [Figure 20.11](#).

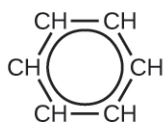
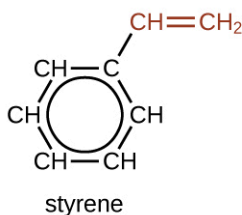
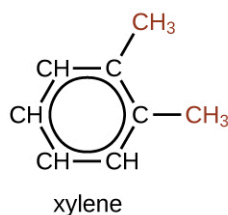
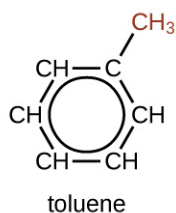


FIGURE 20.11 This condensed formula shows the unique bonding structure of benzene.

There are many derivatives of benzene. The hydrogen atoms can be replaced by many different substituents. Aromatic compounds more readily undergo substitution reactions than addition reactions; replacement of one of the hydrogen atoms with another substituent will leave the delocalized double bonds intact. The following are typical examples of substituted benzene derivatives:



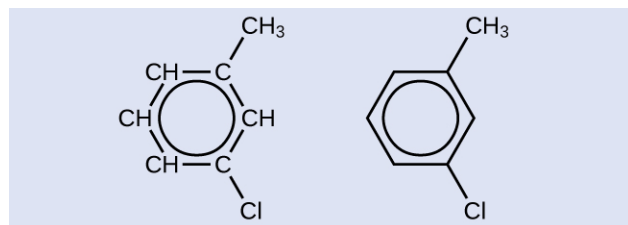
Toluene and xylene are important solvents and raw materials in the chemical industry. Styrene is used to produce the polymer polystyrene.



EXAMPLE 20.7

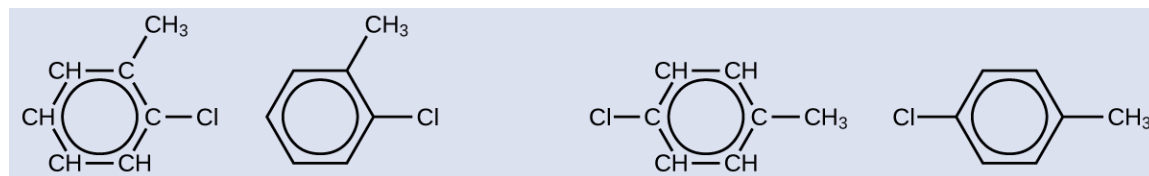
Structure of Aromatic Hydrocarbons

One possible isomer created by a substitution reaction that replaces a hydrogen atom attached to the aromatic ring of toluene with a chlorine atom is shown here. Draw two other possible isomers in which the chlorine atom replaces a different hydrogen atom attached to the aromatic ring:



Solution

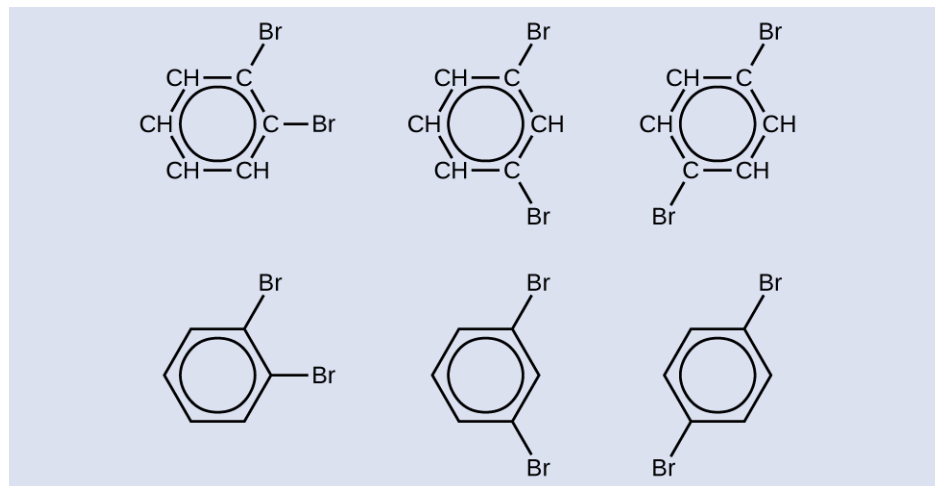
Since the six-carbon ring with alternating double bonds is necessary for the molecule to be classified as aromatic, appropriate isomers can be produced only by changing the positions of the chloro-substituent relative to the methyl-substituent:



Check Your Learning

Draw three isomers of a six-membered aromatic ring compound substituted with two bromines.

Answer:



20.2 Alcohols and Ethers

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the structure and properties of alcohols
- Describe the structure and properties of ethers
- Name and draw structures for alcohols and ethers

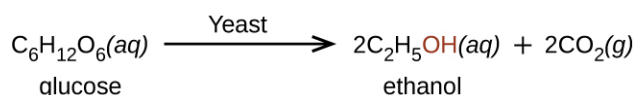
In this section, we will learn about alcohols and ethers.

Alcohols

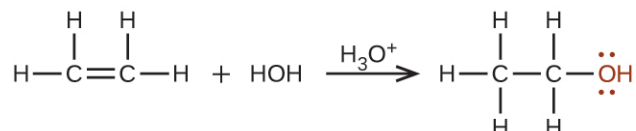
Incorporation of an oxygen atom into carbon- and hydrogen-containing molecules leads to new functional groups and new families of compounds. When the oxygen atom is attached by single bonds, the molecule is either an alcohol or ether.

Alcohols are derivatives of hydrocarbons in which an -OH group has replaced a hydrogen atom. Although all alcohols have one or more hydroxyl (-OH) functional groups, they do not behave like bases such as NaOH and KOH . NaOH and KOH are ionic compounds that contain OH^- ions. Alcohols are covalent molecules; the -OH group in an alcohol molecule is attached to a carbon atom by a covalent bond.

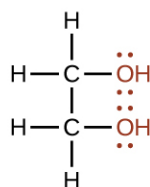
Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$, also called ethyl alcohol, is a particularly important alcohol for human use. Ethanol is the alcohol produced by some species of yeast that is found in wine, beer, and distilled drinks. It has long been prepared by humans harnessing the metabolic efforts of yeasts in fermenting various sugars:



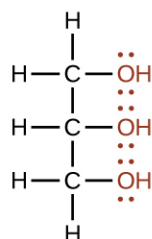
Large quantities of ethanol are synthesized from the addition reaction of water with ethylene using an acid as a catalyst:



Alcohols containing two or more hydroxyl groups can be made. Examples include 1,2-ethanediol (ethylene glycol, used in antifreeze) and 1,2,3-propanetriol (glycerine, used as a solvent for cosmetics and medicines):



1,2-ethanediol



1,2,3-propanetriol

Naming Alcohols

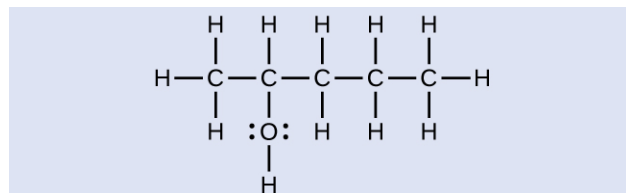
The name of an alcohol comes from the hydrocarbon from which it was derived. The final -e in the name of the hydrocarbon is replaced by -ol , and the carbon atom to which the -OH group is bonded is indicated by a number placed before the name.⁵

⁵ The IUPAC adopted new nomenclature guidelines in 2013 that require this number to be placed as an “infix” rather than a prefix. For example, the new name for 2-propanol would be propan-2-ol. Widespread adoption of this new nomenclature will take some time, and students are encouraged to be familiar with both the old and new naming protocols.

EXAMPLE 20.8

Naming Alcohols

Consider the following example. How should it be named?

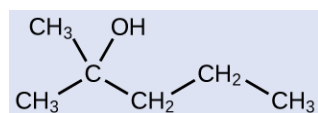


Solution

The carbon chain contains five carbon atoms. If the hydroxyl group was not present, we would have named this molecule pentane. To address the fact that the hydroxyl group is present, we change the ending of the name to *-ol*. In this case, since the -OH is attached to carbon 2 in the chain, we would name this molecule 2-pentanol.

Check Your Learning

Name the following molecule:

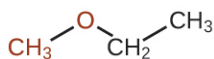


Answer:

2-methyl-2-pentanol

Ethers

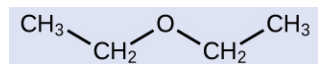
Ethers are compounds that contain the functional group -O- . Ethers do not have a designated suffix like the other types of molecules we have named so far. In the IUPAC system, the oxygen atom and the smaller carbon branch are named as an alkoxy substituent and the remainder of the molecule as the base chain, as in alkanes. As shown in the following compound, the red symbols represent the smaller alkyl group and the oxygen atom, which would be named “methoxy.” The larger carbon branch would be ethane, making the molecule methoxyethane. Many ethers are referred to with common names instead of the IUPAC system names. For common names, the two branches connected to the oxygen atom are named separately and followed by “ether.” The common name for the compound shown in [Example 20.9](#) is ethylmethyl ether:



EXAMPLE 20.9

Naming Ethers

Provide the IUPAC and common name for the ether shown here:



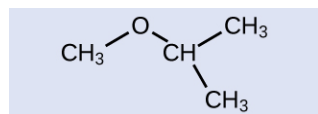
Solution

IUPAC: The molecule is made up of an ethoxy group attached to an ethane chain, so the IUPAC name would be ethoxyethane.

Common: The groups attached to the oxygen atom are both ethyl groups, so the common name would be diethyl ether.

Check Your Learning

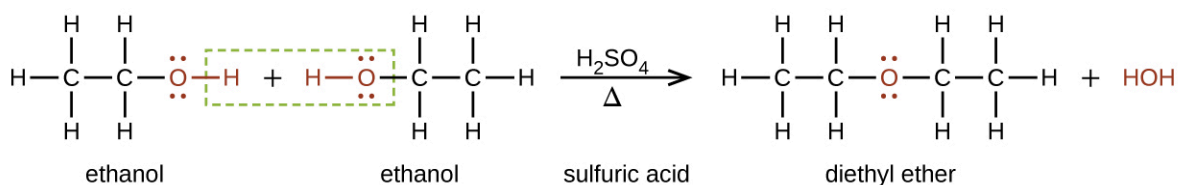
Provide the IUPAC and common name for the ether shown:



Answer:

IUPAC: 2-methoxypropane; common: isopropylmethyl ether

Ethers can be obtained from alcohols by the elimination of a molecule of water from two molecules of the alcohol. For example, when ethanol is treated with a limited amount of sulfuric acid and heated to 140 °C, diethyl ether and water are formed:



In the general formula for ethers, $R-O-R$, the hydrocarbon groups (R) may be the same or different. Diethyl ether, the most widely used compound of this class, is a colorless, volatile liquid that is highly flammable. It was first used in 1846 as an anesthetic, but better anesthetics have now largely taken its place. Diethyl ether and other ethers are presently used primarily as solvents for gums, fats, waxes, and resins. *Tertiary*-butyl methyl ether, $C_4H_9OCH_3$ (abbreviated MTBE—italicized portions of names are not counted when ranking the groups alphabetically—so butyl comes before methyl in the common name), is used as an additive for gasoline. MTBE belongs to a group of chemicals known as oxygenates due to their capacity to increase the oxygen content of gasoline.

LINK TO LEARNING

Want more practice naming ethers? This brief [video review \(http://openstax.org/l/16ethers\)](http://openstax.org/l/16ethers) summarizes the nomenclature for ethers.

CHEMISTRY IN EVERYDAY LIFE

Carbohydrates and Diabetes

Carbohydrates are large biomolecules made up of carbon, hydrogen, and oxygen. The dietary forms of carbohydrates are foods rich in these types of molecules, like pastas, bread, and candy. The name “carbohydrate” comes from the formula of the molecules, which can be described by the general formula $C_m(H_2O)_n$, which shows that they are in a sense “carbon and water” or “hydrates of carbon.” In many cases, m and n have the same value, but they can be different. The smaller carbohydrates are generally referred to as “sugars,” the biochemical term for this group of molecules is “saccharide” from the Greek word for sugar ([Figure 20.12](#)). Depending on the number of sugar units joined together, they may be classified as monosaccharides (one sugar unit), disaccharides (two sugar units), oligosaccharides (a few sugars), or polysaccharides (the polymeric version of sugars—polymers were described in the feature box earlier in this chapter on recycling plastics). The scientific names of sugars can be recognized by the suffix *-ose* at the end of the name (for instance, fruit sugar is a monosaccharide called “fructose” and milk sugar is a disaccharide called lactose composed of two monosaccharides, glucose and galactose, connected together). Sugars contain some of the functional groups we have discussed: Note the alcohol groups present in the structures and how monosaccharide units are linked to form a disaccharide by formation of an ether.

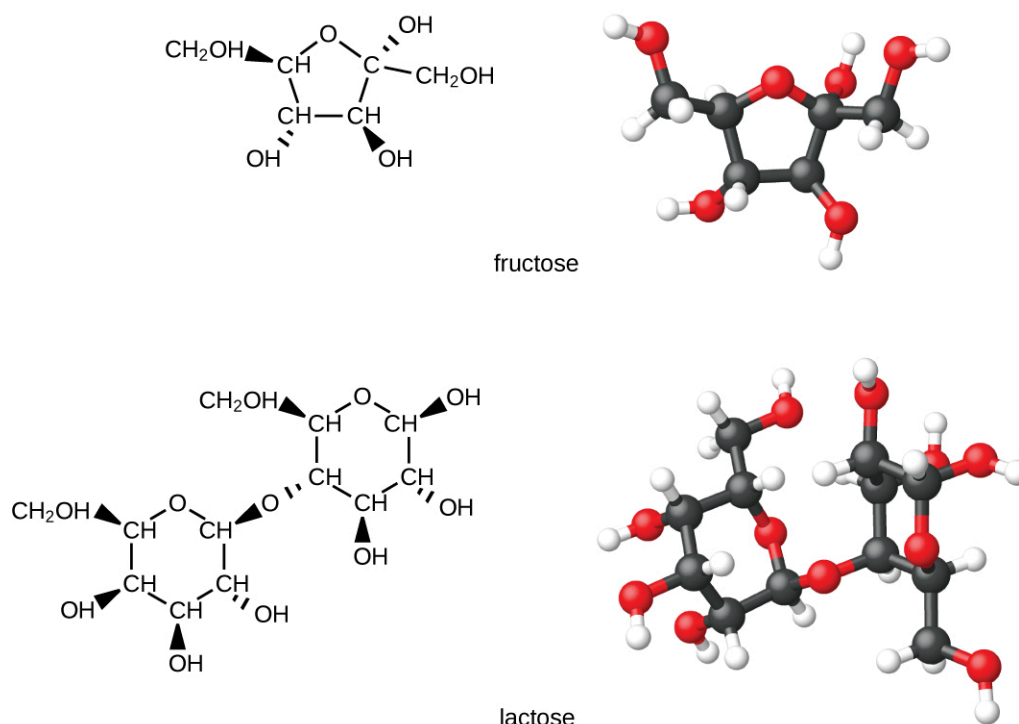


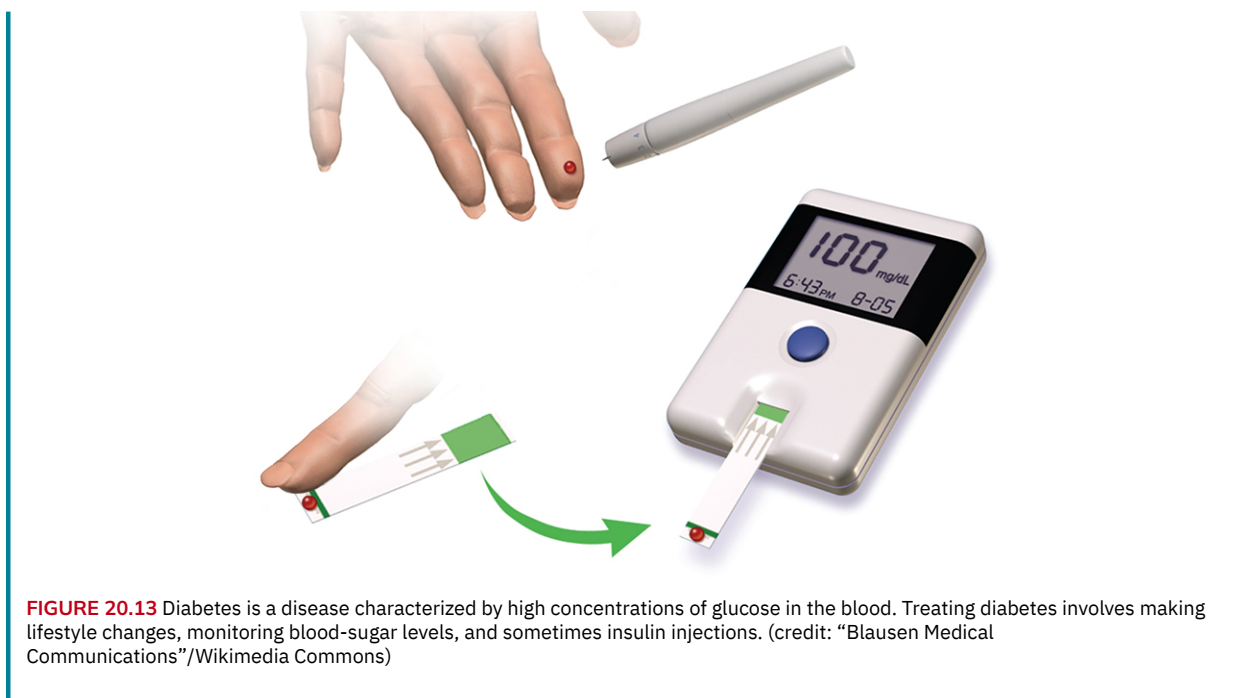
FIGURE 20.12 The illustrations show the molecular structures of fructose, a five-carbon monosaccharide, and of lactose, a disaccharide composed of two isomeric, six-carbon sugars.

Organisms use carbohydrates for a variety of functions. Carbohydrates can store energy, such as the polysaccharides glycogen in animals or starch in plants. They also provide structural support, such as the polysaccharide cellulose in plants and the modified polysaccharide chitin in fungi and animals. The sugars ribose and deoxyribose are components of the backbones of RNA and DNA, respectively. Other sugars play key roles in the function of the immune system, in cell-cell recognition, and in many other biological roles.

Diabetes is a group of metabolic diseases in which a person has a high sugar concentration in their blood ([Figure 20.13](#)). Diabetes may be caused by insufficient insulin production by the pancreas or by the body's cells not responding properly to the insulin that is produced. In a healthy person, insulin is produced when it is needed and functions to transport glucose from the blood into the cells where it can be used for energy. The long-term complications of diabetes can include loss of eyesight, heart disease, and kidney failure.

In 2013, it was estimated that approximately 3.3% of the world's population (~380 million people) suffered from diabetes, resulting in over a million deaths annually. Prevention involves eating a healthy diet, getting plenty of exercise, and maintaining a normal body weight. Treatment involves all of these lifestyle practices and may require injections of insulin.

Even after treatment protocols were introduced, the need to continually monitor their glucose levels posed a challenge for people with diabetes. The first tests required a doctor or lab, and therefore limited access and frequency. Eventually, researchers developed small tablets that would react to the presence of glucose in urine, but these still required a relatively complex process. Chemist Helen Free, who was working on improvements to the tablets, conceived a simpler device: a small test strip. With her husband and research partner, Alfred Free, she produced the first such product for measuring glucose; soon after, she expanded the technology to provide test strips for other compounds and conditions. While very recent advances (such as breath tests, discussed earlier in the text) have shown promise in replacing test strips, they have been widely used for decades and remain a primary method today.



20.3 Aldehydes, Ketones, Carboxylic Acids, and Esters

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the structure and properties of aldehydes, ketones, carboxylic acids and esters

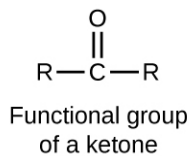
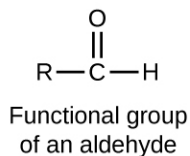
Another class of organic molecules contains a carbon atom connected to an oxygen atom by a double bond, commonly called a carbonyl group. The trigonal planar carbon in the carbonyl group can attach to two other substituents leading to several subfamilies (aldehydes, ketones, carboxylic acids and esters) described in this section.

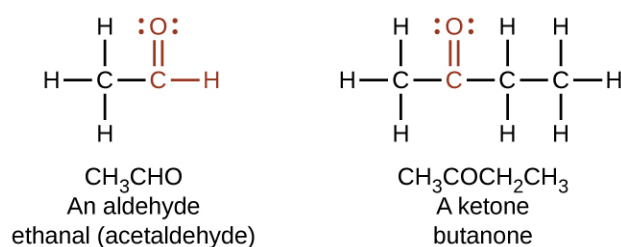
Aldehydes and Ketones

Both **aldehydes** and **ketones** contain a **carbonyl group**, a functional group with a carbon-oxygen double bond. The names for aldehyde and ketone compounds are derived using similar nomenclature rules as for alkanes and alcohols, and include the class-identifying suffixes *-al* and *-one*, respectively:



In an aldehyde, the carbonyl group is bonded to at least one hydrogen atom. In a ketone, the carbonyl group is bonded to two carbon atoms:





As text, an aldehyde group is represented as $-\text{CHO}$; a ketone is represented as $-\text{C}(\text{O})-$ or $-\text{CO}-$.

In both aldehydes and ketones, the geometry around the carbon atom in the carbonyl group is trigonal planar; the carbon atom exhibits sp^2 hybridization. Two of the sp^2 orbitals on the carbon atom in the carbonyl group are used to form σ bonds to the other carbon or hydrogen atoms in a molecule. The remaining sp^2 hybrid orbital forms a σ bond to the oxygen atom. The unhybridized p orbital on the carbon atom in the carbonyl group overlaps a p orbital on the oxygen atom to form the π bond in the double bond.

Like the $\text{C}=\text{O}$ bond in carbon dioxide, the $\text{C}=\text{O}$ bond of a carbonyl group is polar (recall that oxygen is significantly more electronegative than carbon, and the shared electrons are pulled toward the oxygen atom and away from the carbon atom). Many of the reactions of aldehydes and ketones start with the reaction between a Lewis base and the carbon atom at the positive end of the polar $\text{C}=\text{O}$ bond to yield an unstable intermediate that subsequently undergoes one or more structural rearrangements to form the final product (Figure 20.14).

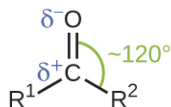
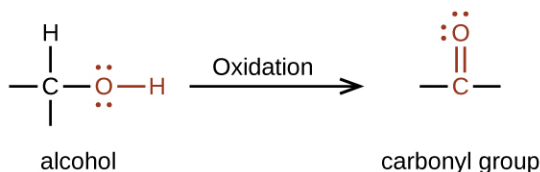


FIGURE 20.14 The carbonyl group is polar, and the geometry of the bonds around the central carbon is trigonal planar.

The importance of molecular structure in the reactivity of organic compounds is illustrated by the reactions that produce aldehydes and ketones. We can prepare a carbonyl group by oxidation of an alcohol—for organic molecules, oxidation of a carbon atom is said to occur when a carbon-hydrogen bond is replaced by a carbon-oxygen bond. The reverse reaction—replacing a carbon-oxygen bond by a carbon-hydrogen bond—is a reduction of that carbon atom. Recall that oxygen is generally assigned a -2 oxidation number unless it is elemental or attached to a fluorine. Hydrogen is generally assigned an oxidation number of $+1$ unless it is attached to a metal. Since carbon does not have a specific rule, its oxidation number is determined algebraically by factoring the atoms it is attached to and the overall charge of the molecule or ion. In general, a carbon atom attached to an oxygen atom will have a more positive oxidation number and a carbon atom attached to a hydrogen atom will have a more negative oxidation number. This should fit nicely with your understanding of the polarity of $\text{C}-\text{O}$ and $\text{C}-\text{H}$ bonds. The other reagents and possible products of these reactions are beyond the scope of this chapter, so we will focus only on the changes to the carbon atoms:



EXAMPLE 20.10

Oxidation and Reduction in Organic Chemistry

Methane represents the completely reduced form of an organic molecule that contains one carbon atom. Sequentially replacing each of the carbon-hydrogen bonds with a carbon-oxygen bond would lead to an alcohol, then an aldehyde, then a carboxylic acid (discussed later), and, finally, carbon dioxide:



What are the oxidation numbers for the carbon atoms in the molecules shown here?

bodies. It is also used to sterilize soil or other materials. Formaldehyde is used in the manufacture of Bakelite, a hard plastic having high chemical and electrical resistance.

Dimethyl ketone, CH_3COCH_3 , commonly called acetone, is the simplest ketone. It is made commercially by fermenting corn or molasses, or by oxidation of 2-propanol. Acetone is a colorless liquid. Among its many uses are as a solvent for lacquer (including fingernail polish), cellulose acetate, cellulose nitrate, acetylene, plastics, and varnishes; as a paint and varnish remover; and as a solvent in the manufacture of pharmaceuticals and chemicals.

Carboxylic Acids and Esters

The odor of vinegar is caused by the presence of acetic acid, a carboxylic acid, in the vinegar. The odor of ripe bananas and many other fruits is due to the presence of esters, compounds that can be prepared by the reaction of a carboxylic acid with an alcohol. Because esters do not have hydrogen bonds between molecules, they have lower vapor pressures than the alcohols and carboxylic acids from which they are derived (see [Figure 20.15](#)).

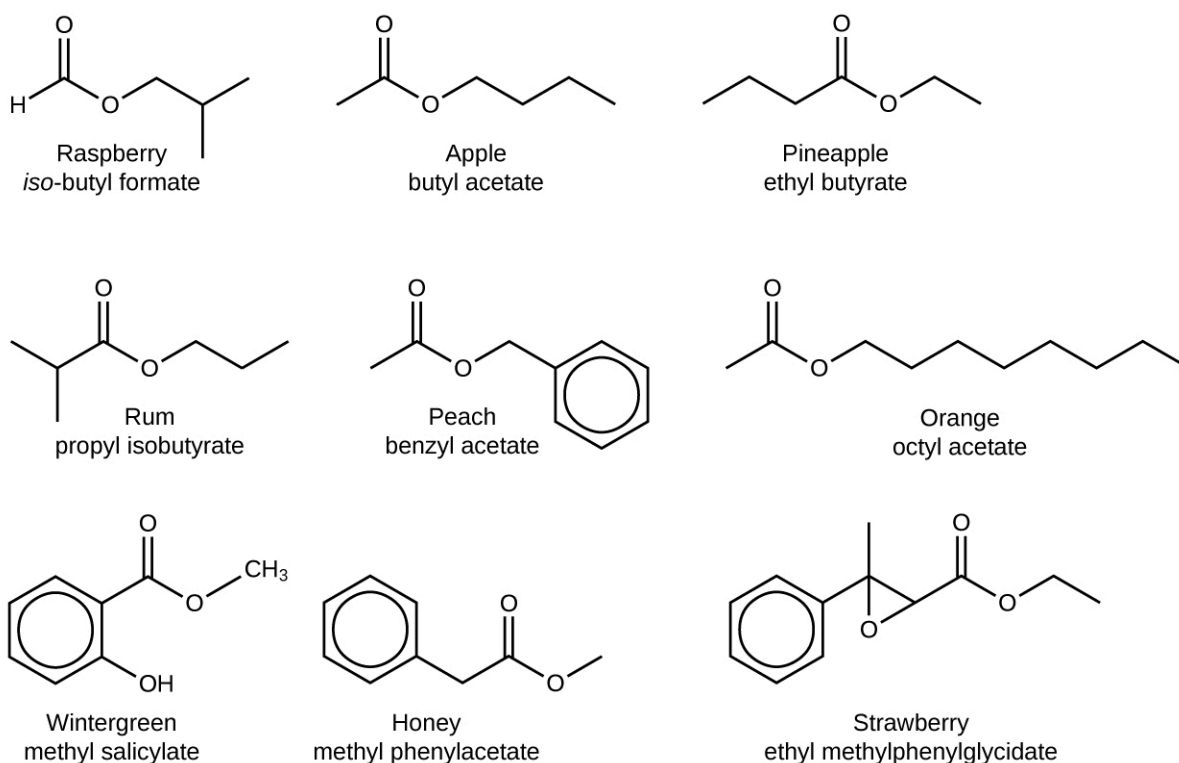
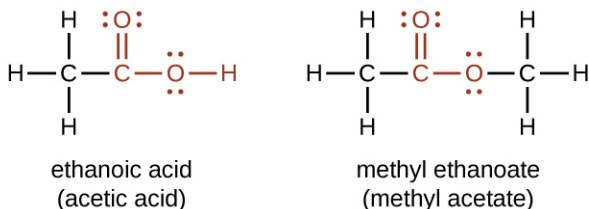


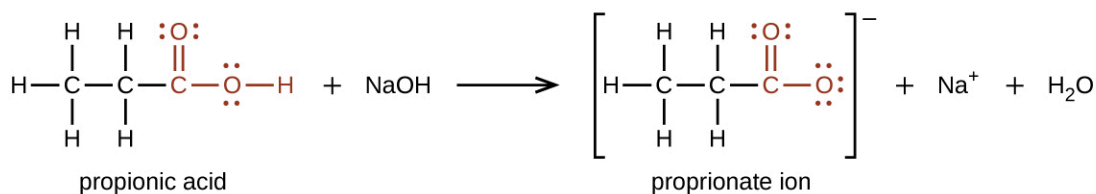
FIGURE 20.15 Esters are responsible for the odors associated with various plants and their fruits.

Both **carboxylic acids** and **esters** contain a carbonyl group with a second oxygen atom bonded to the carbon atom in the carbonyl group by a single bond. In a carboxylic acid, the second oxygen atom also bonds to a hydrogen atom. In an ester, the second oxygen atom bonds to another carbon atom. The names for carboxylic acids and esters include prefixes that denote the lengths of the carbon chains in the molecules and are derived following nomenclature rules similar to those for inorganic acids and salts (see these examples):



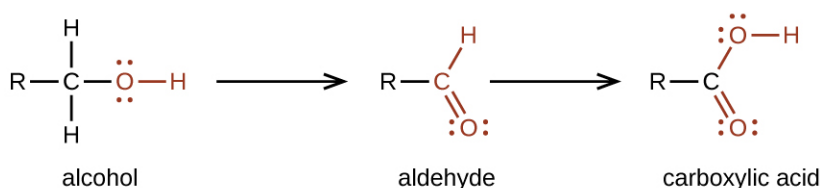
The functional groups for an acid and for an ester are shown in red in these formulas.

The hydrogen atom in the functional group of a carboxylic acid will react with a base to form an ionic salt:

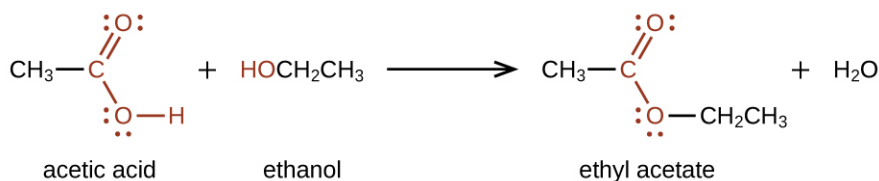


Carboxylic acids are weak acids (see the chapter on acids and bases), meaning they are not 100% ionized in water. Generally only about 1% of the molecules of a carboxylic acid dissolved in water are ionized at any given time. The remaining molecules are undissociated in solution.

We prepare carboxylic acids by the oxidation of aldehydes or alcohols whose $-\text{OH}$ functional group is located on the carbon atom at the end of the chain of carbon atoms in the alcohol:



Esters are produced by the reaction of acids with alcohols. For example, the ester ethyl acetate, $\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_3$, is formed when acetic acid reacts with ethanol:



The simplest carboxylic acid is formic acid, HCO_2H , known since 1670. Its name comes from the Latin word *formicus*, which means “ant”; it was first isolated by the distillation of red ants. It is partially responsible for the pain and irritation of ant and wasp stings, and is responsible for a characteristic odor of ants that can be sometimes detected in their nests.

Acetic acid, $\text{CH}_3\text{CO}_2\text{H}$, constitutes 3–6% vinegar. Cider vinegar is produced by allowing apple juice to ferment without oxygen present. Yeast cells present in the juice carry out the fermentation reactions. The fermentation reactions change the sugar present in the juice to ethanol, then to acetic acid. Pure acetic acid has a penetrating odor and produces painful burns. It is an excellent solvent for many organic and some inorganic compounds, and it is essential in the production of cellulose acetate, a component of many synthetic fibers such as rayon.

The distinctive and attractive odors and flavors of many flowers, perfumes, and ripe fruits are due to the presence of one or more esters (Figure 20.16). Among the most important of the natural esters are fats (such as lard, tallow, and butter) and oils (such as linseed, cottonseed, and olive oils), which are esters of the trihydroxyl alcohol glycerine, $\text{C}_3\text{H}_5(\text{OH})_3$, with large carboxylic acids, such as palmitic acid, $\text{CH}_3(\text{CH}_2)_{14}\text{CO}_2\text{H}$, stearic acid, $\text{CH}_3(\text{CH}_2)_{16}\text{CO}_2\text{H}$, and oleic acid, $\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{CO}_2\text{H}$. Oleic acid is an unsaturated acid; it contains a $\text{C}=\text{C}$ double bond. Palmitic and stearic acids are saturated acids that contain no double or triple bonds.



FIGURE 20.16 Over 350 different volatile molecules (many members of the ester family) have been identified in strawberries. (credit: Rebecca Siegel)

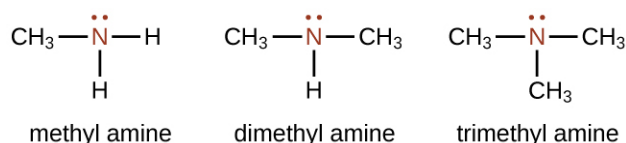
20.4 Amines and Amides

LEARNING OBJECTIVES

By the end of this section, you will be able to:

- Describe the structure and properties of an amine
- Describe the structure and properties of an amide

Amines are molecules that contain carbon-nitrogen bonds. The nitrogen atom in an amine has a lone pair of electrons and three bonds to other atoms, either carbon or hydrogen. Various nomenclatures are used to derive names for amines, but all involve the class-identifying suffix *-ine* as illustrated here for a few simple examples:



In some amines, the nitrogen atom replaces a carbon atom in an aromatic hydrocarbon. Pyridine ([Figure 20.17](#)) is one such heterocyclic amine. A heterocyclic compound contains atoms of two or more different elements in its ring structure.

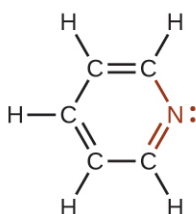


FIGURE 20.17 The illustration shows one of the resonance structures of pyridine.



HOW SCIENCES INTERCONNECT

DNA in Forensics and Paternity

The genetic material for all living things is a polymer of four different molecules, which are themselves a combination of three subunits. The genetic information, the code for developing an organism, is contained in the specific sequence of the four molecules, similar to the way the letters of the alphabet can be sequenced to form words that convey information. The information in a DNA sequence is used to form two other types of polymers, one of which are proteins. The proteins interact to form a specific type of organism with individual characteristics.

A genetic molecule is called DNA, which stands for deoxyribonucleic acid. The four molecules that make up DNA are

called nucleotides. Each nucleotide consists of a single- or double-ringed molecule containing nitrogen, carbon, oxygen, and hydrogen called a nitrogenous base. Each base is bonded to a five-carbon sugar called deoxyribose. The sugar is in turn bonded to a phosphate group ($-\text{PO}_4^{3-}$). When new DNA is made, a polymerization reaction occurs that binds the phosphate group of one nucleotide to the sugar group of a second nucleotide. The nitrogenous bases of each nucleotide stick out from this sugar-phosphate backbone. DNA is actually formed from two such polymers coiled around each other and held together by hydrogen bonds between the nitrogenous bases. Thus, the two backbones are on the outside of the coiled pair of strands, and the bases are on the inside. The shape of the two strands wound around each other is called a double helix (see [Figure 20.18](#)).

It probably makes sense that the sequence of nucleotides in the DNA of a cat differs from those of a dog. But it is also true that the sequences of the DNA in the cells of two individual pugs differ. Likewise, the sequences of DNA in you and a sibling differ (unless your sibling is an identical twin), as do those between you and an unrelated individual. However, the DNA sequences of two related individuals are more similar than the sequences of two unrelated individuals, and these similarities in sequence can be observed in various ways. This is the principle behind DNA fingerprinting, which is a method used to determine whether two DNA samples came from related (or the same) individuals or unrelated individuals.

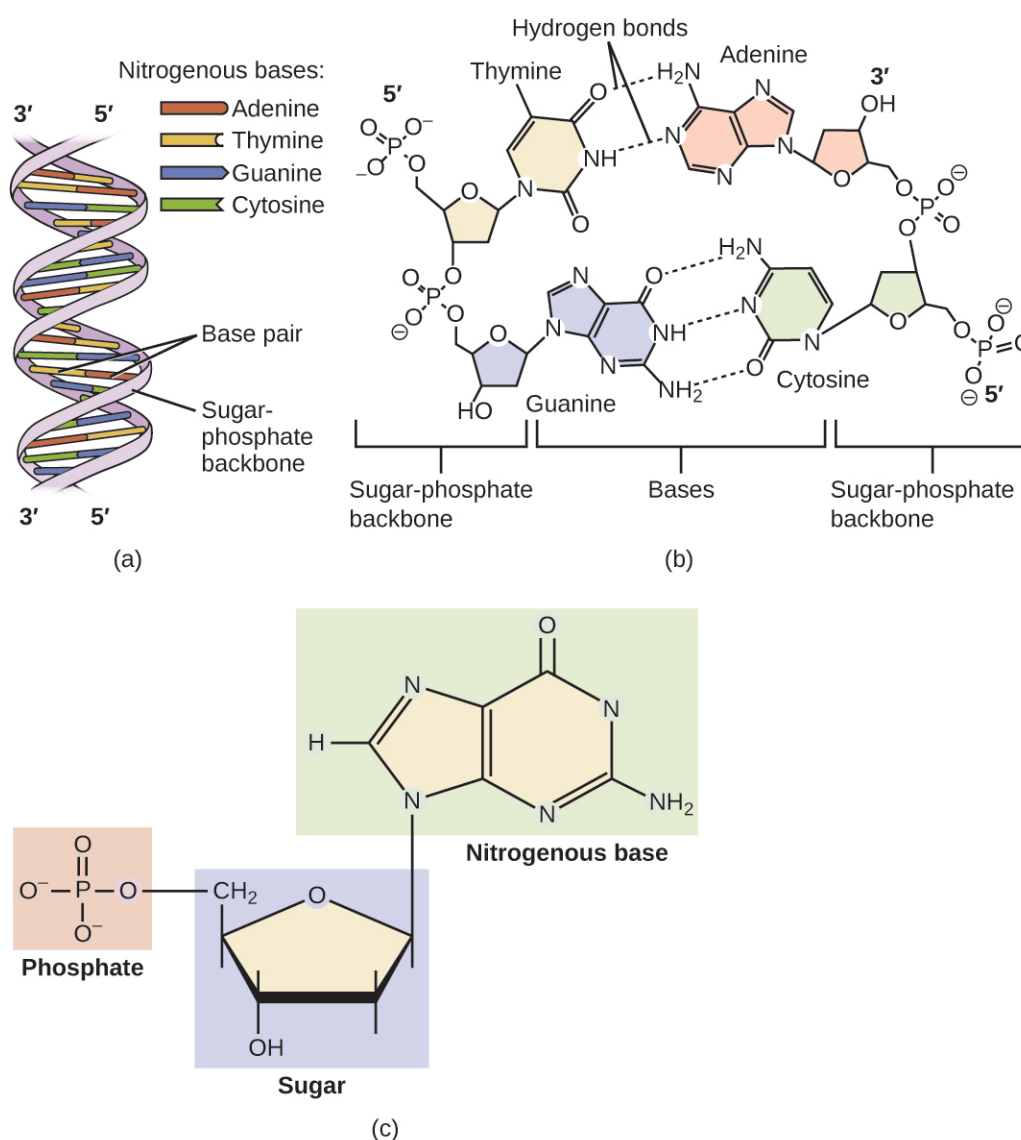


FIGURE 20.18 DNA is an organic molecule and the genetic material for all living organisms. (a) DNA is a double helix consisting of two single DNA strands hydrogen bonded together at each nitrogenous base. (b) This detail shows the hydrogen bonding (dotted lines) between nitrogenous bases on each DNA strand and the way in which each nucleotide is joined to the next, forming a backbone of sugars and

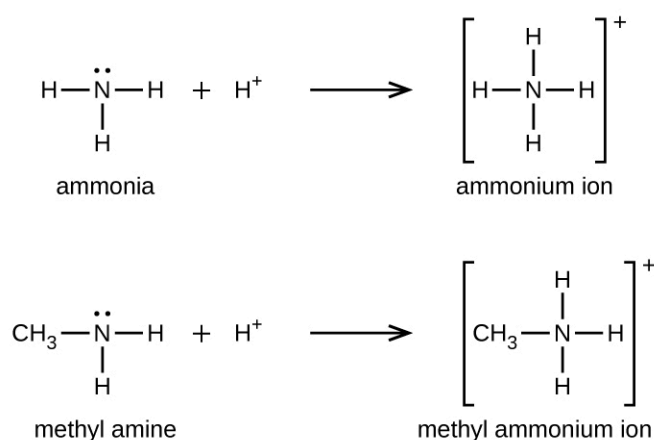
phosphate groups along each strand. (c) This detail shows the structure of one of the four nucleotides that makes up the DNA polymer. Each nucleotide consists of a nitrogenous base (a double-ring molecule, in this case), a five-carbon sugar (deoxyribose), and a phosphate group.

Using similarities in sequences, technicians can determine whether a man is the father of a child (the identity of the mother is rarely in doubt, except in the case of an adopted child and a potential birth mother). Likewise, forensic geneticists can determine whether a crime scene sample of human tissue, such as blood or skin cells, contains DNA that matches exactly the DNA of a suspect.

LINK TO LEARNING

Watch this [video animation \(http://openstax.org/l/16dnapackaging\)](http://openstax.org/l/16dnapackaging) of how DNA is packaged for a visual lesson in its structure.

Like ammonia, amines are weak bases due to the lone pair of electrons on their nitrogen atoms:

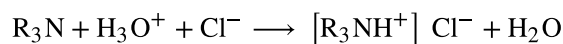


The basicity of an amine's nitrogen atom plays an important role in much of the compound's chemistry. Amine functional groups are found in a wide variety of compounds, including natural and synthetic dyes, polymers, vitamins, and medications such as penicillin and codeine. They are also found in many molecules essential to life, such as amino acids, hormones, neurotransmitters, and DNA.

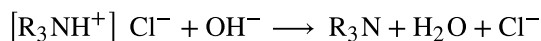
HOW SCIENCES INTERCONNECT

Addictive Alkaloids

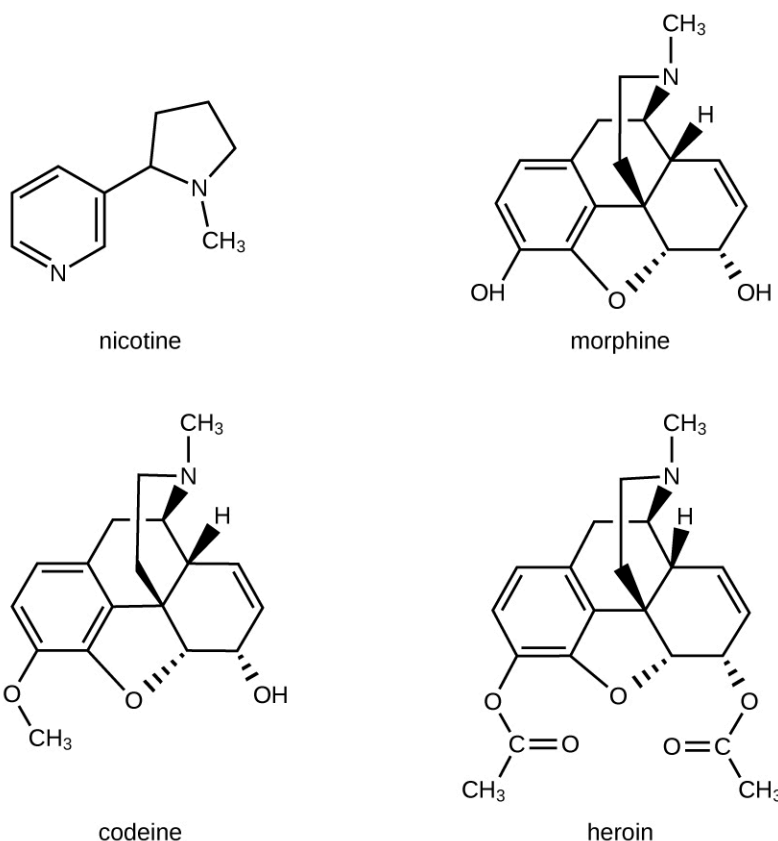
Since ancient times, plants have been used for medicinal purposes. One class of substances, called *alkaloids*, found in many of these plants has been isolated and found to contain cyclic molecules with an amine functional group. These amines are bases. They can react with H_3O^+ in a dilute acid to form an ammonium salt, and this property is used to extract them from the plant:



The name alkaloid means “like an alkali.” Thus, an alkaloid reacts with acid. The free compound can be recovered after extraction by reaction with a base:



The structures of many naturally occurring alkaloids have profound physiological and psychotropic effects in humans. Examples of these drugs include nicotine, morphine, codeine, and heroin. The plant produces these substances, collectively called secondary plant compounds, as chemical defenses against the numerous pests that attempt to feed on the plant:

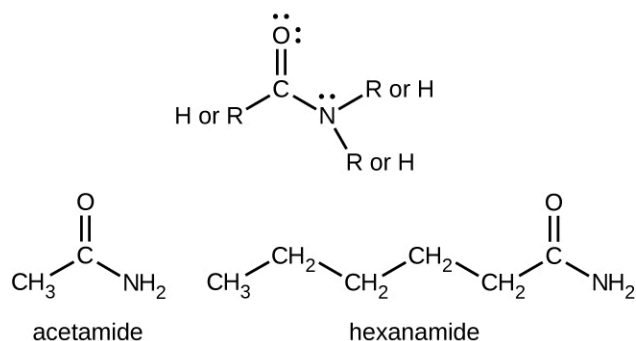


In these diagrams, as is common in representing structures of large organic compounds, carbon atoms in the rings and the hydrogen atoms bonded to them have been omitted for clarity. The solid wedges indicate bonds that extend out of the page. The dashed wedges indicate bonds that extend into the page. Notice that small changes to a part of the molecule change the properties of morphine, codeine, and heroin. Morphine, a strong narcotic used to relieve pain, contains two hydroxyl functional groups, located at the bottom of the molecule in this structural formula. Changing one of these hydroxyl groups to a methyl ether group forms codeine, a less potent drug used as a local anesthetic. If both hydroxyl groups are converted to esters of acetic acid, the powerfully addictive drug heroin results ([Figure 20.19](#)).

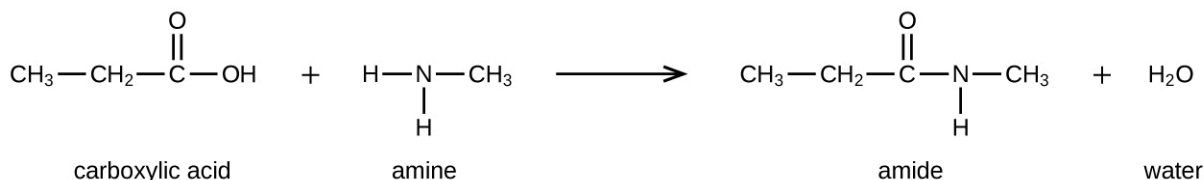


FIGURE 20.19 Poppies can be used in the production of opium, a plant latex that contains morphine from which other opiates, such as heroin, can be synthesized. (credit: Karen Roe)

Amides are molecules that contain nitrogen atoms connected to the carbon atom of a carbonyl group. Like amines, various nomenclature rules may be used to name amides, but all include use of the class-specific suffix *-amide*:



Amides can be produced when carboxylic acids react with amines or ammonia in a process called amidation. A water molecule is eliminated from the reaction, and the amide is formed from the remaining pieces of the carboxylic acid and the amine (note the similarity to formation of an ester from a carboxylic acid and an alcohol discussed in the previous section):



The reaction between amines and carboxylic acids to form amides is biologically important. It is through this reaction that amino acids (molecules containing both amine and carboxylic acid substituents) link together in a polymer to form proteins.



HOW SCIENCES INTERCONNECT

Proteins and Enzymes

Proteins are large biological molecules made up of long chains of smaller molecules called amino acids. Organisms rely on proteins for a variety of functions—proteins transport molecules across cell membranes, replicate DNA, and catalyze metabolic reactions, to name only a few of their functions. The properties of proteins are functions of the combination of amino acids that compose them and can vary greatly. Interactions between amino acid sequences in the chains of proteins result in the folding of the chain into specific, three-dimensional structures that determine the protein's activity.

Amino acids are organic molecules that contain an amine functional group ($-\text{NH}_2$), a carboxylic acid functional group ($-\text{COOH}$), and a side chain (that is specific to each individual amino acid). Most living things build proteins from the same 20 different amino acids. Amino acids connect by the formation of a peptide bond, which is a covalent bond formed between two amino acids when the carboxylic acid group of one amino acid reacts with the amine group of the other amino acid. The formation of the bond results in the production of a molecule of water (in general, reactions that result in the production of water when two other molecules combine are referred to as condensation reactions). The resulting bond—between the carbonyl group carbon atom and the amine nitrogen atom is called a peptide link or peptide bond. Since each of the original amino acids has an unreacted group (one has an unreacted amine and the other an unreacted carboxylic acid), more peptide bonds can form to other amino acids, extending the structure. (Figure 20.20) A chain of connected amino acids is called a polypeptide. Proteins contain at least one long polypeptide chain.

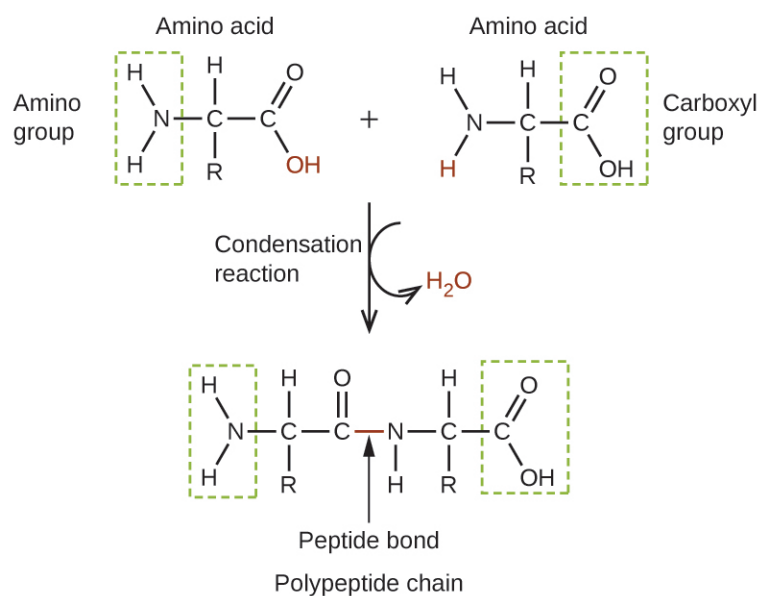


FIGURE 20.20 This condensation reaction forms a dipeptide from two amino acids and leads to the formation of water.

Enzymes are large biological molecules, mostly composed of proteins, which are responsible for the thousands of metabolic processes that occur in living organisms. Enzymes are highly specific catalysts; they speed up the rates of certain reactions. Enzymes function by lowering the activation energy of the reaction they are catalyzing, which can dramatically increase the rate of the reaction. Most reactions catalyzed by enzymes have rates that are millions of times faster than the noncatalyzed version. Like all catalysts, enzymes are not consumed during the reactions that they catalyze. Enzymes do differ from other catalysts in how specific they are for their substrates (the molecules that an enzyme will convert into a different product). Each enzyme is only capable of speeding up one or a few very specific reactions or types of reactions. Since the function of enzymes is so specific, the lack or malfunctioning of an enzyme can lead to serious health consequences. One disease that is the result of an enzyme malfunction is phenylketonuria. In this disease, the enzyme that catalyzes the first step in the degradation of the amino acid phenylalanine is not functional ([Figure 20.21](#)). Untreated, this can lead to an accumulation of phenylalanine, which can lead to intellectual disabilities.

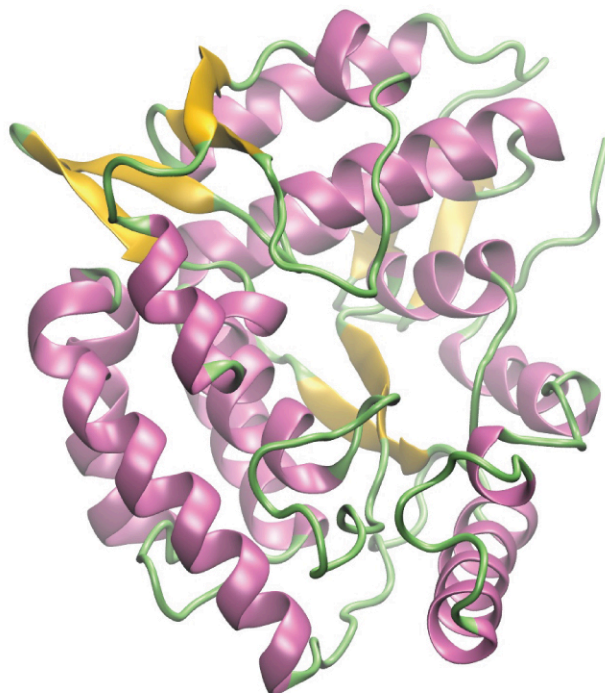


FIGURE 20.21 A computer rendering shows the three-dimensional structure of the enzyme phenylalanine hydroxylase. In the disease phenylketonuria, a defect in the shape of phenylalanine hydroxylase causes it to lose its function in breaking down phenylalanine.

CHEMISTRY IN EVERYDAY LIFE

Kevlar

Kevlar ([Figure 20.22](#)) is a synthetic polymer made from two monomers 1,4-phenylene-diamine and terephthaloyl chloride (Kevlar is a registered trademark of DuPont). The material was developed by Stephanie Kwolek while she worked to find a replacement for steel in tires. Kwolek's work involved synthesizing polyamides and dissolving them in solvents, then spinning the resulting solution into fibers. One of her solutions proved to be quite different in initial appearance and structure. And once spun, the resulting fibers were particularly strong. From this initial discovery, Kevlar was created. The material has a high tensile strength-to-weight ratio (it is about 5 times stronger than an equal weight of steel), making it useful for many applications from bicycle tires to sails to body armor.

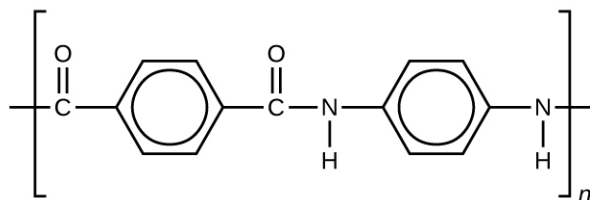


FIGURE 20.22 This illustration shows the formula for polymeric Kevlar.

The material owes much of its strength to hydrogen bonds between polymer chains (refer back to the chapter on intermolecular interactions). These bonds form between the carbonyl group oxygen atom (which has a partial negative charge due to oxygen's electronegativity) on one monomer and the partially positively charged hydrogen atom in the N–H bond of an adjacent monomer in the polymer structure (see dashed line in [Figure 20.23](#)). There is additional strength derived from the interaction between the unhybridized *p* orbitals in the six-membered rings, called aromatic stacking.

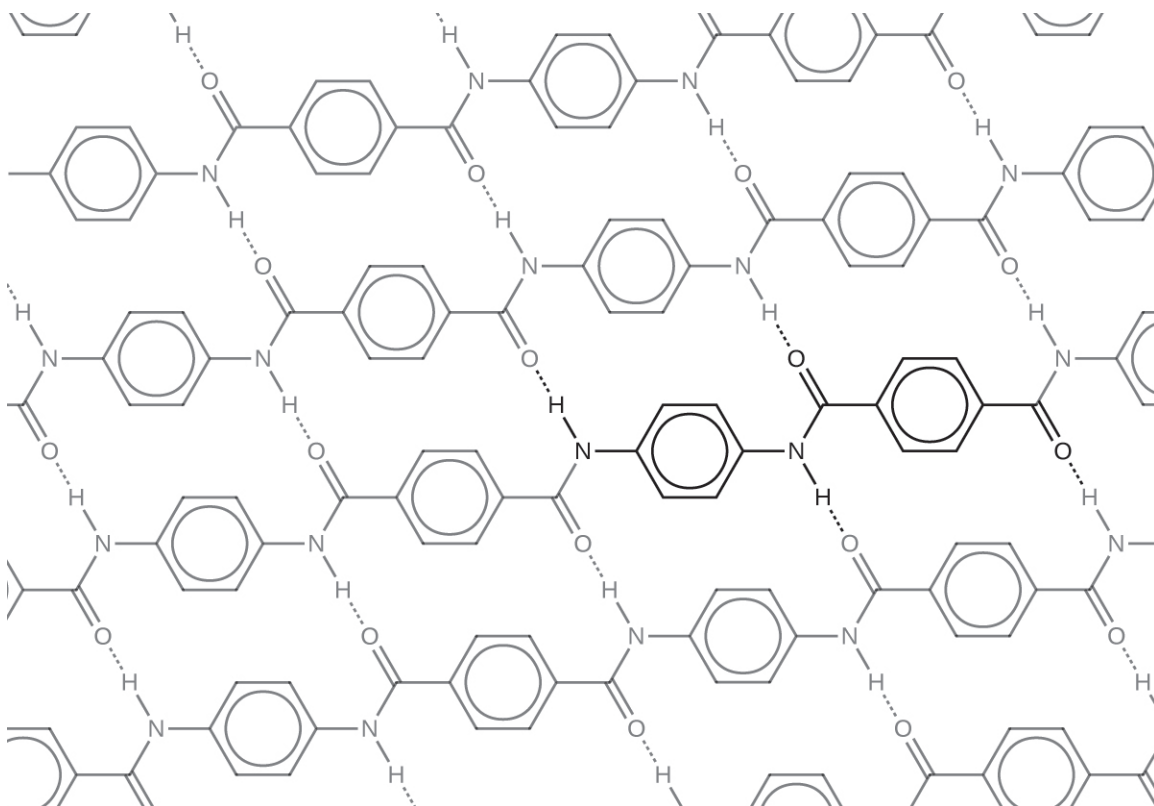


FIGURE 20.23 The diagram shows the polymer structure of Kevlar, with hydrogen bonds between polymer chains represented by dotted lines.

Kevlar may be best known as a component of body armor, combat helmets, and face masks. Since the 1980s, the US military has used Kevlar as a component of the PASGT (personal armor system for ground troops) helmet and vest. Kevlar is also used to protect armored fighting vehicles and aircraft carriers. Civilian applications include protective gear for emergency service personnel such as body armor for police officers and heat-resistant clothing for fire fighters. Kevlar based clothing is considerably lighter and thinner than equivalent gear made from other materials ([Figure 20.24](#)). Beyond Kevlar, Stephanie Kwolek was instrumental in the development of Nomex, a fireproof material, and was also involved in the creation of Lycra. She became just the fourth woman inducted into the National Inventors Hall of Fame, and received a number of other awards for her significant contributions to science and society.



(a)



(b)

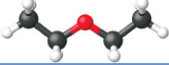
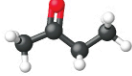
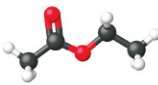


(c)

FIGURE 20.24 (a) These soldiers are sorting through pieces of a Kevlar helmet that helped absorb a grenade blast. Kevlar is also used to make (b) canoes and (c) marine mooring lines. (credit a: modification of work by “Cla68”/Wikimedia Commons; credit b: modification of work by “OakleyOriginals”/Flickr; credit c: modification of work by Casey H. Kyhl)

In addition to its better-known uses, Kevlar is also often used in cryogenics for its very low thermal conductivity (along with its high strength). Kevlar maintains its high strength when cooled to the temperature of liquid nitrogen (-196°C).

The table here summarizes the structures discussed in this chapter:

Compound Name	Structure of Compound and Functional Group (red)	Example	
		Formula	Name
alkene	$\text{C}=\text{C}$	C_2H_4 	ethene
alkyne	$\text{C}\equiv\text{C}$	C_2H_2 	ethyne
alcohol	$\text{R}-\ddot{\text{O}}-\text{H}$	$\text{CH}_3\text{CH}_2\text{OH}$ 	ethanol
ether	$\text{R}-\ddot{\text{O}}-\text{R}'$	$(\text{C}_2\text{H}_5)_2\text{O}$ 	diethyl ether
aldehyde	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\text{H} \end{array}$	CH_3CHO 	ethanal
ketone	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\text{R}' \end{array}$	$\text{CH}_3\text{COCH}_2\text{CH}_3$ 	methyl ethyl ketone
carboxylic acid	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\ddot{\text{O}}-\text{H} \end{array}$	CH_3COOH 	acetic acid
ester	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\ddot{\text{O}}-\text{R}' \end{array}$	$\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_3$ 	ethyl acetate
amine	$\begin{array}{c} \text{R}-\ddot{\text{N}}-\text{H} \\ \\ \text{H} \end{array} \quad \begin{array}{c} \text{R}-\ddot{\text{N}}-\text{H} \\ \\ \text{R}' \end{array} \quad \begin{array}{c} \text{R}-\ddot{\text{N}}-\text{R}'' \\ \\ \text{R}' \end{array}$	$\text{C}_2\text{H}_5\text{NH}_2$ 	ethylamine
amide	$\begin{array}{c} \text{:O:} \\ \\ \text{R}-\text{C}-\ddot{\text{N}}-\text{R}' \\ \\ \text{H} \end{array}$	CH_3CONH_2 	acetamide

Key Terms

addition reaction reaction in which a double carbon-carbon bond forms a single carbon-carbon bond by the addition of a reactant. Typical reaction for an alkene.

alcohol organic compound with a hydroxyl group (-OH) bonded to a carbon atom

aldehyde organic compound containing a carbonyl group bonded to two hydrogen atoms or a hydrogen atom and a carbon substituent

alkane molecule consisting of only carbon and hydrogen atoms connected by single (σ) bonds

alkene molecule consisting of carbon and hydrogen containing at least one carbon-carbon double bond

alkyl group substituent, consisting of an alkane missing one hydrogen atom, attached to a larger structure

alkyne molecule consisting of carbon and hydrogen containing at least one carbon-carbon triple bond

amide organic molecule that features a nitrogen atom connected to the carbon atom in a carbonyl group

amine organic molecule in which a nitrogen atom is bonded to one or more alkyl group

aromatic hydrocarbon cyclic molecule consisting of carbon and hydrogen with delocalized alternating carbon-carbon single and double bonds, resulting in enhanced stability

carbonyl group carbon atom double bonded to an oxygen atom

carboxylic acid organic compound containing a carbonyl group with an attached hydroxyl group

ester organic compound containing a carbonyl group with an attached oxygen atom that is bonded to a carbon substituent

ether organic compound with an oxygen atom that is bonded to two carbon atoms

functional group part of an organic molecule that imparts a specific chemical reactivity to the molecule

ketone organic compound containing a carbonyl group with two carbon substituents attached to it

organic compound natural or synthetic compound that contains carbon

saturated hydrocarbon molecule containing carbon and hydrogen that has only single bonds between carbon atoms

skeletal structure shorthand method of drawing organic molecules in which carbon atoms are represented by the ends of lines and bends in between lines, and hydrogen atoms attached to the carbon atoms are not shown (but are understood to be present by the context of the structure)

substituent branch or functional group that replaces hydrogen atoms in a larger hydrocarbon chain

substitution reaction reaction in which one atom replaces another in a molecule

Summary

20.1 Hydrocarbons

Strong, stable bonds between carbon atoms produce complex molecules containing chains, branches, and rings. The chemistry of these compounds is called organic chemistry. Hydrocarbons are organic compounds composed of only carbon and hydrogen. The alkanes are saturated hydrocarbons—that is, hydrocarbons that contain only single bonds. Alkenes contain one or more carbon-carbon double bonds. Alkynes contain one or more carbon-carbon triple bonds. Aromatic hydrocarbons contain ring structures with delocalized π electron systems.

20.2 Alcohols and Ethers

Many organic compounds that are not hydrocarbons can be thought of as derivatives of hydrocarbons. A hydrocarbon derivative can be formed by replacing one or more hydrogen atoms of a hydrocarbon by a functional group, which contains at least one atom of an element other than carbon or hydrogen. The properties of hydrocarbon derivatives are determined

largely by the functional group. The -OH group is the functional group of an alcohol. The -R-O-R- group is the functional group of an ether.

20.3 Aldehydes, Ketones, Carboxylic Acids, and Esters

Functional groups related to the carbonyl group include the -CHO group of an aldehyde, the -CO- group of a ketone, the $\text{-CO}_2\text{H}$ group of a carboxylic acid, and the $\text{-CO}_2\text{R}$ group of an ester. The carbonyl group, a carbon-oxygen double bond, is the key structure in these classes of organic molecules: Aldehydes contain at least one hydrogen atom attached to the carbonyl carbon atom, ketones contain two carbon groups attached to the carbonyl carbon atom, carboxylic acids contain a hydroxyl group attached to the carbonyl carbon atom, and esters contain an oxygen atom attached to another carbon group connected to the carbonyl carbon atom. All of these compounds contain oxidized carbon atoms relative to the carbon atom of an alcohol group.

20.4 Amines and Amides

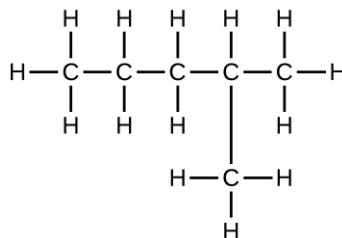
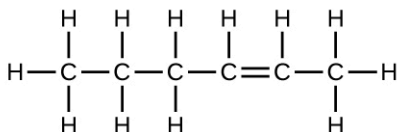
The addition of nitrogen into an organic framework leads to two families of molecules. Compounds containing a nitrogen atom bonded in a hydrocarbon framework are classified as amines. Compounds that

have a nitrogen atom bonded to one side of a carbonyl group are classified as amides. Amines are a basic functional group. Amines and carboxylic acids can combine in a condensation reaction to form amides.

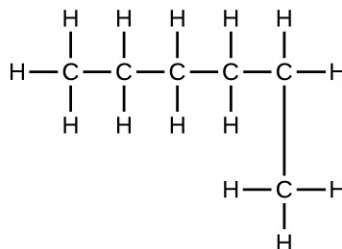
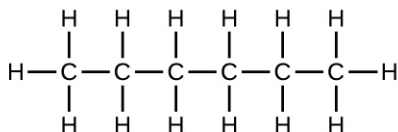
Exercises

20.1 Hydrocarbons

- Write the chemical formula and Lewis structure of the following, each of which contains five carbon atoms:
 - an alkane
 - an alkene
 - an alkyne
- What is the difference between the hybridization of carbon atoms' valence orbitals in saturated and unsaturated hydrocarbons?
- On a microscopic level, how does the reaction of bromine with a saturated hydrocarbon differ from its reaction with an unsaturated hydrocarbon? How are they similar?
- On a microscopic level, how does the reaction of bromine with an alkene differ from its reaction with an alkyne? How are they similar?
- Explain why unbranched alkenes can form geometric isomers while unbranched alkanes cannot. Does this explanation involve the macroscopic domain or the microscopic domain?
- Explain why these two molecules are not isomers:



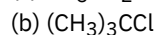
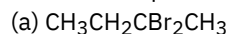
- Explain why these two molecules are not isomers:



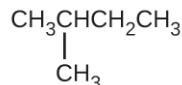
- How does the carbon-atom hybridization change when polyethylene is prepared from ethylene?
- Write the Lewis structure and molecular formula for each of the following hydrocarbons:
 - hexane
 - 3-methylpentane
 - cis*-3-hexene
 - 4-methyl-1-pentene
 - 3-hexyne
 - 4-methyl-2-pentyne
- Write the chemical formula, condensed formula, and Lewis structure for each of the following hydrocarbons:

- (a) heptane
- (b) 3-methylhexane
- (c) *trans*-3-heptene
- (d) 4-methyl-1-hexene
- (e) 2-heptyne
- (f) 3,4-dimethyl-1-pentyne

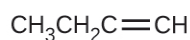
11. Give the complete IUPAC name for each of the following compounds:



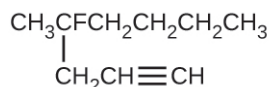
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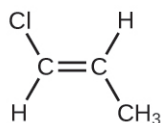
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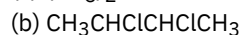
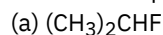
(e)



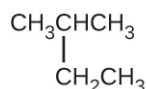
(f)



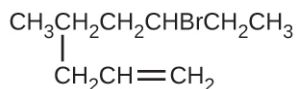
12. Give the complete IUPAC name for each of the following compounds:



(c)



(e)



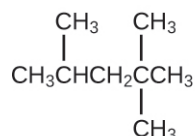
13. Butane is used as a fuel in disposable lighters. Write the Lewis structure for each isomer of butane.

14. Write Lewis structures and name the five structural isomers of hexane.

15. Write Lewis structures for the *cis-trans* isomers of $\text{CH}_3\text{CH}=\text{CHCl}$.

16. Write structures for the three isomers of the aromatic hydrocarbon xylene, $\text{C}_6\text{H}_4(\text{CH}_3)_2$.

17. Isooctane is the common name of the isomer of C_8H_{18} used as the standard of 100 for the gasoline octane rating:



- (a) What is the IUPAC name for the compound?
 - (b) Name the other isomers that contain a five-carbon chain with three methyl substituents.
18. Write Lewis structures and IUPAC names for the alkyne isomers of C_4H_6 .
 19. Write Lewis structures and IUPAC names for all isomers of $\text{C}_4\text{H}_9\text{Cl}$.
 20. Name and write the structures of all isomers of the propyl and butyl alkyl groups.
 21. Write the structures for all the isomers of the $-\text{C}_5\text{H}_{11}$ alkyl group.
 22. Write Lewis structures and describe the molecular geometry at each carbon atom in the following compounds:
 - (a) *cis*-3-hexene
 - (b) *cis*-1-chloro-2-bromoethene
 - (c) 2-pentyne
 - (d) *trans*-6-ethyl-7-methyl-2-octene
 23. Benzene is one of the compounds used as an octane enhancer in unleaded gasoline. It is manufactured by the catalytic conversion of acetylene to benzene:

$$3\text{C}_2\text{H}_2 \longrightarrow \text{C}_6\text{H}_6$$
 Draw Lewis structures for these compounds, with resonance structures as appropriate, and determine the hybridization of the carbon atoms in each.
 24. Teflon is prepared by the polymerization of tetrafluoroethylene. Write the equation that describes the polymerization using Lewis symbols.
 25. Write two complete, balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures.
 - (a) 1 mol of 1-butyne reacts with 2 mol of iodine.
 - (b) Pentane is burned in air.
 26. Write two complete, balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures.
 - (a) 2-butene reacts with chlorine.
 - (b) benzene burns in air.
 27. What mass of 2-bromopropane could be prepared from 25.5 g of propene? Assume a 100% yield of product.
 28. Acetylene is a very weak acid; however, it will react with moist silver(I) oxide and form water and a compound composed of silver and carbon. Addition of a solution of HCl to a 0.2352-g sample of the compound of silver and carbon produced acetylene and 0.2822 g of AgCl.
 - (a) What is the empirical formula of the compound of silver and carbon?
 - (b) The production of acetylene on addition of HCl to the compound of silver and carbon suggests that the carbon is present as the acetylide ion, C_2^{2-} . Write the formula of the compound showing the acetylide ion.
 29. Ethylene can be produced by the pyrolysis of ethane:

$$\text{C}_2\text{H}_6 \longrightarrow \text{C}_2\text{H}_4 + \text{H}_2$$
 How many kilograms of ethylene is produced by the pyrolysis of 1.000×10^3 kg of ethane, assuming a 100.0% yield?

20.2 Alcohols and Ethers

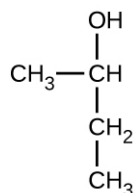
30. Why do the compounds hexane, hexanol, and hexene have such similar names?

31. Write condensed formulas and provide IUPAC names for the following compounds:

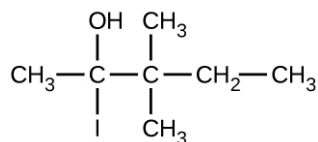
- (a) ethyl alcohol (in beverages)
- (b) methyl alcohol (used as a solvent, for example, in shellac)
- (c) ethylene glycol (antifreeze)
- (d) isopropyl alcohol (used in rubbing alcohol)
- (e) glycerine

32. Give the complete IUPAC name for each of the following compounds:

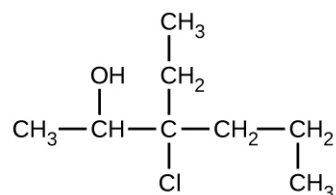
(a)



(b)

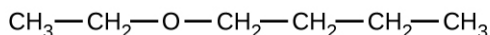


(c)

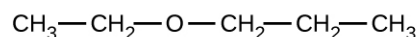


33. Give the complete IUPAC name and the common name for each of the following compounds:

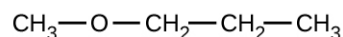
(a)



(b)



(c)



34. Write the condensed structures of both isomers with the formula $\text{C}_2\text{H}_6\text{O}$. Label the functional group of each isomer.

35. Write the condensed structures of all isomers with the formula $\text{C}_2\text{H}_6\text{O}_2$. Label the functional group (or groups) of each isomer.

36. Draw the condensed formulas for each of the following compounds:

- (a) dipropyl ether
- (b) 2,2-dimethyl-3-hexanol
- (c) 2-ethoxybutane

37. MTBE, Methyl *tert*-butyl ether, $\text{CH}_3\text{OC}(\text{CH}_3)_3$, is used as an oxygen source in oxygenated gasolines. MTBE is

manufactured by reacting 2-methylpropene with methanol.

(a) Using Lewis structures, write the chemical equation representing the reaction.

(b) What volume of methanol, density 0.7915 g/mL, is required to produce exactly 1000 kg of MTBE, assuming a 100% yield?

38. Write two complete balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures.

(a) propanol is converted to dipropyl ether

(b) propene is treated with water in dilute acid.

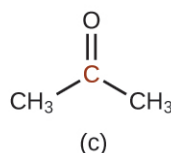
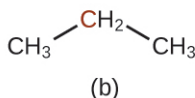
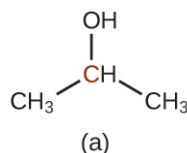
39. Write two complete balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures.

(a) 2-butene is treated with water in dilute acid

(b) ethanol is dehydrated to yield ethene

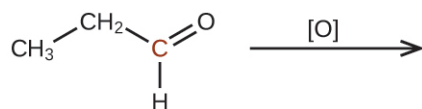
20.3 Aldehydes, Ketones, Carboxylic Acids, and Esters

40. Order the following molecules from least to most oxidized, based on the marked carbon atom:

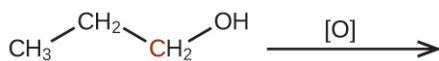


41. Predict the products of oxidizing the molecules shown in this problem. In each case, identify the product that will result from the minimal increase in oxidation state for the highlighted carbon atom:

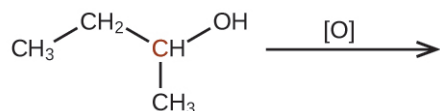
(a)



(b)

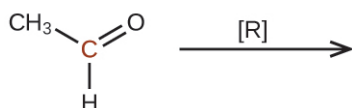


(c)

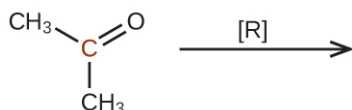


42. Predict the products of reducing the following molecules. In each case, identify the product that will result from the minimal decrease in oxidation state for the highlighted carbon atom:

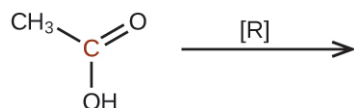
(a)



(b)



(c)



43. Explain why it is not possible to prepare a ketone that contains only two carbon atoms.
44. How does hybridization of the substituted carbon atom change when an alcohol is converted into an aldehyde? An aldehyde to a carboxylic acid?
45. Fatty acids are carboxylic acids that have long hydrocarbon chains attached to a carboxylate group. How does a saturated fatty acid differ from an unsaturated fatty acid? How are they similar?
46. Write a condensed structural formula, such as CH_3CH_3 , and describe the molecular geometry at each carbon atom.
- propene
 - 1-butanol
 - ethyl propyl ether
 - cis*-4-bromo-2-heptene
 - 2,2,3-trimethylhexane
 - formaldehyde
47. Write a condensed structural formula, such as CH_3CH_3 , and describe the molecular geometry at each carbon atom.
- 2-propanol
 - acetone
 - dimethyl ether
 - acetic acid
 - 3-methyl-1-hexene
48. The foul odor of rancid butter is caused by butyric acid, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$.
- Draw the Lewis structure and determine the oxidation number and hybridization for each carbon atom in the molecule.
 - The esters formed from butyric acid are pleasant-smelling compounds found in fruits and used in perfumes. Draw the Lewis structure for the ester formed from the reaction of butyric acid with 2-propanol.
49. Write the two-resonance structures for the acetate ion.
50. Write two complete, balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures:
- ethanol reacts with propionic acid
 - benzoic acid, $\text{C}_6\text{H}_5\text{CO}_2\text{H}$, is added to a solution of sodium hydroxide
51. Write two complete balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures.
- 1-butanol reacts with acetic acid
 - propionic acid is poured onto solid calcium carbonate
52. Yields in organic reactions are sometimes low. What is the percent yield of a process that produces 13.0 g of ethyl acetate from 10.0 g of $\text{CH}_3\text{CO}_2\text{H}$?
53. Alcohols A, B, and C all have the composition $\text{C}_4\text{H}_{10}\text{O}$. Molecules of alcohol A contain a branched carbon chain and can be oxidized to an aldehyde; molecules of alcohol B contain a linear carbon chain and can be oxidized to a ketone; and molecules of alcohol C can be oxidized to neither an aldehyde nor a ketone. Write the Lewis structures of these molecules.

20.4 Amines and Amides

54. Write the Lewis structures of both isomers with the formula $\text{C}_2\text{H}_7\text{N}$.
55. What is the molecular structure about the nitrogen atom in trimethyl amine and in the trimethyl ammonium ion, $(\text{CH}_3)_3\text{NH}^+$? What is the hybridization of the nitrogen atom in trimethyl amine and in the trimethyl ammonium ion?
56. Write the two resonance structures for the pyridinium ion, $\text{C}_5\text{H}_5\text{NH}^+$.
57. Draw Lewis structures for pyridine and its conjugate acid, the pyridinium ion, $\text{C}_5\text{H}_5\text{NH}^+$. What are the hybridizations, electron domain geometries, and molecular geometries about the nitrogen atoms in pyridine and in the pyridinium ion?
58. Write the Lewis structures of all isomers with the formula $\text{C}_3\text{H}_7\text{ON}$ that contain an amide linkage.
59. Write two complete balanced equations for the following reaction, one using condensed formulas and one using Lewis structures.
Methyl amine is added to a solution of HCl.
60. Write two complete, balanced equations for each of the following reactions, one using condensed formulas and one using Lewis structures.
Ethylammonium chloride is added to a solution of sodium hydroxide.
61. Identify any carbon atoms that change hybridization and the change in hybridization during the reactions in [Exercise 20.26](#).
62. Identify any carbon atoms that change hybridization and the change in hybridization during the reactions in [Exercise 20.39](#).
63. Identify any carbon atoms that change hybridization and the change in hybridization during the reactions in [Exercise 20.51](#).